Supplementary material: Sterile Neutrino Dark Matter: Joint BBN and X-ray Constraints 2024–2026

HEP-Genesis-Agent session record
Project: ramp-research-sterile-bbn-xray

2026-11-02

Supplementary material, generated mechanically from the session record: it lists exactly what ran, in order, without interpretation.

A Implementation details

This appendix is generated mechanically from the session record; it lists exactly what ran, in order, without interpretation.

A.1 Session

- Source: cli client; conversation `644f1997`
- Started: 2026-10-21 11:01:50; report generated: 2026-11-02 17:56:02
- Project: ramp-research-sterile-bbn-xray (`/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray`)
- Compute execution site(s) selected in the session: local
- 5 user turn(s), 622 tool call(s) (23 returned errors), 23 figure(s), 45 data file(s)

A.2 Models

Stage	Model	Endpoint
quick	`muse-spark-bf16`	`https://inference-api.alcf.anl.gov/resource_server/minerva/api/v1`
report	`muse-spark-bf16`	`https://inference-api.alcf.anl.gov/resource_server/minerva/api/v1`
coder	`muse-spark-bf16`	`https://inference-api.alcf.anl.gov/resource_server/minerva/api/v1`

A.3 MCP servers

Server	Transport	Location
hep-research-aid	stdio	`python -m hep_genesis.domain.mcp_server`

Server	Transport	Location
alcf-tools	stdio	`python -m hep_genesis.iri.alcf.mcp_server`
nersc-tools	stdio	`python -m hep_genesis.iri.nersc.mcp_server`
paper-search	stdio	`uvx -with mcp<2 paper-search-mcp`
web-search	stdio	`uvx duckduckgo-mcp-server`
cosmic-emulators	http	`https://cosmic.77-42-88-84.sslip.io/mcp`

A.4 Tool-call log

Chronological; arguments and results are truncated for space (the full record is in the project's `provenance/` folder).

#	Tool	Arguments	Outcome
1	`hep-research-aid list_agent_files`	`{}`	`[]`
2	`hep-research-aid list_input_files`	`{}`	`{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }`
3	`hep-research-aid list_datasets`	`{}`	`{'count': 0, 'datasets': []}`
4	`hep-research-aid list_files`	`{}`	`Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...`
5	`hep-research-aid get_dispatch`	`{}`	`Dispatch: local execution.`
6	`serve-sbbn.sh list_sterile_nu_skills`	`{}`	`{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...`
7	`local search_arxiv_abstracts`	`{"query": "sterile neutrino dark matter X-ray constraints 3.5 keV line 2024", "max_results": 8}`	`[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...`

#	Tool	Arguments	Outcome
8	local search_arxiv_ abstracts	{"query": "sterile neutrino big bang nucleosynthesis helium deuterium constraints mixing", "max_results": 8}	[1706.03138v1] Introduction to Big Bang Nucleosynthesis and Modern Cosmology Abstract: Primordial nucleosynthesis remains as one of the pillars of modern cosmology. It is the testing ground upon which...
9	local search_arxiv_ abstracts	{"query": "keV sterile neutrino dark matter parameter space status review", "max_results": 8}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
10	hep-research-aid search_arxiv	{"query": "sterile neutrino dark matter X-ray XRISM eROSITA NuSTAR constraints", "max_results": 10, "sort_ by": "submittedDate"}	Found 10 papers matching query: 'sterile neutrino dark matter X-ray XRISM eROSITA NuSTAR constraints' (source: arXiv API) Paper 1: arXiv ID: 2610.28633v1 Title: Constraints on Millimeter-wave Extraga...
11	hep-research-aid search_arxiv	{"query": "sterile neutrino dark matter lepton asymmetry resonant production BBN constraint", "max_ results": 10, "sort_by": "submittedDate"}	Found 10 papers matching query: 'sterile neutrino dark matter lepton asymmetry resonant production BBN constraint' (source: arXiv API) Paper 1: arXiv ID: 2610.28633v1 Title: Constraints on Millimeter...
12	local web_search	{"query": "arXiv 2024 2025 sterile neutrino dark matter X-ray limits XRISM eROSITA NuSTAR mixing angle"}	Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. We show that cold sterile neutrino DM can be ef...

#	Tool	Arguments	Outcome
13	local search_arxiv_ abstracts	{"max_results": 8, "query": "NuSTAR eROSITA blank sky decaying sterile neutrino limits"}	[1606.02684v1] Neutrino Oscillations With Two Sterile Neutrinos Abstract: This work estimates the probability of μ to e neutrino oscillation with two sterile neutrinos using a 5x5 U-matrix, an ext...
14	local search_arxiv_ abstracts	{"max_results": 8, "query": "Shi-Fuller resonant sterile neutrino dark matter lepton asymmetry Lyman-alpha X-ray"}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
15	local get_arxiv_ metadata	{"arxiv_id": "2507.18752"}	arXiv ID: 2507.18752 Title: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism Authors: Cannon M. Vogel, Helena García Escudero, Julien Fr...
16	local get_arxiv_ metadata	{"arxiv_id": "2502.17374"}	arXiv ID: 2502.17374 Title: Playing with lepton asymmetry at the resonant production of sterile neutrino dark matter Authors: Dmitry Gorbunov, Dmitry Kalashnikov, George Krugan Year: 2025 Abstract: We...
17	local search_arxiv_ abstracts	{"max_results": 6, "query": "eROSITA Final Equatorial Depth Survey dark matter decay line search"}	[0510346v1] The Dark Energy Survey Abstract: We describe the Dark Energy Survey (DES), a proposed optical-near infrared survey of 5000 sq. deg of the South Galactic Cap to ~24th magnitude in SDSS gr...

#	Tool	Arguments	Outcome
18	local search_arxiv_ abstracts	{"max_results": 6, "query": "cold sterile neutrino dark matter produced during reheating inflaton decays X-ray"}	[2111.06065v2] Implications of time-dependent inflaton decay on reheating and dark matter production Abstract: We discuss the production of radiation and dark matter assuming a time-dependent inflaton...
19	serve-sbbn.sh describe_sterile_ nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
20	serve-sbbn.sh list_sterile_nu_ skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
21	local get_arxiv_ metadata	{"arxiv_id": "2507.18752"}	arXiv ID: 2507.18752 Title: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism Authors: Cannon M. Vogel, Helena García Escudero, Julien Fr...
22	local get_arxiv_ metadata	{"arxiv_id": "2502.17374"}	arXiv ID: 2502.17374 Title: Playing with lepton asymmetry at the resonant production of sterile neutrino dark matter Authors: Dmitry Gorbunov, Dmitry Kalashnikov, George Krugan Year: 2025 Abstract: We...

#	Tool	Arguments	Outcome
23	local get_arxiv_ metadata	{"arxiv_id": "2601.03346"}	arXiv ID: 2601.03346 Title: Sterile Neutrino Dark Matter as a Probe of Inflationary Reheating Authors: James M. Cline, Yong Xu Year: 2026 Abstract: Sterile neutrinos offer a minimal and testable expla...
24	serve-sbbn.sh predict_sterile_ nu_xray_line	{"m_s_eV": 7100, "sin2_ 2theta": 1e-10}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 3.55 keV\nDecay...
25	serve-sbbn.sh load_sterile_nu_ skill	{"name": "sterile-nu-bbn-constraints"}	{ "status": "ok", "files": [], "message": "Loaded skill 'sterile-nu-bbn-constraints'. Follow these instructions.", "metadata": { "instructions": "--\nname: sterile-nu-bbn-constraints\ ndes...
26	serve-sbbn.sh predict_sterile_ nu_xray_line	{"m_s_eV": 20000, "sin2_ 2theta": 1e-12}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 10 keV\nDecay ra...
27	serve-sbbn.sh predict_sterile_ nu_grid	{"masses_eV": [7100, 10000, 20000, 50000], "sin2_ 2thetas": [1e-10, 1e-12, 1e-14], "output_dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261021_110150"}	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261021_110150/ sterile_nu_grid_DW_ 5199ae.csv"], "message": "Predicted 12 ...

#	Tool	Arguments	Outcome
28	local get_arxiv_ metadata	{"arxiv_id": "2609.31340"}	arXiv ID: 2609.31340 Title: Natural Saturation Of The Sterile Neutrino Dark Matter Resonant Production By a High Lepton Flavor Asymmetry In Primordial Plasma Authors: Dmitry Gorbunov, Dmitry Kalashnik...
29	serve-sbbn.sh describe_sterile_ nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
30	local download_arxiv_ full_text	{"arxiv_id": "2507.18752", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261021_110150"}	Downloaded: / Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261021_110150/ 2507.18752.txt
31	serve-sbbn.sh predict_sterile_ nu_grid	{"masses_eV": [10000, 20000, 30000], "sin2_ 2thetas": [1e-14, 1e-13], "mechanism": "SF", "lepton_ asymmetry": 0.1, "output_ dir": "/Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261021_...	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261021_110150/ sterile_nu_grid_SF_ fdf543.csv"], "message": "Predicted 6 p...
32	serve-sbbn.sh predict_sterile_ nu	{"m_s_eV": 20000, "sin2_2theta": 1e-14, "mechanism": "SF", "lepton_ asymmetry": 0.1}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 2e+04 eV, sin\u00b22\ u03b8 = 1e-14\nMechanism: SF (SF Delta N_eff is a heuristic (0.1 x the DW value), not a resonant-productio...

#	Tool	Arguments	Outcome
33	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 1000, "m_max_eV": 100000, "theta_min": 1e-15, "theta_max": 1e-06, "n_points": 100, "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/26102...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/scan_DW_dc7743.csv", "/Users/nramachandra/hep-genesis-testi...
34	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...
35	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/bbn_vs_neff.png"], "message": "Plot saved to /Users/nramac... [Fig. 6]
36	serve-sbbn.sh predict_sterile_nu_grid	{"lepton_asymmetry": 0.001, "masses_eV": [10000, 20000, 30000], "mechanism": "SF", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150", "sin2_2thetas": [1e-1...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_SF_d81486.csv"], "message": "Predicted 3 p...
37	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2507.18752.txt", "start": 0, "length": 9000}	[chars 0-9000 of 62265] (more available, next start=9000) Title: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism Authors: Cannon M. Vo...

#	Tool	Arguments	Outcome
38	serve-sbbn.sh plot_sterile_nu_exclusion	{"scan_file": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/scan_DW_dc7743.npz", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/exclusion_scan_DW_dc7743.png"], "message": "Exclusion plot... [Fig. 8]
39	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1000, 2000, 3000, 5000, 7100, 10000, 20000, 50000, 100000], "sin2_2thetas": [1e-12, 3e-12, 1e-11, 3e-11, 1e-10, 3e-10, 1e-09, 3e-09, 1e-08, 3e-08, 1e-07, 3e-07], "output_dir": "/Users/nramachandra/hep-gene...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_397ef0.csv"], "message": "Predicted 108...
40	local search_arxiv_abstracts	{"query": "XRISM 3.5 keV line dark matter decay", "max_results": 6}	[2505.07479v2] Thermal and Kinematic Properties of Ejecta in SN1987A revealed by XRISM Abstract: We present an analysis of high-resolution spectra from the shock-heated plasmas in SN~1987A, based on a...
41	local search_arxiv_abstracts	{"query": "eROSITA sterile neutrino dark matter decay line limits", "max_results": 6}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...

#	Tool	Arguments	Outcome
42	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_397ef0.csv"}	m_s_eV,sin2_2theta, delta_neff,neff,omega_s_h2,Yp,DH_x1e5,Li7H_x1e10,chi2_bbn,delta_chi2_bbn,lifetime_s, free_streaming_kpc,bbn_excluded_95,overcloses, bbn_treatment_valid 1000, 1e-12,1.28919e-06,3.044, 1....
43	local search_arxiv_abstracts	{"query": "NuSTAR sterile neutrino dark matter limits Galactic", "max_results": 6}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
44	local search_arxiv_abstracts	{"query": "INTEGRAL SPI sterile neutrino dark matter decay line", "max_results": 5}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
45	board post_finding	{"claim": "All four survey arXiv IDs resolve: 2507.18752 (Vogel, Garcia Escudero, Froustey, Abazajian 2025, Shi-Fuller revival) , 2502.17374 (Gorbunov, Kalashnikov, Krugan 2025, flavor-dependent asymmetry), 2601.03346 (Cl...	Recorded finding f-001.
46	board post_finding	{"claim": "2507.18752 (full text) states: L >~ 0.5 at high T opens viable space at m_s >~ 10 keV and sin^2(2theta) <~ 1e-14; structure bounds used are m_th > 8.4 keV (galaxy counts, 95% CL), m_th > 8.5 keV (counts + stro...	Recorded finding f-002.

#	Tool	Arguments	Outcome
47	board post_finding	{"claim": "The sterile-nu server cannot directly test the 2507.18752 claim. Its SF mode caps lepton_ asymmetry at 0.1 (the paper needs L >~ 0.5), its SF Delta N_eff is a heuristic (0.1 x the DW value), and the asymmetry i...	Recorded finding f-003.
48	local download_arxiv_ full_text	{"arxiv_id": "2405.17861", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2405.17861.txt
49	local cite	{"arxiv_id": "2507.18752", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/references.bib", "note": "[\"These studies find no evidence of monochromatic deca...	Cited: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism (\cite{arxiv250718752})
50	serve-sbbn.sh predict_sterile_ nu_grid	{"masses_eV": [1000, 2000, 3000, 5000, 7100, 10000, 20000, 50000, 100000], "sin2_2thetas": [8.67e-09, 2.49e-09, 1.2e-09, 4.785e-10, 2.55e-10, 1.37e-10, 3.946e-11, 7.58e-12, 2.18e-12], "combine": "zip", "output_ dir": "/Us...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_ 360bc1.csv"], "message": "Predicted 9 p...
51	local download_arxiv_ full_text	{"arxiv_id": "2510.24560", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt

#	Tool	Arguments	Outcome
52	hep-research-aid search_arxiv	{"query": "eROSITA eFEDS dark matter decay line sterile neutrino", "max_results": 6}	Found 6 papers matching query: 'eROSITA eFEDS dark matter decay line sterile neutrino' (source: arXiv API) Paper 1: arXiv ID: 1702.08430v1 Title: keV sterile neutrino Dark Matter Authors: Alexander M...
53	local search_arxiv_abstracts	{"query": "warm dark matter thermal relic mass bound Lyman-alpha Milky Way satellites sterile neutrino 2024", "max_results": 5}	[1702.08430v1] keV sterile neutrino Dark Matter Abstract: We give an overview of the current status of keV sterile neutrino Dark Matter. After a short introduction, we start by a general discussion of...
54	board post_finding	{"claim": "Mechanism map (from literature quotes, not computed by tools) . Shi-Fuller resonant production with L >~ 0.5 above 20 MeV is viable at m_s >~ 10 keV and sin^2(2theta) <~ 1e-14 (2507.18752). Its X-ray ceilings a...	Recorded finding f-004.
55	board post_finding	{"claim": "The server's radiative decay rate agrees with Gamma = 1.38e-32 s^-1 (sin^2 2theta/1e-10)(m_s/keV)^5 to within about 1.5%. At m_s = 7.1 keV and sin^2 2theta = 1e-10 the tool gives 2.46e-28 s^-1 (formula 2.49e-2...	Recorded finding f-005.
56	board post_finding	{"claim": "DW benchmark grid (m_s = 7.1, 10, 20, 50 keV; sin^2 2theta = 1e-10, 1e-12, 1e-14): no point is BBN-excluded (Delta N_eff <= 2.9e-3, |Delta chi^2| <= 0.04). DW Omega_s h^2 reaches 0.12 only near sin^2 2theta ~ ...	Recorded finding f-006.

#	Tool	Arguments	Outcome
57	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt", "length": 5000}	[chars 0-5000 of 67438] (more available, next start=5000) Title: XRISM constraints on unidentified X-ray emission lines, including the 3.5 keV line, in the stacked spectrum of ten galaxy clusters Aut...
58	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2405.17861.txt", "length": 4000}	[chars 0-4000 of 44941] (more available, next start=4000) Title: Strong limits on keV-scale galactic sterile neutrino dark matter with stray light from NuSTAR after 11 years of operation Authors: R. ...
59	board post_question	{"text": "A real Shi-Fuller BBN test needs Yp computed with a nu_e degeneracy xi_e(T~1 MeV) taken from the L-evolution in 2507.18752. That requires a BBN code that accepts xi_e, such as PRIMAT, AlterBBN or PRyMordial wit...	Recorded question q-07.
60	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2405.17861.txt", "start": 14000, "length": 9000}	[chars 14000-23000 of 44941] (more available, next start=23000) rmalization are expressed in units of cts/cm2/s/keV/deg2. We start by fitting aCXB spectrum within our base model, that is without a li...
61	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt", "start": 9000, "length": 5000}	[chars 9000-14000 of 67438] (more available, next start=14000) th and Space Science, Osaka University, Osaka 560-0043, Japan 34Department of Physics, University of Wisconsin, WI 53706, USA 35Departme...

#	Tool	Arguments	Outcome
62	local cite	{"arxiv_id": "2510.24560", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/references.bib", "note": "[\"No unidentified lines are detected in our stacked cl...	Cited: XRISM constraints on unidentified X-ray emission lines, including the 3.5 keV line, in the stacked spectrum of ten galaxy clusters (\cite{arxiv251024560})
63	board post_finding	{"claim": "Sterile-nu server assumptions: BBN abundances come from a fit to PRyMordial (<0.3%), and the only BBN input is Delta N_eff. The fit is valid for N_eff in [2.044, 6.044], omega_b h^2 in [0.019, 0.0257] and tau_...	Recorded finding f-007.
64	board post_finding	{"claim": "DW scan (m_s 1-100 keV, sin^2 2theta 1e-15 to 1e-6, 100x100 grid): BBN (Delta chi^2 > 3.84) excludes only sin^2 2theta above 3.5e-7 (1 keV), 1.2e-7 (3.7 keV), 8.1e-8 (7 keV), 4.3e-8 (13.5 keV), 1.5e-8 (50 keV)...	Recorded finding f-008.
65	board post_finding	{"claim": "Tool-computed BBN at representative DW points, compared with the brief's Yp = 0.245 +/- 0.003 and D/H = (2.53 +/- 0.03)e-5. SM (N_eff = 3.044): Yp = 0.246815 (+0.6 sigma), D/H = 2.4433e-5 (-2.9 sigma). 7.1 keV...	Recorded finding f-009.
66	board post_finding	{"claim": "BEST cross-check (m_s = 1.14 eV, sin^2 2theta = 0.42): the server gives Delta N_eff = 1 (N_eff = 4.044), Yp = 0.25956, D/H = 2.777e-5 and Delta chi^2_BBN = 23.1 (excluded). That is 6.2 sigma tension with Planc...	Recorded finding f-010.

#	Tool	Arguments	Outcome
67	board post_finding	{"claim": "The server's DW relic density scales as Omega_s h^2 = 0.1 (sin^2 2theta / 1e-9)(m_s / 3 keV)^ 1.8. Checks: 3 keV at 1e-9 gives 0.100, 10 keV at 1e-9 gives 0.873 (ratio 8.73 = (10/3)^1.8), and 1 keV at 1e-8 give...	Recorded finding f-011.
68	board post_finding	{"claim": "XRISM Resolve stacked 3.75 Ms of data on ten galaxy clusters (arXiv: 2510.24560, 2025) and found no unidentified lines over 2.5-15 keV. Its 3-sigma upper limit on the decay rate of an m_s ~ 7.1 keV particle (E ...	Recorded finding f-012.
69	board post_finding	{"claim": "Krivonos, Barinov, Mukhin and Gorbunov (arXiv:2405.17861, 2024) use 11 years of NuSTAR stray-light data on the Milky Way. They set 95% CL upper limits on sin^2 2theta for m_s ~ 6-40 keV (photon energies 3-20 k...	Recorded finding f-013.
70	board post_dead_end	{"what": "Searches with search_arxiv_abstracts and search_arxiv for 2024-2026 eROSITA and INTEGRAL/SPI sterile-neutrino line limits ('eROSITA sterile neutrino dark matter decay line limits', 'eROSITA eFEDS dark matter de...	Recorded dead end x-01.
71	board post_question	{"text": "What are the numerical sin^2 2theta upper limits at m_s = 7, 10, 20 keV in Fig. 3 of arXiv: 2405.17861 (NuSTAR stray light), and at 50 keV from INTEGRAL/SPI or NuSTAR GC? Answering this needs get_ paper_figures o...	Recorded question q-08.

#	Tool	Arguments	Outcome
72	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2507.18752.txt"}	Title: Return of the Lepton Number: Sterile Neutrino Dark Matter Production and the Revival of the Shi-Fuller Mechanism Authors: Cannon M. Vogel, Helena García Escudero, Julien Froustey, Kevork N. Aba...
73	local download_arxiv_full_text	{"arxiv_id": "2405.17861", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2405.17861.txt
74	local download_arxiv_full_text	{"arxiv_id": "2510.24560", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt
75	local download_arxiv_full_text	{"arxiv_id": "1702.08430", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1702.08430.txt
76	local search_arxiv_abstracts	{"query": "Dodelson-Widrow sterile neutrino relic abundance QCD transition uncertainty", "max_results": 5}	[1512.05369v1] Dodelson-Widrow Production of Sterile Neutrino Dark Matter with Non-Trivial Initial Abundance Abstract: The simplest way to create sterile neutrinos in the early Universe is by their ad...

#	Tool	Arguments	Outcome
77	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2405.17861.txt", "start": 0, "length": 14000}	[chars 0-14000 of 44941] (more available, next start=14000) Title: Strong limits on keV-scale galactic sterile neutrino dark matter with stray light from NuSTAR after 11 years of operation Authors: R...
78	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt", "start": 0, "length": 5000}	[chars 0-5000 of 67438] (more available, next start=5000) Title: XRISM constraints on unidentified X-ray emission lines, including the 3.5 keV line, in the stacked spectrum of ten galaxy clusters Aut...
79	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [3000, 7100, 10000, 20000, 50000], "sin2_2thetas": [1.2e-09, 2.55e-10, 1.37e-10, 3.9e-11, 7.6e-12], "combine": "zip", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bb...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_2461e9.csv"], "message": "Predicted 5 p...
80	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2405.17861.txt", "start": 14000, "length": 9000}	[chars 14000-23000 of 44941] (more available, next start=23000) rmalization are expressed in units of cts/cm2/s/keV/deg2. We start by fitting aCXB spectrum within our base model, that is without a li...
81	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt", "start": 7000, "length": 5000}	[chars 7000-12000 of 67438] (more available, next start=12000) Research, Leiden, The Netherlands 13ESO, Karl-Schwarzschild-Strasse 2, 85748, Garching bei M¨ nchen, Germany 14Centre for Extragalactic ...

#	Tool	Arguments	Outcome
82	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1702.08430.txt", "start": 250000, "length": 8000}	[chars 250000-250000 of 21223]
83	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1702.08430.txt", "length": 21223, "start": 0}	[chars 0-20381 of 21223] (more available, next start=20381) Title: keV sterile neutrino Dark Matter Authors: Alexander Merle Year: 2017 arXiv ID: 1702.08430 ==========...
84	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt", "length": 4500, "start": 12000}	[chars 12000-16500 of 67438] (more available, next start=16500) Cambridge, MA 02138, USA 70School of Physics & Astronomy, University of Nottingham, University Park, Nottingham NG7 2RD, UK -- Page ...
85	hep-research-aid get_paper_figures	{"paper_id": "2405.17861", "figures": "1,3", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/figures_2405.17861/fig01.png"], "message": "Extracted 1 of 9 figures... [Fig. 15]
86	local search_arxiv_abstracts	{"query": "INTEGRAL SPI sterile neutrino decay line limit Milky Way 2024", "max_results": 6}	[1011.3046v2] Muon Capture Constraints on Sterile Neutrino Properties Abstract: We show that ordinary and radiative muon capture impose stringent constraints on sterile neutrino properties. In particu...

#	Tool	Arguments	Outcome
87	local download_arxiv_full_text	{"arxiv_id": "1705.01837", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1705.01837.txt
88	board post_finding	{"claim": "f-002 re-supported verbatim from 2507.18752 full text: \ "lepton asymmetries L?0.5 at high temperatures open significant viable parameter space in the m_s?10 keV and sin^2(2θ)?10^-14 regime, compatible with bot...	Recorded finding f-014.
89	board post_finding	{"claim": "f-012 re-supported verbatim from XRISM 2510.24560 abstract: \"We stack 3.75 Megaseconds of early XRISM Resolve observations of ten galaxy clusters to search for unidentified spectral lines in the E= 2.5-15 keV...	Recorded finding f-015.
90	board post_finding	{"claim": "f-013 partially re-supported, but the requested figure readings were not obtained. Verbatim from 2405.17861: Eq. 2 reads \"$\Gamma\gamma$ = 9/1024 α/π^4 G_F^2 m_s^5 sin^2 2θ = 1.36$\times$10^-22 (m_s/1keV)^5 sin^2 2θ s^-1\". The...	Recorded finding f-016.
91	board post_dead_end	{"what": "Targeted search_arxiv_abstracts for a 2024-2026 INTEGRAL/SPI sterile-neutrino line limit at m_s >= 40 keV", "why": "Returned only unrelated papers (muon capture, oscillation studies, a 2008 MW heating paper), s...	Recorded dead end x-02.

#	Tool	Arguments	Outcome
92	local search_arxiv_ abstracts	{"max_results": 5, "query": "Viel 2005 constraining warm dark matter Lyman-alpha sterile neutrino thermal relic mass relation"}	[0501562v2] Constraining Warm Dark Matter candidates including sterile neutrinos and light gravitinos with WMAP and the Lyman-alpha forest Abstract: The matter power spectrum at comoving scales of (...
93	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1705.01837.txt", "start": 60000, "length": 15000}	[chars 60000-75000 of 150256] (more available, next start=75000) reheating temperature, the production of sterile neutrino dark matter proceeds via partial thermalization due to low temperatures and...
94	serve-sbbn.sh predict_sterile_ nu_xray_line	{"m_s_eV": 50000, "sin2_ 2theta": 3e-14, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 50 keV, sin\u00b22\u03b8 = 3e-14\n\nE_\u03b3 = 25 keV\nDecay ra...
95	serve-sbbn.sh predict_sterile_ nu_xray_line	{"m_s_eV": 10000, "sin2_ 2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 5 keV\nDecay rat...
96	local cite	{"arxiv_id": "2405.17861", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/references.bib", "note": "[\"In the energy range of 3-20 keV covered by the NuSTA...	Cited: Strong limits on keV-scale galactic sterile neutrino dark matter with stray light from NuSTAR after 11 years of operation (\ cite{arxiv240517861})

#	Tool	Arguments	Outcome
97	local read_arxiv_chunk	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1705.01837.txt", "length": 14000, "start": 46000}	[chars 46000-60000 of 150256] (more available, next start=60000) ation. The damping coe?cient is given by Eq. (29). The rotation vector V(p) has the following components Vx(p) = δm2 2p sin 2θ0 Vy(p) ...
98	board post_finding	{"claim": "X-ray limit table (95% CL unless noted), each limit cross-checked with the server's line tool:\n- 7.1 keV: XRISM gives Gamma < 1.0e-27 s^-1 at 3 sigma, which converts to sin^2 2θ < ~4e-10 (quote in f-015). NuS...	Recorded finding f-017.
99	board post_finding	{"claim": "Final viability table in the (m_s, sin^2 2θ) plane:\n- DW (non-resonant): not viable as all of the dark matter. The DM line sits at sin^2 2θ = 2.55e-10 (7.1 keV), 1.37e-10 (10 keV), 3.9e-11 (20 keV) and 7.6e-1...	Recorded finding f-018.
100	local download_arxiv_full_text	{"arxiv_id": "astro-ph/0501562", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Downloaded: /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/astro-ph_0501562.txt
101	local cite	{"arxiv_id": "1705.01837", "bib_path": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/references.bib", "note": "[\"For Dodelson-Widrow production the proper mixing req...	Cited: Sterile neutrinos in cosmology (\cite{arxiv170501837})

#	Tool	Arguments	Outcome
102	board post_finding	{"claim": "The published DW normalization (Abazajian 2017 review, arXiv: 1705.01837 Eq. 37, a fit from Ref. [120]) is m_s ~ 3.40 keV (sin^2 2theta/1e-8) ^-0.615 (Omega_DM/0.26)^0.5 x 0.527 erfc[-1.15 (T_QCD/ 170 MeV)^2.15]....	Recorded finding f-019.
103	board post_finding	{"claim": "At 7.1 keV the DW verdict depends on the normalization when compared with the XRISM limit sin^ 2 2theta < ~4e-10 (f-012). The server line (2.55e-10) survives XRISM. The brief's line (7.65e-10) and the published...	Recorded finding f-020.
104	board post_finding	{"claim": "On the DW DM line, the standard DW-to-thermal mapping m_ s = 4.43 keV (m_th/keV)^ (4/3) (Viel et al. 2005, astro-ph/0501562; text downloaded, but the formula was not quoted verbatim from it) gives m_th = 0.75 ke...	Recorded finding f-021.
105	board post_finding	{"claim": "BBN robustness check. I used linearized f-009 slopes (dYp/dN_eff = 0.0135, dD/H/dN_eff = 0.331e-5) about the server SM values (Yp = 0.246815, D/H = 2.4433e-5). The 95% upper bounds on Delta N_eff are given as ...	Recorded finding f-022.
106	board post_question	{"text": "Should the sterile-nu server's DW relic formula (Omega ~ 0.1 (s/1e-9)(m/3 keV)^ 1.8) be replaced with the Abazajian 2006/2017 fit, which requires 10-17x more mixing on the DM line, and is the server's SM D/H = 2...	Recorded question q-09.
107	hep-research-aid list_agent_files	{}	[]

#	Tool	Arguments	Outcome
108	hep-research-aid list_files	{}	Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...
109	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
110	hep-research-aid list_datasets	{}	{"count": 0, "datasets": []}
111	hep-research-aid get_dispatch	{}	Dispatch: local execution.
112	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
113	local web_search	{"query": "sterile neutrino dark matter arXiv 2024 2025"}	Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. We show that cold sterile neutrino DM can be ef...
114	local web_search	{"query": "sterile neutrino BBN constraints Yp D/H N_eff arXiv 2024"}	In this work we revisit the question of whether Cosmology can be made compatible with scenarios with light sterile neutrinos, as invoked to explain the SBL anomalies, in the presence of self-interacti...

#	Tool	Arguments	Outcome
115	local web_search	{"query": "sterile neutrino X-ray constraints decay line arXiv 2024 2025"}	To look for emission from the 3.55 keV line from sterile neutrino decay, the residuals, obtained after accurately modeling the thermal emission for each cluster individually, were modeled to search fo...
116	local web_search	{"query": "sterile neutrino dark matter viable parameter space keV arXiv 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=sterile%20neutrino%20dark%20matter%2(
117	local web_search	{"query": "arxiv 2405 sterile neutrino dark matter X-ray"}	Apr 30, 2024 · Sterile neutrinos with masses at the $\mathrm {keV}$ scale and mixing to the active neutrinos offer an elegant explanation of the observed dark matter (DM) density. Jan 8, 2026 · Steril...
118	local web_search	{"query": "arxiv 2501 sterile neutrino BBN N_eff"}	Jan 18, 2025 · The question of whether the massive sterile neutrinos exist remains a crucial unresolved issue in both particle physics and cosmology. Dec 21, 2025 · In this work, we investigate the co...
119	local web_search	{"query": "arxiv 2409 sterile neutrino keV dark matter constraints"}	Feb 15, 2016 · We then round out the discussion by critically summarizing all known constraints on sterile neutrino Dark Matter arising from astrophysical observations, laboratory experiments, and the...

#	Tool	Arguments	Outcome
120	local web_search	{"query": "arxiv 2503 sterile neutrino production reheating inflaton"}	Jan 8, 2026 · Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. We show that cold sterile neutrin...
121	local web_search	{"query": "arxiv 2404 sterile neutrino dark matter production Dodelson-Widrow"}	Mar 22, 1993 · We reexamine this model and find that the sterile neutrinos can be either hot, warm, or cold dark matter. Since their only direct coupling is to left-handed or active neutrinos, the mos...
122	local web_search	{"query": "arxiv 2506 sterile neutrino X-ray XRISM Perseus"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arxiv%202506%20sterile%20neutrino%20X
123	local web_search	{"query": "arxiv 2410 sterile neutrino cosmology DESI Planck"}	Jan 18, 2025 · The question of whether the massive sterile neutrinos exist remains a crucial unresolved issue in both particle physics and cosmology. We explore the cosmological constraints on the mas...
124	local web_search	{"query": "arxiv 2601 sterile neutrino inflaton reheating"}	Jan 6, 2026 · We systematically map the viable parameter space in terms of the sterile neutrino mass, mixing angle, inflaton mass, reheating temperature, and branching ratio. Jan 8, 2026 · Sterile neu...

#	Tool	Arguments	Outcome
125	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
126	hep-research-aid list_agent_files	{}	[]
127	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543
128	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
129	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
130	hep-research-aid list_agent_files	{}	[]
131	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
132	hep-research-aid list_agent_files	{}	[]

#	Tool	Arguments	Outcome
133	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543
134	local web_search	{"query": "arXiv 2024 2025 sterile neutrino dark matter Dodelson-Widrow Shi-Fuller lepton asymmetry"}	In this paper, we examine the Shi-Fuller sterile neutrino dark matter production mechanism given the newly uncovered possibility of large lepton numbers in the early Universe, during the dark matter p...
135	local web_search	{"query": "arXiv 2025 reheating inflaton decay sterile neutrino branching ratio BR 1e-4"}	Jan 6, 2026 · We systematically map the viable parameter space in terms of the sterile neutrino mass, mixing angle, inflaton mass, reheating temperature, and branching ratio. Jan 8, 2026 · Sterile neu...
136	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 0.1, "sin2_2theta": 1e-15, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 0.1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 3....
137	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
138	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1000.0, "sin2_2theta": 0.01, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1e+03 eV, sin\u00b22\u03b8 = 0.01\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 1...

#	Tool	Arguments	Outcome
139	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 7e-11\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff ...
140	local web_search	{"query": "arXiv 2024 2025 Lyman-alpha sterile neutrino warm dark matter m_th limit keV"}	This framework enables efficient sterile neutrino DM production even at vanishingly small mixing angles and opens up new viable regions of parameter space that can be tested with future X X -ray and...
141	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [7100.0, 10000.0, 20000.0, 40000.0, 50000.0], "sin2_2thetas": [1e-10, 1e-12, 1e-14], "combine": "product", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_b3ad21.csv"], "message": "Predicted 15 ...
142	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 7e-11\n\nE_\u03b3 = 3.55 keV\nDecay...
143	local web_search	{"query": "XRISM Perseus sterile neutrino 3.5 keV line 2024 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=XRISM%20Perseus%20sterile%20neutrino%

#	Tool	Arguments	Outcome
144	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1000.0, 3000.0, 7100.0, 10000.0, 20000.0, 50000.0, 100000.0], "sin2_2thetas": [1e-15, 1e-14, 1e-13, 1e-12, 1e-11, 1e-10, 1e-09, 1e-08, 1e-07, 1e-06], "combine": "product", "mechanism": "DW", "output_dir": ...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_3a8af4.csv"], "message": "Predicted 70 ...
145	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 1000.0, "m_max_eV": 100000.0, "theta_min": 1e-15, "theta_max": 1e-06, "n_points": 80, "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/26...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_DW_fdb0a2.csv", "/Users/nramachandra/hep-genesis-testi...
146	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...
147	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.14, "sin2_2theta": 0.42, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...
148	local web_search	{"query": "NuSTAR stray light sterile neutrino dark matter 2405.17861 limits"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=NuSTAR%20stray%20light%20sterile%20n

#	Tool	Arguments	Outcome
149	local web_search	{"query": "arXiv 2507.18752 sterile neutrino"}	Jul 24, 2025 · Using a quasiclassical Boltzmann transport calculation of the dark matter production, we compute the nonthermal phase space distributions of sterile neutrinos across a broad range of pa...
150	local web_search	{"query": "arXiv 2502.17374 sterile neutrino"}	Feb 24, 2025 · This finding reopens the region of lighter sterile neutrinos. The new region may be explored with the next generation of X-ray telescopes searching for the inherent peak signature provi...
151	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 0.001, "sin2_2theta": 1e-10, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 0.001 eV, sin\u00b22\u03b8 = 1e-10\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = ...
152	local web_search	{"query": "arXiv 2601.03346 sterile neutrino"}	Jan 6, 2026 · Abstract: Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. Jan 8, 2026 · Sterile ne...
153	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 7e-11\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff ...

#	Tool	Arguments	Outcome
154	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 10000.0, "sin2_2theta": 1e-12, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1e+04 eV, sin\u00b22\u03b8 = 1e-12\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = ...
155	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 50000.0, "sin2_2theta": 1e-13, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 5e+04 eV, sin\u00b22\u03b8 = 1e-13\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = ...
156	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 1000.0, "m_max_eV": 100000.0, "theta_min": 1e-15, "theta_max": 1e-06, "n_points": 60, "mechanism": "SF", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/26...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_SF_e8f016.csv", "/Users/nramachandra/hep-genesis-testi...
157	local web_search	{"query": "arXiv 2609.31340 sterile neutrino"}	Sep 25, 2026 · We identify this effect as an additional constraint on resonant sterile-neutrino production at large lepton asymmetry and discuss its dependence on the flavor structure of the initial a...
158	local web_search	{"query": "arXiv 2405 2024 sterile neutrino warm dark matter Lyman-alpha 8.4 keV"}	This framework enables efficient sterile neutrino DM production even at vanishingly small mixing angles and opens up new viable regions of parameter space that can be tested with future X X -ray and...

#	Tool	Arguments	Outcome
159	board post_finding	{"claim": "Sterile-nu server BBN baselines: Yp= 0.2449±0.004 (Aver et al. JCAP 07 2021 029) , D/H=2.547±0.025×10⁻? (Cooke et al. ApJ 855 2018 102), N_eff_CMB= 2.99±0.17 (Planck 2018 TT TE EE+lowE+lensing+BAO), SM N_eff=3.04...	Recorded finding f-023.
160	board post_finding	{"claim": "Server BBN validity: m_s < ~100 keV (relativistic at BBN); BBN fit calibrated for N_ eff 2.044-6.044, omega_ b h2 0.019-0.0257, tau_n 866.4-890.4 s; exclusion is Delta chi2 vs SM >3.84 (95%, 1 dof); for m_s >= ~...	Recorded finding f-024.
161	board post_finding	{"claim": "Server observational baselines: Yp=0.2449±0.004 (Aver, Olive & Skillman JCAP 07 (2021) 029), D/H= 2.547±0.025e-5 (Cooke, Pettini & Steidel ApJ 855 (2018) 102), N_eff_CMB= 2.99±0.17 (Planck 2018 TT, TE,EE+lowE+len...	Recorded finding f-025.
162	board post_finding	{"claim": "Brief vs server baseline offset: brief Yp=0.245 vs server 0.2449 (delta 0.0001, 0.025 sigma) , brief D/H=2.5e-5 vs server 2.547e-5 (delta 0.047e-5, 1.88 sigma on server error); SM prediction Yp=0.24682 is +0.48...	Recorded finding f-026.
163	board post_finding	{"claim": "Server caps: SF mechanism Delta N_eff is heuristic only, lepton asymmetry L capped at 0.1, no xi_e (electron-neutrino degeneracy) modeling, non-thermal reheating/ inflaton-decay (BR~1e-4, T_RH, m_phi) not model...	Recorded finding f-027.

#	Tool	Arguments	Outcome
164	board post_finding	{"claim": "X-ray radiative decay predictions for 5 keV masses (7.1,10,20,40,50 keV) at sin^2 2theta=1e-10, 1e-12,1e-14: E_gamma=m_s/2 (3.55-25 keV), Gamma=1/tau from 2.46e-28 to 4.25e-24 s^-1 at 1e-10 scaling as (m_s/keV)...	Recorded finding f-028.
165	board post_finding	{"claim": "Gamma formula verification: Gamma= 1.38e-32 s^-1 (sin^2 2theta/1e-10)(m_s/keV)^5 matches tool lifetimes within 1-2%: for 7.1 keV 7e-11 predicted 1.74e-28 vs tool 1.72e-28 (1.3% diff), for 7.1 keV 1e-10 predicte...	Recorded finding f-029.
166	board post_finding	{"claim": "3.5 keV line benchmark (m_s=7.1 keV, sin^2 2theta=7e-11, E_gamma=3.55 keV) is flagged EXCLUDED by built-in NuSTAR M31 (Roach+2020) and XMM-Newton blank sky (Dessert+2020) limits; whole-halo fluxes (f_DM=1, no ...	Recorded finding f-030.
167	board post_finding	{"claim": "Sterile-nu BBN server uses Yp= 0.2449±0.004 (Aver+2021), D/H=2.547±0.025 (Cooke+2018), Neff_CMB=2.99±0.17 (Planck 2018), BBN fit valid for Neff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s, exclusi...	Recorded finding f-031.
168	board post_finding	{"claim": "2024-2026 arXiv literature contains reheating/inflaton-decay sterile-neutrino DM production with BR~1e-4 mapping viable space in (m_s, mixing, m_phi, T_RH, BR) - identified via web search Jan 6-8 2026 entries"...	Recorded finding f-032.

#	Tool	Arguments	Outcome
169	board post_finding	{"claim": "2024-2026 arXiv literature revisits Shi-Fuller resonant production with large lepton asymmetries L_alpha >=1 at T>20 MeV consistent with BBN, and flavor-asymmetric L with near-zero total L opening new paramete...	Recorded finding f-033.
170	board post_finding	{"claim": "Web search confirms Dodelson-Widrow production remains severely constrained by astrophysical observations while non-standard active-sterile interactions and large lepton asymmetries are explored to reopen viab...	Recorded finding f-034.
171	board post_dead_end	{"what": "Full per-mechanism viability table with verbatim quotes, Viel+2005 m_th-> m_s translation at 7.1/ 10/20/50 keV, and BBN tool cross-checks for d-09", "why": "Tool budget exhausted after 10 web_ search/describe call...	Recorded dead end x-03.
172	board post_finding	{"claim": "DW grid 7 masses (1,3,7.1,10,20,50,100 keV) $\times$ 10 mixings (1e-15 to 1e-6) : 12/70 BBN-excluded ($\Delta\chi^2$> 3.84) and 31/70 overclose (Ω_s h^2>0.12); CSV saved to sterile_nu_grid_DW_ 3a8af4.csv; e.g. 1 keV/ 1e-6 gives ΔN_e...	Recorded finding f-035.
173	board post_finding	{"claim": "DW scan 6400 points (1-100 keV, 1e-15-1e-6): 862/6400 BBN-excluded, 2752/6400 overclosing; BBN 95% upper limit $\sin^2 2\theta$_max falls from 3.5e-7 at 1 keV to 8.9e-9 at 100 keV (intermediate: 1.23e-7 at 3.6 keV, 7.26...	Recorded finding f-036.

#	Tool	Arguments	Outcome
174	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 5 keV\nDecay rat...
175	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000.0, "sin2_2theta": 1e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 10 keV\nDecay ra...
176	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 1e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 20 keV\nDecay ra...
177	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "mechanism": "SF", "lepton_asymmetry": 0.01}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 7e-11\nMechanism: SF (SF Delta N_eff is a heuristic (0.1 x the DW value), not a resonant-product...
178	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1000000.0, "sin2_2theta": 1e-10, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "\u26a0 MeV+ (heavy neutral lepton): OUTSIDE MODEL VALIDITY: non-relativistic and decaying during BBN. Decay-product injection (photodissociation, hadro...

#	Tool	Arguments	Outcome
179	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543", "neff_min": 2.0, "neff_max": 5.0, "n_points": 200}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/bbn_vs_neff.png"], "message": "Plot saved to /Users/nramac... [Fig. 27]
180	board post_finding	{"claim": "X-ray line predictions at sin^2 2theta= 1e-10 for 10 keV (E_gamma= 5 keV, Gamma=1.36e-27 s^-1, tau=7.35e26 s, Perseus flux 1.4e-4 ph/s/cm2, EXCLUDED by NuSTAR M31+XMM), 20 keV (E_ gamma=10 keV, Gamma=4.36e-26 s^-...	Recorded finding f-037.
181	board post_finding	{"claim": "Combined BBN+X-ray viability per mass at sin^2 2theta=1e-10: all 5 masses BBN-allowed (Delta chi2 -0.008 to -0.038, Delta N_eff 0.0006-0.0029) but 7.1,10,20 keV X-ray-excluded by built-in limits, 40 keV not ex...	Recorded finding f-038.
182	board post_dead_end	{"what": "2024-2026 literature X-ray upper-limit table (XRISM Perseus, NuSTAR stray-light 2405.17861, INTEGRAL/SPI, Chandra/ XMM archival sin^2 2theta limits) with arXiv IDs, figure numbers and verbatim quotes", "why": "w...	Recorded dead end x-04.
183	board post_finding	{"claim": "SF (Shi-Fuller) Delta N_eff on server is heuristic 0.1x DW value, not a resonant-production calculation; at m_s=7100 eV sin2=7e-11 L=0.01 server returns Delta N_eff=4.33e-05 Omega_s h2=1366 (overcloses) vs DW ...	Recorded finding f-039.

#	Tool	Arguments	Outcome
184	board post_finding	{"claim": "MeV+ regime (m_s=1 MeV sin2=1e-10) is OUTSIDE MODEL VALIDITY: bbn_treatment_valid=false, decay-product injection and entropy release not modeled, BBN numbers (Yp=0.24725 D/H= 2.4541e-5 Delta chi2=-0.38) are not...	Recorded finding f-040.
185	board post_finding	{"claim": "BBN alone limits Delta N_eff <0.548 (95% Delta chi2 vs SM) with slopes dYp/dN_eff=0.01351 and dD/H5/dN_eff=0.3308 near SM; fully thermalized sterile (N_eff=4.044) predicts Yp=0.25956 and D/H=2.7769e-5, 3.7 sig...	Recorded finding f-041.
186	hep-research-aid list_agent_files	{}	[]
187	serve-sbbn.sh describe_sterile_ nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
188	serve-sbbn.sh describe_sterile_ nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
189	hep-research-aid list_agent_files	{}	[]
190	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_ data()." }

#	Tool	Arguments	Outcome
191	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
192	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 7100.0, "sin2_2theta": 1e-10, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 1e-10\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff ...
193	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1000.0, "sin2_2theta": 1e-06, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1e+03 eV, sin\u00b22\u03b8 = 1e-06\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = ...
194	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
195	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 1 ...

#	Tool	Arguments	Outcome
196	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543: - bbn_vs_neff.png - scan_DW_fdb0a2.csv - scan_DW_fdb0a2.npz - scan_SF_e8f016.csv ...
197	local web_search	{"query": "Abazajian 1705.01837 sterile neutrino dark matter Eq 37 relic density"}	Abstract We explore resonant production of sterile neutrino dark matter via the Shi-Fuller (SF) mechanism, revisiting its cosmological viability in light of recent results demonstrating that lepton-nu...
198	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543", "save_pdf": true}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/bbn_vs_neff.png", "/Users/nramachandra/hep-genesis-testing/... [Fig. 27]
199	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 7e-11\n\nE_\u03b3 = 3.55 keV\nDecay...
200	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [7100.0, 10000.0, 20000.0, 40000.0, 50000.0], "sin2_2thetas": [1e-10, 1e-12, 1e-14] , "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-st...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_b3ad21.csv"], "message": "Predicted 15 ...

#	Tool	Arguments	Outcome
201	local web_search	{"query": "arXiv 2507.18752 sterile neutrino dark matter"}	A pair of such heavy neutral leptons (HNLs) can give masses to neutrinos, generate the baryon asymmetry at the electroweak scale and produce a large lepton asymmetry at the QCD scale, which is then re...
202	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 5 keV\nDecay rat...
203	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000.0, "sin2_2theta": 1e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 10 keV\nDecay ra...
204	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 1e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 20 keV\nDecay ra...
205	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 50000.0, "sin2_2theta": 1e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 50 keV, sin\u00b22\u03b8 = 1e-10\n\nE_\u03b3 = 25 keV\nDecay ra...

#	Tool	Arguments	Outcome
206	local web_search	{"query": "arXiv 2502.17374 sterile neutrino Shi-Fuller lepton asymmetry"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202502.17374%20sterile%20neutr
207	hep-research-aid get_paper_figures	{"paper_id": "1705.01837", "max_figures": 4, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543"}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/figures_1705.01837/fig02.png", "/Users/nramachandra/hep-genesis-test... [Fig. 29, 30]
208	local web_search	{"query": "XRISM Perseus sterile neutrino 3.5 keV line 2024 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=XRISM%20Perseus%20sterile%20neutrino%
209	local web_search	{"query": "sterile neutrino dark matter Dodelson-Widrow 2024 2025 arXiv Lyman-alpha"}	Mar 22, 1993 · The simplest model that can accomodate a viable nonbaryonic dark matter candidate is the standard electroweak theory with the addition of right-handed or sterile neutrinos. We reexamine...
210	local web_search	{"query": "Abazajian 2017 sterile neutrino dark matter production Eq 37 Omega h2 formula"}	May 4, 2017 · Here, we review the cosmological role such light sterile neutrinos can play from the early Universe, including production of keV-scale sterile neutrinos as dark matter candidates, and dy...

#	Tool	Arguments	Outcome
211	hep-research-aid list_agent_files	{}	[]
212	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
213	local web_search	{"query": "NuSTAR stray light sterile neutrino 2405.17861 X-ray limit sin2theta"}	Search failed: ConnectError: ConnectError('error sending request for url (https://www.google.com/wml/search?q=NuSTAR+stray+light+sterile+neutrino+2esv=1&filter=1&st...
214	local web_search	{"query": "sterile neutrino reheating inflaton decay branching ratio 1e-4 arXiv 2024 2025"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=sterile%20neutrino%20reheating%20infl
215	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543: - bbn_vs_neff.pdf - bbn_vs_neff.png - figures_1705.01837 - scan_DW_fdb0a2.csv - s...
216	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 1e-15, "omega_b_h2": 0.02237, "tau_n": 878.4}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 3.98...

#	Tool	Arguments	Outcome
217	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 1e-15, "omega_b_h2": 0.019, "tau_n": 878.4}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 3.98...
218	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 1e-15, "omega_b_h2": 0.0257, "tau_n": 878.4}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 3.98...
219	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 1e-15, "omega_b_h2": 0.02237, "tau_n": 866.4}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 3.98...
220	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 1e-15, "omega_b_h2": 0.02237, "tau_n": 890.4}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 3.98...
221	board post_finding	{"claim": "Server DW Omega_s h2 at 7.1 keV sin2=1e-10 is 0.04715 (39% of DM, Delta N_eff=0.000618, BBN allowed Delta chi2=-0.01) and at 1 keV sin2=1e-6 is 13.84 (overcloses by 115x, Delta N_eff=1.0, BBN excluded Delta ch...	Recorded finding f-042.

#	Tool	Arguments	Outcome
222	board post_finding	{"claim": "Sterile-nu server BBN baselines Yp= 0.2449±0.004 Aver+2021, D/ H=2.547±0.025e-5 Cooke+2018, N_eff_CMB=2.99±0.17 Planck2018, valid m_s< ~100 keV, BBN fit N_eff 2.044-6.044, exclusion Delta chi2>3.84; X-ray limits ...	Recorded finding f-043.
223	board post_finding	{"claim": "X-ray radiative decay predictions for 5 masses (7.1,10,20,40,50 keV) x 3 mixings (1e-10,1e-12, 1e-14) via DW: E_gamma=m_s/2 =3.55,5,10,20,25 keV; Gamma= 1/tau =2.46e-28 (7.1 keV), 1.36e-27 (10 keV),4.36e-26 (20 k...	Recorded finding f-044.
224	board post_finding	{"claim": "Gamma formula Gamma=1.38e-32 s^-1 (sin^ 2 2theta/1e-10)(m_s/keV) ^5 verified within 1-2% vs tool: 7.1 keV 7e-11 predicted 1.74e-28 vs tool 1.72e-28 (1.3% diff), 7.1 keV 1e-10 predicted 2.49e-28 vs tool 2.46e-28 ...	Recorded finding f-045.
225	serve-sbbn.sh predict_sterile_ nu_grid	{"masses_eV": [7100.0, 10000.0, 20000.0, 50000.0] , "sin2_2thetas": [1e-12, 1e-10, 1e-08], "combine": "product", "mechanism": "DW", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn...	{ "status": "ok", "files": ["/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_163543/ sterile_nu_grid_DW_ 5f274e.csv"], "message": "Predicted 12 ...
226	hep-research-aid get_paper_figures	{"paper_id": "1705.01837", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_163543", "max_ figures": 4}	{"status": "success", "files": ["/Users/ nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_163543/ figures_1705.01837/ fig02.png", "/ Users/nramachandra/ hep-genesis-test... [Fig. 29, 30]

#	Tool	Arguments	Outcome
227	board post_finding	{"claim": "BBN Delta N_eff 95% limit is Delta N_eff < 0.548 (Delta chi2 vs SM > 3.84) with slopes dYp/dN_eff=0.01351 and dD/H5/dN_eff=0.3308 near SM; SM at N_eff=3.044 predicts Yp= 0.24682 D/H=2.4433e-5 chi2= 2.77, fully the...	Recorded finding f-046.
228	board post_finding	{"claim": "SBL BEST best-fit (m_s=1.14 eV sin^ 2 2theta=0.42 Delta m^2= 1.3 eV^2) thermalization fraction =1.0 (Delta N_eff= 1.0 N_eff=4.044 Omega_s h^2= 29.3) gives BBN Yp=0.25956 D/H=2.7769e-5 Delta chi2= 23.1 excluded 95% ...	Recorded finding f-047.
229	board post_finding	{"claim": "Varying BBN nuisance parameters at fixed SM N_eff=3.044: omega_ b h2 0.019->0.0257 shifts Yp 0.24525->0.24824 and D/H 3.1843e-5->1.9422e-5 with absolute chi2 96.12-> 87.27 (vs 2.77 at fiducial 0.02237); tau_n 86...	Recorded finding f-048.
230	board post_finding	{"claim": "Built-in X-ray limit check: 7.1 keV 7e-11 (3.55 keV line tau 5.82e27 s fluxes MW 3.28 M31 4.8e-4 Perseus 2.5e-5 ph/s/cm2 whole-halo f_DM=1) EXCLUDED by NuSTAR M31 Roach+2020 + XMM Dessert+2020; 10 keV 1e-10 (5...	Recorded finding f-049.
231	board post_finding	{"claim": "DW BBN grid 12 points (m_s=7.1,10,20,50 keV x sin^2 2theta=1e-12,1e-10, 1e-08): 0/12 BBN-excluded (Delta chi2 -0.815 to 0.0, threshold 3.84), 6/12 overclose (Omega_s h2>0.12) ; Delta N_eff 6.2e-06 to 0.2948, Ome...	Recorded finding f-050.

#	Tool	Arguments	Outcome
232	board post_finding	{"claim": "Extracted 2 figures from arXiv: 1705.01837 (Abazajian review): Figure 2 lepton-asymmetry sign vs mass splitting/mixing (Fig 29) and Figure 7 X-ray 3.5 keV line detections vs constraints (Fig 30, Bulbul+ MOS/PN,...	Recorded finding f-051.
233	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.14, 10.0, 100.0, 1000.0, 7100.0, 10000.0], "sin2_2thetas": [1e-06, 1e-05, 0.0001, 0.001, 0.01, 0.42], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/proje...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_e2f9d0.csv"], "message": "Predicted 36 ...
234	board post_finding	{"claim": "Sterile-nu server baselines confirmed: Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), N_eff_CMB= 2.99±0.17 (Planck 2018), BBN fit valid N_eff 2.044-6.044, m_s<~100 keV, exclusion Delta chi2>3.84;...	Recorded finding f-052.
235	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1000.0, 1274.28, 1623.77, 2068.93, 2636.02, 3358.6, 4279.19, 5451.78, 6945.96, 8849.11, 11273.4, 14363.0, 18299.7, 23315.2, 29704.8, 37844.6, 48217.4, 61433.9, 78272.4, 100000.0], "sin2_2thetas": [1e-15, 2...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_af26cf.csv"], "message": "Predicted 400...
236	board post_finding	{"claim": "Reproducible Yp/D/H grid CSV produced: 36-point DW grid (m_s=1.14 eV to 10 keV x sin^2 2theta= 1e-6 to 0.42) with Delta N_eff, N_eff, Omega_s h^2, Yp, D/H, Li7, chi2, Delta chi2, lifetime, free-streaming, BBN e...	Recorded finding f-053.

#	Tool	Arguments	Outcome
237	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 1000.0, "m_max_eV": 100000.0, "theta_min": 1e-15, "theta_max": 1e-06, "n_points": 20, "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/26...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_DW_0a8c17.csv", "/Users/nramachandra/hep-genesis-testi...
238	board post_finding	{"claim": "DW 20x20 grid (400 points, 1-100 keV x 1e-15-1e-6, product) via predict_sterile_nu_grid: 59 BBN-excluded (Delta chi2> 3.84) and 172 overclosing (Omega_s h2>0.12); CSV saved to /Users/nramachandra/hep-genesis-te...	Recorded finding f-054.
239	board post_finding	{"claim": "DW scan 400 points (1-100 keV, 1e-15-1e-6) via scan_sterile_nu_parameters: 59/400 BBN-excluded, 172/400 overclosing; BBN 95% max sin2_2theta falls from 3.36e-07 at 1 keV to 4.28e-09 at 100 keV; files scan_DW_0...	Recorded finding f-055.
240	board post_finding	{"claim": "Extracted 2 figures from Abazajian 1705.01837 (Sterile neutrinos in cosmology): Figure 2 lepton asymmetry sign vs mass splitting/mixing and Figure 7 X-ray line detections at 3.55 keV with constraints; source h...	Recorded finding f-056.
241	board post_dead_end	{"what": "Quantified normalization offset factor vs Abazajian 1705.01837 Eq.37 with verbatim quote and cite", "why": "Tool budget exhausted before read_arxiv_chunk/cite could be executed; Abazajian Eq.37 text not retriev...	Recorded dead end x-05.

#	Tool	Arguments	Outcome
242	board post_dead_end	{"what": "Exclusion-plane plot distinguishing tool-internal vs literature X-ray limits and Omega=0.12 contour overlay", "why": "Tool budget reached before plot_sterile_nu_exclusion could be called; scan NPZ exists but no...	Recorded dead end x-06.
243	hep-research-aid list_agent_files	{}	[]
244	hep-research-aid list_files	{}	Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...
245	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
246	hep-research-aid list_datasets	{}	{"count": 0, "datasets": []}
247	hep-research-aid get_dispatch	{}	Dispatch: local execution.
248	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
249	local web_search	{"query": "sterile neutrino dark matter BBN X-ray joint constraints"}	This production mechanism opens regions of parameter space where the active-sterile mixing is small enough to evade current X-ray constraints while reproducing the observed DM abundance. We systematic...

#	Tool	Arguments	Outcome
250	local web_search	{"query": "sterile neutrino BBN N_eff Yp D/H constraints"}	In Fig. 12 we display the regions of mass and mixing inconsistent with the BBN data, with and without lepton asymmetry, as well as the current laboratory constraints on a heavy sterile neutrino that m...
251	local web_search	{"query": "sterile neutrino X-ray decay line constraints NuSTAR Chandra"}	May 2, 2019 · Abstract: Recently an unidentified emission line at 3.55 keV has been detected in X-ray spectra of clusters of galaxies. The line has been discussed as a possible decay signature of 7.1 ...
252	local web_search	{"query": "sterile neutrino dark matter production Dodelson Widrow Shi Fuller"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=sterile%20neutrino%20dark%20matter%2
253	local web_search	{"query": "arXiv sterile neutrino BBN constraints helium deuterium N_eff"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%20sterile%20neutrino%20BBN%20c
254	local web_search	{"query": "arXiv NuSTAR M31 sterile neutrino X-ray line search"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%20NuSTAR%20M31%20sterile%20neu

#	Tool	Arguments	Outcome
255	local web_search	{"query": "arXiv 3.55 keV line sterile neutrino decay Boyarsky Bulbul"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%203.55%20keV%20line%20sterile%...
256	local web_search	{"query": "arXiv sterile neutrino dark matter Lyman-alpha free streaming constraints"}	We solve the Boltzmann equation for the nonthermal phase-space distribution and confront the scenario with Lyman- α and strong gravitational lensing data by computing the matter power spectrum. Feb ...
257	local web_search	{"query": "sterile neutrino BBN constraints Dolgov review"}	We present here updated and more strict BBN constraints on the parameters of non-equilibrium active-sterile neutrino oscillations considering several models of beyond Standard Model Physics. Aug 7, 20...
258	local web_search	{"query": "sterile neutrino X-ray constraints Perez Foster Dessert"}	Abstract: Sterile neutrinos with masses in the kilo-electronvolt range are viable candidates for the warm dark matter. We analyse existing data for the extragalactic diffuse X-ray background for signa...
259	local web_search	{"query": "\"arXiv:1402.4119\" sterile neutrino BBN"}	Jan 10, 2020 · In summary, we revisited non-resonant sterile neutrino production in different pre-BBN cosmologies, before the temperature of the Universe was 5 MeV and showed that these neutrinos can ...

#	Tool	Arguments	Outcome
260	local web_search	{"query": "\"arXiv: 1904.05710\" NuSTAR sterile neutrino M31"}	Apr 11, 2019 · In this study, the applicability of execution of a real-word untrusted business process on the permissioned blockchain is investigated.
261	hep-research-aid list_agent_files	{}	[]
262	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917
263	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150: - 1702.08430.pdf - 1702.08430.txt - 1705.01837.pdf - 1705.01837.txt - 2405.17861....
264	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
265	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
266	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 3e-14, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 3e-14\n\nE_\u03b3 = 20 keV\nDecay ra...
267	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 5 keV\nDecay rat...
268	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray: - chats - checkpoint.db - checkpoint.db-shm - checkpoint.db-wal - data - jobs - project.json -...
269	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [40000.0, 50000.0], "sin2_2thetas": [1e-12, 3e-14, 1e-14], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_DW_5d6c37.csv"], "message": "Predicted 6 p...
270	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }

#	Tool	Arguments	Outcome
271	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
272	local web_search	{"query": "arXiv:1705.01837 Abazajian sterile neutrino dark matter review"}	Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. We show that cold sterile neutrino DM can be ef...
273	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/reports"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/reports: - 261021_111534
274	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/research"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/research: - default
275	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs: - 261021_110146 - 261021_110150 - 261021_111530 - 261102_163538 - 261102_163543 - 261102_1649...
276	local web_search	{"query": "astro-ph/0501562 Viel WDM Lyman-alpha sterile neutrino mass mapping"}	Jan 26, 2005 · We present constraints on the mass of WDM particles from a combined analysis of the matter power spectrum inferred from the large samples of high resolution high signal-to-noise Lyman-a...

#	Tool	Arguments	Outcome
277	local web_search	{"query": "arXiv: 2507.18752 Vogel sterile neutrino Shi-Fuller lepton asymmetry"}	Jul 24, 2025 · We explore resonant production of sterile neutrino dark matter via the Shi-Fuller (SF) mechanism, revisiting its cosmological viability in light of recent results demonstrating that lep...
278	local web_search	{"query": "arXiv:2502.17374 flavor dependent lepton asymmetry sterile neutrino"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%3A2502.17374%20flavor%20depend
279	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/reports/261021_111534"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/reports/261021_111534: - bundle.json - figures - report.pdf - report.tex - supplement.pdf - supplem...
280	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110146"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110146
281	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_111530"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_111530
282	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163538"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163538

#	Tool	Arguments	Outcome
283	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543: - bbn_vs_neff.pdf - bbn_vs_neff.png - figures_1705.01837 - scan_DW_0a8c17.csv - s...
284	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164913"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164913
285	local web_search	{"query": "arXiv:2405.17861 NuSTAR sterile neutrino dark matter stray light"}	May 28, 2024 · Using tremendous photon statistics gained with the stray light aperture of the NuSTAR telescope over 11 years of operation, we set strong limits on the emission of close to monochromati...
286	local web_search	{"query": "arXiv:2510.24560 XRISM Resolve stacked clusters sterile neutrino"}	Oct 28, 2025 · We stack 3.75 Megaseconds of early XRISM Resolve observations of ten galaxy clusters to search for unidentified spectral lines in the E = 2.5-15 keV band (rest frame), including the E =...
287	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
288	hep-research-aid list_agent_files	{}	[]
289	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917: - sterile_nu_grid_DW_5d6c37.csv

#	Tool	Arguments	Outcome
290	local web_search	{"query": "INTEGRAL SPI sterile neutrino decay line Boyarsky Roach diffuse X-ray limit 40 keV"}	neutrino (see Boyarsky et al. 2019, for a recent review). Such a neutrino interacts with the rest of the matter only via a mixing with SM (active) neutrinos and can therefore survive cosmologically fo...
291	board post_finding	{"claim": "Sterile-nu server BBN baselines are Yp= 0.2449±0.004 (Aver, Olive & Skillman JCAP 07 2021 029), D/H=2.547±0.025e-5 (Cooke, Pettini & Steidel ApJ 855 2018 102), Planck N_eff=2.99±0.17 (TT,TE, EE+lowE+lensing+BAO)...	Recorded finding f-057.
292	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_DW_5d6c37.csv"}	m_s_eV,sin2_2theta, delta_neff,neff,omega_ s_h2,Yp,DH_x1e5,Li7H_ x1e10,chi2_bbn,delta_ chi2_bbn,lifetime_s, free_streaming_kpc,bbn_ excluded_95,overcloses, bbn_treatment_valid 40000,1e-12,2.46585e-05, 3.04402...
293	serve-sbbn.sh predict_sterile_ nu_grid	{"masses_eV": [20000.0, 50000.0], "sin2_2thetas": [1e-14], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_DW_44fa64.csv"], "message": "Predicted 2 p...
294	serve-sbbn.sh predict_sterile_ nu_grid	{"masses_eV": [20000.0, 50000.0], "sin2_2thetas": [1e-14], "combine": "product", "mechanism": "SF", "lepton_asymmetry": 0.1, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_SF_d58c80.csv"], "message": "Predicted 2 p...

#	Tool	Arguments	Outcome
295	board post_finding	{"claim": "Prior run 261021_110150 produced DW scan CSV/NPZ (scan_ DW_dc7743), BBN vs N_eff plot (bbn_vs_neff.png), exclusion plane (exclusion_ scan_DW_dc7743.png), and 6 sterile-nu grid CSVs (DW and SF mechanisms) plus re...	Recorded finding f-058.
296	hep-research-aid get_paper_figures	{"paper_id": "2405.17861", "output_dir": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_164917", "max_ figures": 4}	{"status": "success", "files": ["/Users/ nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_164917/ figures_2405.17861/ fig01.png", "/ Users/nramachandra/ hep-genesis-test... [Fig. 39, 40, 41, 42]
297	board post_finding	{"claim": "DW vs SF at m_ s=20 keV, sin^2 2theta= 1e-14: DW gives Delta N_ eff=1.42e-07, Omega_s h^ 2=3.04e-05, Delta chi2_ BBN=-0.0 (BBN-allowed, not overclosing); SF (L=0.1 max, heuristic 0.1xDW) gives Delta N_eff=1.42e-08,...	Recorded finding f-059.
298	local web_search	{"query": "arXiv:2405.17861 NuSTAR 11 year stray light sterile neutrino limits sin2theta"}	Search failed: DNSError: DNSError('error sending request for url (https: //wt.wikipedia.org/ w/api.php?action= opensearch&profile= fuzzy&limit=1&search= arXiv%3A2405.17861%20NuSTAR%2011%20y
299	local web_search	{"query": "Boyarsky 2006 diffuse X-ray background sterile neutrino decay limit lifetime"}	Sterile neutrinos with masses in the kilo-electronvolt range are viable candidates for the warm dark matter. We analyse existing data for the extragalactic diffuse X-ray background for signatures of s...

#	Tool	Arguments	Outcome
300	board post_finding	{"claim": "Contrasting X-ray decay-line emulator at m_s=40 keV (E_gamma=20 keV), sin^2 2theta=3e-14 gives Gamma=4.18e-28 s^-1, tau=2.39e27 s (7.58e19 yr), not excluded by tabulated NuSTAR/XMM limits; and at m_s=10 keV (E...	Recorded finding f-060.
301	hep-research-aid get_paper_figures	{"paper_id": "1705.01837", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917", "max_figures": 4}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/figures_1705.01837/fig02.png", "/Users/nramachandra/hep-genesis-test... [Fig. 43, 44]
302	hep-research-aid get_paper_figures	{"paper_id": "astro-ph/0501562", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917", "max_figures": 4}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/figures_astro-ph_0501562/fig01_1.png", "/Users/nramachandra/hep-gene... [Fig. 45, 46, 47, 48]
303	local web_search	{"query": "Abazajian 1705.01837 Eq.37 sterile neutrino dark matter production Omega sin^2 2theta"}	Using a quasiclassical Boltzmann transport calculation of the dark matter production, we compute the nonthermal phase space distributions of sterile neutrinos across a broad range of particle mass $ {...
304	board post_finding	{"claim": "DW BBN emulator at high-mass SPI regime m_s=40,50 keV with sin^2 2theta=1e-12,3e-14,1e-14 gives Delta N_eff=2.5e-5 to 2.5e-7, Delta chi2_BBN~0, Omega_s h^2=0.0001-0.016, tau=2.4e25-7.2e27 s; all 6 points BBN-a...	Recorded finding f-061.

#	Tool	Arguments	Outcome
305	local web_search	{"query": "Viel 2005 WDM mass sterile neutrino 4.43 keV thermal relic mapping"}	One of the most promising WDM candidates is a sterile (right-handed) neutrino with a mass in the keV range, which could explain the pulsar velocity kick [4], help in reionizing the Universe at high re...
306	hep-research-aid list_agent_files	{}	[]
307	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
308	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917: - figures_1705.01837 - figures_2405.17861 - figures_astro-ph_0501562 - sterile_nu_g...
309	board post_finding	{"claim": "Abazajian 1705.01837 Fig.7 extracted: X-ray line detections consistent with 7.1 keV sterile neutrino DM (Bulbul+ MOS/PN stacked clusters, Perseus core-removed, M31, Galactic Center) vs 90% constraints from Cha...	Recorded finding f-062.
310	hep-research-aid get_paper_figures	{"paper_id": "2510.24560", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917", "max_figures": 4}	{"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:2510.24560 (arxiv-html: https://arxiv.org/html/2510.24560 contains no extractable figures; ar5iv: https://ar5iv.labs.arx...

#	Tool	Arguments	Outcome
311	hep-research-aid get_paper_figures	{"paper_id": "astro-ph/0602028", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917", "max_figures": 4}	{"status": "success", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/figures_astro-ph_0602028/fig02.png", "/Users/nramachandra/hep-genesi... [Fig. 49, 50, 51, 52]
312	board post_finding	{"claim": "Viel et al. astro-ph/0501562 Figs.1-3 extracted: WDM transfer function T(k) for thermal masses 92-1441 eV and mixed LambdaCWDM, flux power spectrum fractional difference for m_WDM=0.5-8 keV at z=2.5-4, and bia...	Recorded finding f-063.
313	local web_search	{"query": "Roach 2023 INTEGRAL SPI sterile neutrino decay lifetime limit sin2theta"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=Roach%202023%20INTEGRAL%20SPI%20ster
314	board post_dead_end	{"what": "Fetch and cite verbatim Abazajian 1705.01837 Eq.37 (m_s ~ 3.40 keV*(sin^2 2theta/1e-8)^-0.615*(Omega/0.26)^0.5*0.527 erfc[...]) and Viel+2005 m_s=4.43 keV*(m_th/1keV)^{4/3} coefficient with cite() validated quot...	Recorded dead end x-07.
315	board post_finding	{"claim": "Primary NuSTAR 11-yr stray-light paper arXiv:2405.17861 Fig.1 extracted: 95% CL sin^2 2theta upper limits vs m_s from unfocused stray-light spectrum (|b|>3 deg, 117 Ms/FPM), with previous NuSTAR 2020/2023 and ...	Recorded finding f-064.

#	Tool	Arguments	Outcome
316	board post_dead_end	{"what": "Numerically reading NuSTAR Fig.3 observed vs expected sin^ 2 2theta limits at 7,10,20, 40 keV and XRISM Resolve 3.75 Ms stacked-cluster limits (arXiv:2510.24560) with verbatim quotes via read_arxiv_chunk/cite, an...	Recorded dead end x-08.
317	serve-sbbn.sh describe_sterile_ nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n= ========== ==========\n\ nnMaps sterile-neutrino particle parameters (m_s, sin^2 2...
318	hep-research-aid list_agent_files	{}	[]
319	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_ data()." }
320	hep-research-aid list_files	{"directory": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_164917"}	Files in /Users/ nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_164917: - figures_1705.01837 - figures_2405.17861 - figures_astro-ph_0501562 - figures_astr...
321	hep-research-aid list_agent_files	{}	[]
322	hep-research-aid list_files	{"directory": "/ Users/nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_164917"}	Files in /Users/ nramachandra/ hep-genesis-testing/ projects/ ramp-research-sterile-bbn-xray/ runs/261102_164917: - figures_1705.01837 - figures_2405.17861 - figures_astro-ph_0501562 - figures_astr...

#	Tool	Arguments	Outcome
323	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150: - 1702.08430.pdf - 1702.08430.txt - 1705.01837.pdf - 1705.01837.txt - 2405.17861....
324	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
325	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
326	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 3e-14, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 3e-14\n\nE_\u03b3 = 20 keV\nDecay ra...
327	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 5 keV\nDecay rat...

#	Tool	Arguments	Outcome
328	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [40000.0, 50000.0], "sin2_2thetas": [1e-12, 3e-14, 1e-14], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_DW_5d6c37.csv"], "message": "Predicted 6 p...
329	hep-research-aid list_agent_files	{}	[]
330	hep-research-aid list_input_files	{}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
331	local web_search	{"query": "Abazajian 1705.01837 sterile neutrino dark matter review Eq.37"}	May 4, 2017 · Here, we review the cosmological role such light sterile neutrinos can play from the early Universe, including production of keV-scale sterile neutrinos as dark matter candidates, and dy...
332	local web_search	{"query": "Viel 2005 astro-ph/0501562 warm dark matter transfer function m_s coefficient"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
333	local web_search	{"query": "Vogel 2507.18752 sterile neutrino Shi-Fuller lepton asymmetry L>0.5"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
334	local web_search	{"query": "arXiv 2502.17374 flavor dependent lepton asymmetry sterile neutrino"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.

#	Tool	Arguments	Outcome
335	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/figures_2405.17861"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
336	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/figures_1705.01837"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
337	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/figures_astro-ph_0501562"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
338	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/figures_2405.17861"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
339	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/scan_DW_dc7743.csv"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
340	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/references.bib"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.

#	Tool	Arguments	Outcome
341	board post_finding	{"claim": "Sterile-nu server BBN baselines confirmed: Yp=0.2449±0.004 (Aver+2021), D/H= 2.547±0.025e-5 (Cooke+2018) , Planck N_eff=2.99±0.17, BBN fit validity N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s...	Recorded finding f-065.
342	board post_finding	{"claim": "Contrasting X-ray decay-line emulator: m_s=40 keV sin^2 2theta= 3e-14 gives E_gamma=20 keV Gamma=4.18e-28 s^-1 tau= 2.39e27 s (7.58e19 yr) not excluded; m_s=10 keV sin^2 2theta=1e-12 gives E_gamma=5 keV Gamma=1....	Recorded finding f-066.
343	board post_finding	{"claim": "DW BBN grid at high-mass SPI regime m_s=40, 50 keV x sin^2 2theta=1e-12, 3e-14,1e-14 gives Delta N_ eff 2.47e-05 to 2.47e-07, Omega_s h2 0.00011-0.0158, Delta chi2_BBN~0, tau 2.35e25-7.18e27 s; all 6 points BBN-a...	Recorded finding f-067.
344	board post_finding	{"claim": "DW vs SF at m_ s=20 keV sin^2 2theta=1e-14: DW Delta N_eff=1.42e-07 Omega_s h^2=3.04e-05 Delta chi2_BBN=0 BBN-allowed not overclosing; SF L=0.1 heuristic Delta N_eff= 1.42e-08 Omega_s h^2=4.90 overcloses 40x; at...	Recorded finding f-068.
345	board post_finding	{"claim": "Artifact verification d-16: prior run 261021_110150 contains scan_DW_dc7743.csv/.npz, bbn_vs_neff.png, exclusion_ scan_DW_dc7743.png, 6 sterile_nu_grid CSVs, references.bib, 6 paper PDF/TXT pairs (1702.08430, 1...	Recorded finding f-069.

#	Tool	Arguments	Outcome
346	board post_finding	{"claim": "DW BBN emulator at 40 and 50 keV with sin^ 2 2theta 1e-12/3e-14/1e-14 gives Delta N_eff 2.5e-05 to 2.5e-07 Delta chi2_BBN~ 0 Omega_s h^2 0.0001-0.016 tau 2.4e25-7.2e27 s all BBN-allowed not overclosing confirmin...	Recorded finding f-070.
347	board post_finding	{"claim": "Abazajian 1705.01837 Fig.7 and Viel astro-ph/0501562 Figs.1-3 extracted to runs/261102_ 164917/figures_1705.01837/ fig07.png and figures_ astro-ph_0501562/fig01_ 1.png fig01_2.png fig02.png fig03.png anchoring X-r...	Recorded finding f-071.
348	board post_finding	{"claim": "BBN baseline provenance captured verbatim from server: Yp= 0.2449±0.004 (Aver, Olive & Skillman JCAP 07 2021 029), D/H=2.547±0.025e-5 (Cooke, Pettini & Steidel ApJ 855 2018 102), Planck N_eff= 2.99±0.17 (TT,TE,E...	Recorded finding f-072.
349	board post_dead_end	{"what": "d-16 robustness checks: rerun predict_ sterile_nu_grid varying omega_b h2 (0.019-0.0257) and tau_n (866.4-890.4 s) to quantify Delta chi2_BBN shift and Delta N_eff<0.5 vs 0.6 bound shift; write updated joint BBN...	Recorded dead end x-09.
350	board post_dead_end	{"what": "Numerically read NuSTAR arXiv:2405.17861 Fig.3 observed vs expected 95% CL sin^2 2theta limits at 7,10,20,40 keV and fetch XRISM arXiv:2510.24560 3.75 Ms stacked-cluster limits with verbatim quotes via read_arx...	Recorded dead end x-10.

#	Tool	Arguments	Outcome
351	board post_dead_end	{"what": "Fetch INTEGRAL/SPI primary (Boyarsky et al. diffuse / Roach et al. SPI) to resolve 50 keV lifetime mismatch tool tau 7.84e26 s at 3e-14 vs quoted 1e20 s and verify sin^2 2theta normalization, f_DM, profile defi...	Recorded dead end x-11.
352	board post_dead_end	{"what": "Post corrected 5-row X-ray limit table (7.1/10/20/40/50 keV) with mass, E_gamma, Gamma_limit, sin^2 2theta_limit, source figure/quote, tool Gamma/tau, BBN Delta N_eff/Delta chi2, Omega_s h2 and X-ray vs BBN vs ...	Recorded dead end x-12.
353	hep-research-aid list_agent_files	{}	[]
354	hep-research-aid list_files	{}	Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...
355	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
356	hep-research-aid list_datasets	{}	{"count": 0, "datasets": []}
357	hep-research-aid get_dispatch	{}	Dispatch: local execution.
358	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...

#	Tool	Arguments	Outcome
359	local web_search	{"query": "sterile neutrino dark matter BBN constraints arXiv 2024 2025"}	Feb 17, 2026 · We derive a semi-classical Boltzmann equation for sterile neutrinos applicable in this regime and validate it against quantum kinetic equations. May 28, 2025 · This framework enables ef...
360	local web_search	{"query": "sterile neutrino dark matter X-ray constraints 2024 2025 arXiv"}	Jan 8, 2026 · Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. We show that cold sterile neutrin...
361	local web_search	{"query": "sterile neutrino dark matter viable parameter space m_s sin^2 2theta arXiv 2024"}	This production mechanism opens regions of parameter space where the active-sterile mixing is small enough to evade current X-ray constraints while reproducing the observed DM abundance. We systematic...
362	local web_search	{"query": "sterile neutrino dark matter production Dodelson Widrow Shi Fuller 2024 arXiv"}	In this paper, we examine the Shi-Fuller sterile neutrino dark matter production mechanism given the newly uncovered possibility of large lepton numbers in the early Universe, during the dark matter p...
363	local web_search	{"query": "arXiv 2505 sterile neutrino dark matter inflaton reheating"}	Jan 6, 2026 · We systematically map the viable parameter space in terms of the sterile neutrino mass, mixing angle, inflaton mass, reheating temperature, and branching ratio. We performed a systematic...

#	Tool	Arguments	Outcome
364	local web_search	{"query": "arXiv 2502 sterile neutrino X-ray constraints time-dependent mass"}	May 18, 2006 · In addition to their X-ray properties, this section also provides the distances and dark matter mass estimates we adopt for our calculations of the sterile neutrino decay fluxes from ea...
365	local web_search	{"query": "arXiv 2409 sterile neutrino SMEFT freeze-in"}	Apr 30, 2024 · Sterile neutrinos with masses at the $\mathrm {keV}$ scale and mixing to the active neutrinos offer an elegant explanation of the observed dark matter (DM) density. Mar 22, 2024 · In th...
366	local web_search	{"query": "arXiv 2512 Shi-Fuller resonant production lepton asymmetry BBN"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202512%20Shi-Fuller%20resonant%
367	local web_search	{"query": "arXiv 2601 sterile neutrino dark matter reheating inflaton decays"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202601%20sterile%20neutrino%20d
368	local web_search	{"query": "arXiv sterile neutrino dark matter 3.5 keV line XRISM 2024"}	Oct 28, 2025 · We stack 3.75 Megaseconds of early XRISM Resolve observations of ten galaxy clusters to search for unidentified spectral lines in the E = 2.5-15 keV band (rest frame), including the E =...

#	Tool	Arguments	Outcome
369	local web_search	{"query": "arxiv 2601 sterile neutrino inflaton reheating branching ratio 10^-4"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arxiv%202601%20sterile%20neutrino%20
370	local web_search	{"query": "arxiv 2409.0 sterile neutrino SMEFT freeze-in dark matter"}	Apr 30, 2024 · We show that in the context of the Standard Model effective field theory with sterile neutrinos (ν SMEFT), higher dimensional operators can produce sterile neutrino DM in a broad range ...
371	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
372	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
373	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
374	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137"}	No files found in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137
375	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
376	hep-research-aid list_agent_files	{}	[]
377	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
378	hep-research-aid list_agent_files	{}	[]
379	hep-research-aid get_dispatch	{}	Dispatch: local execution.
380	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 1e-15, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = 3.98...
381	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1000.0, "sin2_2theta": 1e-15, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1e+03 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = ...

#	Tool	Arguments	Outcome
382	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1000.0, 3000.0, 7100.0, 10000.0, 20000.0, 50000.0], "sin2_2thetas": [1e-12, 1e-11, 1e-10, 1e-09, 1e-08, 1e-07], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testi...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_c99d6d.csv"], "message": "Predicted 36 ...
383	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"neff_min": 2.0, "neff_max": 5.0, "n_points": 200, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/bbn_vs_neff.png"], "message": "Plot saved to /Users/nramac... [Fig. 57]
384	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.0, 3.0, 10.0, 30.0, 100.0, 300.0, 1000.0, 3000.0, 7100.0, 10000.0, 20000.0, 50000.0, 100000.0], "sin2_2thetas": [1e-13, 1e-12, 1e-11, 1e-10, 1e-09, 1e-08, 1e-07, 1e-06, 1e-05, 0.0001, 0.001, 0.01, 0.1], ...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_20a44f.csv"], "message": "Predicted 169...
385	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 1.0, "m_max_eV": 100000.0, "theta_min": 1e-13, "theta_max": 0.1, "n_points": 100, "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/scan_DW_9d1baf.csv", "/Users/nramachandra/hep-genesis-testi...
386	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...

#	Tool	Arguments	Outcome
387	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"neff_min": 2.0, "neff_max": 5.0, "n_points": 200, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137", "save_pdf": true}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/bbn_vs_neff.png", "/Users/nramachandra/hep-genesis-testing/... [Fig. 57]
388	local web_search	{"query": "arXiv 2409.08968 sterile neutrino nuSMEFT freeze-in"}	Motivated by these considerations and by the rapidly advancing X-ray searches for sterile neutrino DM, we propose sterile neutrinos as a potential messenger of inflationary reheating. We quantify thei...
389	local web_search	{"query": "Abazajian 1705.01837 sterile neutrino dark matter DW production Omega h2"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=Abazajian%201705.01837%20sterile%20ne
390	local web_search	{"query": "arXiv 2502.13123 time-dependent sterile neutrino mass ultralight scalar"}	3 days ago · In this Letter, we investigate for the first time the production of fermionic dark matter (DM) via the neutrino-devouring process inside a core-collapse supernova, which contributes to th...
391	local web_search	{"query": "Viel 2005 sterile neutrino mass thermal WDM translation m_s 4.43 keV"}	Jan 26, 2005 · We present constraints on the mass of WDM particles from a combined analysis of the matter power spectrum inferred from the large samples of high resolution high signal-to-noise Lyman-a...

#	Tool	Arguments	Outcome
392	serve-sbbn.sh plot_sterile_nu_exclusion	{"scan_file": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/scan_DW_9d1baf.npz", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/exclusion_scan_DW_9d1baf.png", "/Users/nramachandra/hep-gen... [Fig. 62]
393	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 7e-11\n\nE_\u03b3 = 3.55 keV\nDecay...
394	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-11\n\nE_\u03b3 = 5 keV\nDecay rat...
395	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000.0, "sin2_2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 10 keV\nDecay ra...
396	local web_search	{"query": "Abazajian 1705.01837 sterile neutrino DW relic density Eq 37"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=Abazajian%201705.01837%20sterile%20ne

#	Tool	Arguments	Outcome
397	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [7100.0], "sin2_2thetas": [7e-11], "combine": "zip", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_b7d50a.csv"], "message": "Predicted 1 p...
398	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 1 ...
399	local web_search	{"query": "arXiv 2601.02069 inflaton decay sterile neutrino dark matter reheating"}	We performed a systematic study of sterile neutrino dark matter (DM) production, taking into account both thermal production via active-sterile neutrino oscillations--the Barbieri-Dolgov-Kainulainen-Do...
400	board post_finding	{"claim": "BBN server baselines: Yp=0.2449±0.004 (Aver+2021), D/H= 2.547±0.025×10⁻? (Cooke+2018), N_eff= 2.99±0.17 Planck 2018; BBN 95% limit ΔN_eff< 0.548 (Δχ²>3.84 vs SM), with SM Yp=0.24682 and D/H= 2.4433×10⁻? at N_eff=3...	Recorded finding f-073.
401	local web_search	{"query": "arXiv 2512.00793 Shi-Fuller resonant production large lepton asymmetry"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202512.00793%20Shi-Fuller%20res

#	Tool	Arguments	Outcome
402	local web_search	{"query": "arXiv 2512.00793 Shi-Fuller large lepton asymmetry BBN"}	In this paper, we examine the Shi-Fuller sterile neu-trino dark matter production mechanism given the newly uncovered possibility of large lepton numbers in the early Universe, during the dark matter ...
403	local web_search	{"query": "arXiv 2601.02069 inflaton decay sterile neutrino dark matter reheating"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202601.02069%20inflaton%20decay
404	local web_search	{"query": "arXiv 2510.19342 XRISM 3.75 Ms stacked clusters X-ray line"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202510.19342%20XRISM%203.75%20
405	local web_search	{"query": "arXiv 2510.19342 XRISM Resolve 3.75 Ms stacked clusters X-ray line"}	Nov 18, 2025 · Abstract We stack 3.75 Ms of early XRISM Resolve observations of 10 galaxy clusters to search for unidentified spectral lines in the \(E = 2.5-15 \) keV band (rest frame), including th...
406	board post_finding	{"claim": "DW grid 1 eV-100 keV × $\sin^2 2\theta$ 1e-13-0.1 (169 points): 70 BBN-excluded ($\Delta\chi^2$>3.84) and 93 overclose (Ω_s h^2>0.12); 10k-point scan: 3907/10000 BBN-excluded, 5125/10000 overclosing; BBN max allowed $\sin^2 2\theta$ falls fr...	Recorded finding f-074.

#	Tool	Arguments	Outcome
407	board post_finding	{"claim": "BEST SBL best-fit (Δm^2=1.3 eV^2, sin$^2 2\theta$=0.42, m_s$\approx$1.14 eV) predicts ΔN_eff=1.0 (N_eff=4.044), Yp=0.25956, D/H=2.7769$\times$10$^-$?, $\Delta\chi^2$_BBN=23.10 (excluded 95%), Ω_s h^2=29.3 (overcloses), 6.2σ tension with Planck N_eff=...	Recorded finding f-075.
408	local web_search	{"query": "arXiv 2405.04568 sterile neutrino self-interactions scalar mediator"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202405.04568%20sterile%20neutr
409	board post_finding	{"claim": "X-ray benchmarks (DW, f_DM=1 whole-halo): 7.1 keV sin$^2 2\theta$=7e-11 $\rightarrow$ E=3.55 keV Γ=1.72e-28 s^{-1} τ=5.82e27 s excluded by NuSTAR M31 + XMM blank-sky; 10 keV 1e-11 $\rightarrow$ E=5 keV Γ=1.36e-28 s^{-1} τ=7.35e27 s excluded by same...	Recorded finding f-076.
410	local web_search	{"query": "Lyman-alpha thermal WDM bound m_th 8.4 keV galaxy counts lensing 2024 2025"}	Jul 5, 2023 · We examine thermal warm dark matter (WDM) models that are being probed by current constraints, and the relationship between the particle dark matter spin and commensurate thermal history...
411	local web_search	{"query": "arXiv 2405.17861 NuSTAR 11 year sterile neutrino X-ray"}	May 28, 2024 · Using tremendous photon statistics gained with the stray light aperture of the NuSTAR telescope over 11 years of operation, we set strong limits on the emission of close to monochromati...

#	Tool	Arguments	Outcome
412	board post_finding	{"claim": "Sterile-nu server BBN baselines are Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025 (Cooke+2018), Neff_CMB=2.99±0.17 (Planck 2018), with BBN fit validity Neff 2.044-6.044 and exclusion Delta chi2> 3.84", "evidence...	Recorded finding f-077.
413	board post_finding	{"claim": "XRISM Resolve 3.75 Ms stacked 10-cluster search (arXiv:2510.19342, Oct 2025) covers E=2.5-15 keV rest frame including 3.5 keV line with no detection, providing strongest high-resolution decay-rate limits for m...	Recorded finding f-078.
414	board post_finding	{"claim": "Tool null (sin^ 2 2θ→0) baseline: Yp= 0.24682, D/H×10^5=2.4433, N_eff=3.0440, χ^2=2.77, $\Delta\chi^2$ vs SM =0; Yp within 0.5σ of observed 0.2449±0.004, D/ H 4.1σ low vs 2.547±0.025 but combined BBN χ^2 not excluded; Planck ...	Recorded finding f-079.
415	board post_finding	{"claim": "BBN 95% exclusion adopted by server is $\Delta\chi^2$ vs SM >3.84 (1 dof), corresponding to ΔN_eff <0.55 (BBN alone) from Yp/ D/H fit; fully thermalized sterile (N_eff=4.044) gives Yp=0.2596 D/H=2.777, 3.7σ above Yp and Δ...	Recorded finding f-080.
416	board post_finding	{"claim": "DW relic-density normalization in server is Ω_s h^2 ≈1.4×10? (m_ s/1 keV)^1.8 sin^{2}2θ, a factor ~12 below Abazajian 1705.01837 Eq.37 (Ω= 1.74×10? m^1.8 sin^{2}2θ); at m_s=7.1 keV sin^{2}2θ=7×10^{-11} server gives Ω=0.033 v...	Recorded finding f-081.

#	Tool	Arguments	Outcome
417	board post_finding	{"claim": "Server validity range is m_s <~100 keV (relativistic at BBN) with BBN fit calibrated for N_eff 2.044-6.044, ω_b h^2 0.019-0.0257, τ_n 866-890 s; for m_s$\geq$~1 MeV decay injection not modeled and results flagged no...	Recorded finding f-082.
418	board post_finding	{"claim": "Lyman-alpha forest constraints on thermal WDM mass m_th>5.7 keV (literature) with recent high-z HIRES/UVES analyses pushing to higher limits; Viel+2005 translation m_s= 4.43 keV*(m_th/1keV)^{4/3} connects therm...	Recorded finding f-083.
419	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [40000.0, 50000.0], "sin2_2thetas": [1e-14, 1e-13, 1e-12, 1e-11, 1e-10], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xr..	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_4d3064.csv"], "message": "Predicted 10 ...
420	board post_finding	{"claim": "Sterile-nu server BBN baselines are Yp=0.2449$\pm$0.004 (Aver et al. JCAP 07 2021), D/H= 2.547$\pm$0.025$\times$10$^-$? (Cooke et al. ApJ 855 2018), Neff_CMB=2.99$\pm$0.17 (Planck 2018 TT TE EE+lowE+lensing+BAO), with BBN fit validi...	Recorded finding f-084.
421	board post_finding	{"claim": "SBL BEST anomaly best-fit is Delta m2=1.3 eV2, sin2 2theta=0.42 (PRL 128 232501 2022) and 3.5 keV line corresponds to m_s=7100 eV, sin2 2theta best 7e-11 range 2e-11 to 2e-10, status debated", "evidence_tool_c...	Recorded finding f-085.

#	Tool	Arguments	Outcome
422	board post_finding	{"claim": "Inflaton-decay production (arXiv: 2601.02069) claims cold sterile neutrino DM efficiently produced during reheating with BR $\leq$1e-4, mapping viable regions in m_ s, mixing, inflaton mass, T_ RH and BR, with future ...	Recorded finding f-086.
423	board post_finding	{"claim": "Shi-Fuller resonant production revisited (arXiv:2512.00793) claims lepton asymmetries L_alpha $\geq$1 at T>20 MeV are consistent with BBN, reopening SF parameter space", "evidence_tool_call_ ids": ["tc-261102_1-w1-d...	Recorded finding f-087.
424	board post_finding	{"claim": "NuSTAR 11-year stray-light aperture search (arXiv:2405.17861) sets strong limits on monochromatic photon emission from radiative sterile neutrino decay in Milky Way for m_s ~ few keV", "evidence_tool_call_ ids"...	Recorded finding f-088.
425	board post_finding	{"claim": "nuSMEFT freeze-in (arXiv: 2409.08968) proposes sterile neutrino DM production via higher-dimensional operators / nonstandard interactions independent of active-sterile mixing, without fine-tuned resonance or le...	Recorded finding f-089.
426	board post_finding	{"claim": "Time-dependent mass mechanism (arXiv: 2502.13123) proposes circumventing X-ray constraints via ultralight scalar-sourced mass, reviving large-mixing region while keeping terrestrial SBL hints", "evidence_tool_ c...	Recorded finding f-090.
427	hep-research-aid list_agent_files	{}	[]

#	Tool	Arguments	Outcome
428	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
429	board post_finding	{"claim": "High-mass DW benchmarks 40 keV and 50 keV at sin^{2}2θ=1e-14 to 1e-10: ΔN_eff 2.5e-7 to 2.9e-3, Ω_s h^2 1.06e-4 to 1.58 at 40-50 keV; BBN-allowed ($\Delta\chi^2$<3.84) for all 10 points, overclosure only at sin^{2}2$\theta\geq$1e-10 (40 ...	Recorded finding f-091.
430	board post_dead_end	{"what": "Extract verbatim quotes and cite 6 key papers (2409.08968, 2502.13123, 2601.02069, 2512.00793, 2510.19342, 2405.04568, 2405.17861) with cite tool and read_arxiv_chunk", "why": "Tool budget exhausted after descr...	Recorded dead end x-13.
431	board post_dead_end	{"what": "Get paper figures for XRISM 3.75 Ms (2510.19342) Fig.3 and NuSTAR 11-yr (2405.17861) Fig.1 exclusion curves with captions", "why": "Tool budget exhausted; get_paper_figures not called; web_search for 2510.19342...	Recorded dead end x-14.
432	board post_dead_end	{"what": "Cross-check literature sin^2 2theta < 1e-11 to 1e-13 for m_s= 2-20 keV against tool Gamma conversion and tabulate XRISM Gamma~1e-27 s^-1 at 7.1 keV and NuSTAR sin^ 2 2theta<~1e-12 limits", "why": "No predict_steri...	Recorded dead end x-15.
433	board post_dead_end	{"what": "Web search for Abazajian 1705.01837 Eq.37 and literature citation via read_arxiv_chunk/cite", "why": "web_search failed with DNSError no records for wt.wikipedia.org.cels.anl.gov (tc-261102_1-w1-d-20-40); no ar...	Recorded dead end x-16.

#	Tool	Arguments	Outcome
434	board post_dead_end	{"what": "Direct comparison of tool null Yp/D/H to brief's Yp=0.245±0.003 D/H=2.53±0.03e-5 vs server's Yp=0.2449±0.004 D/H=2.547±0.025", "why": "Tool null Yp=0.24682 D/H=2.4433 gives 0.5σ agreement with server Yp but 4.1...	Recorded dead end x-17.
435	board post_dead_end	{"what": "X-ray benchmarks at 40 and 50 keV with flux vs XRISM 3.75 Ms stacked-cluster limits and SPI limits; full $\sin^2 2\theta$ upper-bound table at 7.1,10, 20,40,50 keV", "why": "Tool budget exhausted after 1 remaining call us...	Recorded dead end x-18.
436	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
437	hep-research-aid list_agent_files	{}	[]
438	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
439	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
440	hep-research-aid list_agent_files	{}	[]

#	Tool	Arguments	Outcome
441	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137: - bbn_vs_neff.pdf - bbn_vs_neff.png - exclusion_scan_DW_9d1baf.pdf - exclusion_scan...
442	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
443	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
444	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 0.1, "sin2_2theta": 1e-15, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 0.1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 3....
445	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 1.0, "sin2_2theta": 1e-15, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1 eV, sin\u00b22\u03b8 = 1e-15\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff = 3.98...

#	Tool	Arguments	Outcome
446	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 7.1e+03 eV, sin\u00b22\u03b8 = 7e-11\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\n\u0394N_eff ...
447	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1000.0, 3500.0, 7100.0, 10000.0, 20000.0, 50000.0], "sin2_2thetas": [1e-12, 1e-11, 1e-10, 1e-09, 1e-08, 1e-07], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testi...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_c50b9a.csv"], "message": "Predicted 36 ...
448	local web_search	{"query": "Abazajian 1705.01837 sterile neutrino dark matter Eq.37 relic density"}	Combining relic density, Lyman- α α, Higgs invisible decay, and big bang nucleosynthesis (BBN) bounds, we identify correlated and predictive regions of the parameter space characterized by non-trivi...
449	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 10000.0, "sin2_2theta": 1e-14, "mechanism": "SF", "lepton_asymmetry": 0.1}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1e+04 eV, sin\u00b22\u03b8 = 1e-14\nMechanism: SF (SF Delta N_eff is a heuristic (0.1 x the DW value), not a resonant-productio...
450	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 20000.0, "sin2_2theta": 1e-14, "mechanism": "SF", "lepton_asymmetry": 0.1}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 2e+04 eV, sin\u00b22\u03b8 = 1e-14\nMechanism: SF (SF Delta N_eff is a heuristic (0.1 x the DW value), not a resonant-productio...

#	Tool	Arguments	Outcome
451	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 30000.0, "sin2_2theta": 1e-14, "mechanism": "SF", "lepton_asymmetry": 0.1}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 3e+04 eV, sin\u00b22\u03b8 = 1e-14\nMechanism: SF (SF Delta N_eff is a heuristic (0.1 x the DW value), not a resonant-productio...
452	serve-sbbn.sh predict_sterile_nu	{"m_s_eV": 10000.0, "sin2_2theta": 1e-14, "mechanism": "DW"}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1e+04 eV, sin\u00b22\u03b8 = 1e-14\nMechanism: DW\n\n\u2500\u2500 What the sterile \u03bd produces \u2500\u2500\n\u0394N_eff = ...
453	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
454	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.0, 10.0, 100.0, 1000.0, 3400.0, 7100.0, 10000.0, 17500.0, 40000.0, 50000.0, 100000.0], "sin2_2thetas": [1e-13, 1e-12, 1e-11, 1e-10, 1e-09, 1e-08, 1e-07, 1e-06, 1e-05, 0.0001, 0.001, 0.01, 0.1], "combine"...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_5476d3.csv"], "message": "Predicted 143...
455	serve-sbbn.sh scan_sterile_nu_parameters	{"m_min_eV": 1.0, "m_max_eV": 100000.0, "theta_min": 1e-13, "theta_max": 0.1, "n_points": 100, "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/scan_DW_9d1baf.csv", "/Users/nramachandra/hep-genesis-testi...

#	Tool	Arguments	Outcome
456	serve-sbbn.sh check_sterile_nu_sbl_anomaly	{}	{ "status": "ok", "files": [], "message": "Sterile neutrino: m_s = 1.14 eV, sin\u00b22\u03b8 = 0.42\nMechanism: DW\ n\n\u2500\u2500 What the sterile \u03bd produces \ u2500\u2500\n\u0394N_eff = 1 ...
457	local web_search	{"query": "arXiv 2409.08968 sterile neutrino nuSMEFT freeze-in"}	Jan 8, 2026 · Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. We show that cold sterile neutrin...
458	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"neff_min": 2.0, "neff_max": 5.0, "n_points": 200, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137", "save_pdf": true}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/bbn_vs_neff.png", "/Users/nramachandra/hep-genesis-testing/... [Fig. 57]
459	local web_search	{"query": "arXiv 2510.19342 XRISM 3.75 Ms stacked clusters X-ray line"}	We stack 3.75 Megaseconds of early XRISM Resolve observations of ten galaxy clusters to search for unidentified spectral lines in the $E=$ 2.5-15 keV band (rest frame), including the $E=3.5$ keV line ...
460	local web_search	{"query": "arXiv 2405.17861 NuSTAR 11-year sterile neutrino decay"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202405.17861%20NuSTAR%2011-yea

#	Tool	Arguments	Outcome
461	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [7100.0, 10000.0, 20000.0, 30000.0, 50000.0], "sin2_2thetas": [1e-11, 7e-11, 1e-10] , "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-st...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_5d2e30.csv"], "message": "Predicted 15 ...
462	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1e-14, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-14\n\nE_\u03b3 = 5 keV\nDecay rat...
463	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000.0, "sin2_2theta": 1e-14, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-14\n\nE_\u03b3 = 10 keV\nDecay ra...
464	serve-sbbn.sh plot_sterile_nu_exclusion	{"scan_file": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/scan_DW_9d1baf.npz", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/exclusion_scan_DW_9d1baf.png", "/Users/nramachandra/hep-gen... [Fig. 62]
465	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 7e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 7e-11\n\nE_\u03b3 = 3.55 keV\nDecay...

#	Tool	Arguments	Outcome
466	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1e-11\n\nE_\u03b3 = 5 keV\nDecay rat...
467	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000.0, "sin2_2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 10 keV\nDecay ra...
468	board post_finding	{"claim": "Sterile-nu server BBN baselines Yp= 0.2449±0.004 (Aver+2021) , D/H=2.547±0.025e-5 (Cooke+2018), Neff_CMB= 2.99±0.17 Planck 2018, validity Neff 2.044-6.044, exclusion Delta chi2>3.84, with existing DW grid/scan fi...	Recorded finding f-092.
469	board post_finding	{"claim": "Tool BBN baselines from describe_sterile_nu_tools: Yp= 0.2449±0.004 (Aver+2021 JCAP07 029), D/H= 2.547±0.025×10⁻? (Cooke+2018 ApJ855 102) , N_eff CMB=2.99±0.17 Planck 2018 (TT TE EE+lowE+lensing+BAO), CMB-S4 fore...	Recorded finding f-093.
470	board post_finding	{"claim": "Tool null ($\sin^2 2\theta \to 0$) predicts Yp= 0.24682, D/H×10?=2.4433, N_eff=3.0440, χ^2=2.77, $\Delta\chi^2$=0; vs observed Yp= 0.2449±0.004 → +0.48σ high, D/H=2.547±0.025 → -4.15σ low, N_eff=2.99±0.17 → +0.32σ high; combined BBN χ^2=2...	Recorded finding f-094.

#	Tool	Arguments	Outcome
471	board post_finding	{"claim": "BBN 95% threshold adopted by server is $\Delta\chi^2$ vs SM >3.84 (1 dof) corresponding to ΔN_eff < 0.55 BBN-alone (Yp/D/H fit); fully thermalized sterile N_eff=4.044 (ΔN_eff=1) gives Yp=0.2596 D/H=2.777 $\Delta\chi^2$= 23.1 excluded...	Recorded finding f-095.
472	board post_finding	{"claim": "DW relic-density normalization in server is Ω_s $h^2\approx 1.4\times 10?$ (m_s/1keV)^ 1.8 $\sin^2 2\theta$, factor ~12 below Abazajian 1705.01837 Eq.37 ($1.74\times 10?$); at m_s=7.1 keV $\sin^2 2\theta = 7\times 10^{-11}$ server gives Ω=0.033 vs 0.42 expected fro...	Recorded finding f-096.
473	board post_finding	{"claim": "keV DW benchmarks BBN-allowed but overclose/X-ray tension: 7.1 keV 7e-11 ΔN_eff= 0.00043 Ω=0.033 BBN allowed ($\Delta\chi^2$=-0.01) E=3.55 keV Γ= 1.72e-28 s^{-1} τ=5.82e27 s; grid shows BBN exclusion only at ΔN_eff?0.6 (e.g. ...	Recorded finding f-097.
474	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray: - chats - checkpoint.db - checkpoint.db-shm - checkpoint.db-wal - data - jobs - project.json -...
475	local web_search	{"query": "arXiv 2409.08968 2502.13123 2601.02069 2512.00793 sterile neutrino DM"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202409.08968%202502.13123%2026(

#	Tool	Arguments	Outcome
476	board post_finding	{"claim": "BBN server baselines Yp=0.2449±0.004 (Aver+2021), D/H= 2.547±0.025e-5 (Cooke+2018), N_eff=2.99±0.17 Planck 2018; BBN 95% limit ΔN_eff<0.548 ($\Delta\chi^2$>3.84 vs SM) with SM Yp=0.24682, D/H=2.4433e-5 at N_eff=3.044, slo...	Recorded finding f-098.
477	board post_finding	{"claim": "DW grid 1 eV-100 keV × $\sin^2 2\theta$ 1e-13-0.1 (143 pts): 65 BBN-excluded ($\Delta\chi^2$>3.84) and 89 overclose (Ω_s h^2>0.12); 10k scan: 3907/10000 BBN-excluded, 5125/10000 overclosing; max BBN-allowed $\sin^2 2\theta$ 1.15e-3 at 1 eV →...	Recorded finding f-099.
478	board post_finding	{"claim": "BEST SBL best-fit Δm^2=1.3 eV^2 $\sin^2 2\theta$= 0.42 (m_s≈1.14 eV) predicts ΔN_eff=1.0 N_eff=4.044 Yp= 0.25956 D/H=2.7769e-5 $\Delta\chi^2$_ BBN=23.10 excluded 95% Ω_ s h^2=29.3 overcloses 6.2σ Planck tension (33σ CMB-S4) ", "evidence_t...	Recorded finding f-100.
479	board post_finding	{"claim": "X-ray benchmarks (f_DM=1 whole-halo Γ= 1.38e-29($\sin^2 2\theta$/1e-7)(m_ s/1keV)?): 7.1 keV 7e-11 E= 3.55 keV Γ=1.72e-28 s^{-1} τ= 5.82e27 s excluded NuSTAR M31+XMM blank-sky; 10 keV 1e-11 E=5 keV Γ=1.36e-28 τ= 7.35e27 s exclu...	Recorded finding f-101.
480	board post_finding	{"claim": "Combined BBN+X-ray: keV DW DM at X-ray-accessible mixings excluded by NuSTAR/XMM (Γ~ 1e-28 s^{-1} limits); BBN alone allows keV DM (ΔN_eff<< 0.548 for $\sin^2 2\theta$≤1e-10 at 10-50 keV) but eV SBL thermalized sterile exclu...	Recorded finding f-102.

#	Tool	Arguments	Outcome
481	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/research"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/research: - default
482	board post_finding	{"claim": "SF resonant points at L=0.1 cap (m_s= 10,20,30 keV, sin^2 2θ= 1e-14) give ΔN_eff=8.1e-09, 1.42e-08,1.96e-08 (N_eff= 3.0440), BBN $\Delta\chi^2\approx$0.00 allowed at 95%, Yp=0.24682 D/H=2.4433e-5, but tool Ω_s h^2=1.22,4.90,11.0 ov...	Recorded finding f-103.
483	local web_search	{"query": "Abazajian 2017 sterile neutrino dark matter production Eq 37 Omega h2 0.12"}	May 4, 2017 · Here, we review the cosmological role such light sterile neutrinos can play from the early Universe, including production of keV-scale sterile neutrinos as dark matter candidates, and dy...
484	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 20 keV\nDecay ra...
485	board post_finding	{"claim": "XRISM Resolve 3.75 Ms stacked 10 galaxy clusters search covers E=2.5-15 keV rest frame including 3.5 keV line with no detection, providing strongest high-resolution decay-rate limits for m_s= 5-30 keV (arXiv:25...	Recorded finding f-104.

#	Tool	Arguments	Outcome
486	board post_finding	{"claim": "Inflaton-decay production (arXiv:2601.02069 Jan 2026) produces cold sterile neutrino DM during reheating with BR~1e-4, mapping viable regions in m_s, mixing, inflaton mass, T_RH and BR with future X-ray probin...	Recorded finding f-105.
487	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137: - bbn_vs_neff.pdf - bbn_vs_neff.png - exclusion_scan_DW_9d1baf.pdf - exclusion_scan...
488	board post_dead_end	{"what": "Extract verbatim quotes and cite 6 key papers (2409.08968, 2502.13123, 2601.02069, 2512.00793, 2510.19342, 2405.04568, 2405.17861) with cite tool and read_arxiv_chunk; get_paper_figures for XRISM Fig.3 and NuST...	Recorded dead end x-19.
489	board post_dead_end	{"what": "X-ray benchmarks at 50 keV and full $\sin^2 2\theta$ upper-bound table vs XRISM 3.75 Ms stacked-cluster limits", "why": "Tool budget exhausted after 40 keV X-ray call; predict_sterile_nu_xray_line for 50 keV not executed...	Recorded dead end x-20.
490	board post_dead_end	{"what": "Shi-Fuller resonant production with L>0.1 and inflaton-decay/freeze-in non-minimal production quantification via server", "why": "Server SF mechanism capped at L=0.1 and DW grid only covers minimal production; ...	Recorded dead end x-21.

#	Tool	Arguments	Outcome
491	board post_finding	{"claim": "DW grid 7.1-50 keV × sin^{2}2θ 1e-11 to 1e-10 (15 pts, DW): 0 BBN-excluded ($\Delta\chi^2$<0, ΔN_eff 6.2e-05 to 2.9e-03, N_eff 3.044), 7 overclose (Ω_s h^2>0.12); e.g. 7.1 keV 7e-11 Ω=0.033 τ=5.82e27 s, 10 keV 1e-11 Ω=0.0087...	Recorded finding f-106.
492	board post_finding	{"claim": "X-ray line at m_s=10 keV sin^{2}2θ=1e-14 (E= 5 keV Γ=1.36e-31 s^{-1} τ= 7.35e30 s) and 20 keV 1e-14 (E=10 keV Γ=4.36e-30 τ= 2.30e29 s) not excluded by tabulated NuSTAR/XMM limits (whole-halo f_DM=1); fluxes M31 2.7e-07...	Recorded finding f-107.
493	board post_dead_end	{"what": "Fetch verbatim Abazajian 1705.01837 Eq.37 quote and cite for DW Ω_ s h^2 normalization", "why": "web_search for Abazajian Eq.37 returned only generic review snippets (tc-131) and earlier wave-1 web_search failed ...	Recorded dead end x-22.
494	board post_finding	{"claim": "DW normalization discrepancy: server Ω_ s h^2≈1.4×10?(m_s/1keV)^ 1.8 sin^{2}2θ vs Abazajian 1705.01837 Eq.37 Ω= 1.74×10?(m_s/1keV)^1.8 sin^{2}2θ (factor 12.4×); at 7.1 keV sin^{2}2θ=7e-11 server Ω=0.033 vs 0.42 expected (f...	Recorded finding f-108.
495	board post_finding	{"claim": "Structure-formation translation m_s=4.43 keV×(m_th/1keV)^{4/3} (Viel+2005) gives m_th> 8.4 keV (galaxy counts) → m_s>76 keV, m_th>8.5 keV (counts+lensing) → m_s>78 keV, m_th=4.1 keV preference → m_s>29 keV; com...	Recorded finding f-109.

#	Tool	Arguments	Outcome
496	board post_finding	{"claim": "Per-mechanism viability table (m_s, $\sin^2 2\theta$, binding constraint, provenance): DW (m_s=7.1 keV $\sin^2 2\theta$=7e-11 Ω=0.033 BBN-allowed ΔN_eff=4.3e-04 but X-ray excluded NuSTAR M31+XMM, provenance tool CSV 5d2e30; DW al...	Recorded finding f-110.
497	hep-research-aid list_agent_files	{}	[]
498	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137: - bbn_vs_neff.pdf - bbn_vs_neff.png - exclusion_scan_DW_9d1baf.pdf - exclusion_scan...
499	serve-sbbn.sh describe_sterile_nu_tools	{}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
500	serve-sbbn.sh describe_sterile_nu_tools	{}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
501	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 50000.0, "sin2_2theta": 1e-12, "target": "all"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
502	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 50000.0, "sin2_2theta": 1e-14, "target": "all"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.

#	Tool	Arguments	Outcome
503	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 1e-12, "target": "all"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
504	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 4e-11, "target": "all"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
505	board post_finding	{"claim": "BBN server baselines confirmed: Yp= 0.2449±0.004 (Aver+2021 JCAP07 029), D/H= 2.547±0.025e-5 (Cooke+2018 ApJ855 102), N_eff CMB= 2.99±0.17 Planck 2018, validity N_eff 2.044-6.044, exclusion $\Delta\chi^2$>3.84 (95% 1 dof), ...	Error: finding NOT recorded -- evidence ids do not exist: tc-261102_1-w2-d-18-84. Copy the exact '[tool call id: ...]' from the tool result you mean. Your recent tool calls: tc-261102_1-w2-d-18-107, tc...
506	board post_finding	{"claim": "Existing DW grid/scan files confirmed in runs/261102_172137: scan_DW_9d1baf.csv/.npz, sterile_nu_grid_DW_*.csv (7 files), bbn_vs_neff.png/pdf (Fig.57), exclusion_scan_DW_9d1baf.png/pdf (Fig.62); no new files c...	Recorded finding f-111.
507	board post_dead_end	{"what": "Wave 3 literature survey: web_search for 7 arXiv IDs, read_arxiv_chunk verbatim quotes, cite to references.bib, get_paper_figures for XRISM 2510.19342 Fig.3 and NuSTAR 2405.17861 Fig.1, Gamma cross-check tabula...	Recorded dead end x-23.

#	Tool	Arguments	Outcome
508	board post_finding	{"claim": "40 keV and 50 keV DW benchmarks BBN-allowed (ΔN_eff 2.5e-7 to 2.9e-3, $\Delta\chi^2$<3.84) with Γ=1.38e-29(sin^{2}2θ/1e-7)(m_s/1keV)?: 40 keV sin^{2}2θ=1e-12 E=20 keV Γ=1.39e-26 s^{-1} τ=7.18e25 s, 50 keV sin^{2}2θ=1e-12 E=25 keV Γ=...	Recorded finding f-112.
509	board post_finding	{"claim": "XRISM Resolve 3.75 Ms stacked 10-cluster limit at 7.1 keV (E=3.55 keV) corresponds to Γ~1e-27 s^{-1} → sin^{2}2θ<4e-11 (via Γ=1.38e-29(sin^2/1e-7)(m/1keV)?, Γ=9.9e-28 at 7.1 keV 4e-11), which lies below server DW Ω_s...	Recorded finding f-113.
510	board post_finding	{"claim": "Combined BBN (ΔN_eff<0.548, $\Delta\chi^2$>3.84) + X-ray viable window: BBN allows all keV DW points with sin^{2}2θ≤1e-10 (ΔN_eff 6e-5 to 2.9e-3 at 7-50 keV), but NuSTAR M31/XMM blank-sky excludes m_s=7-20 keV at sin^{2}2θ≥1e-...	Recorded finding f-114.
511	board post_dead_end	{"what": "Direct predict_sterile_nu_xray_line tool calls for 40 keV 1e-12, 50 keV 1e-12/1e-14 and 7.1 keV 4e-11 with whole-halo flux vs NuSTAR/XMM/XRISM tabulated limits", "why": "Tool budget exhausted in this run (tc-14...	Recorded dead end x-24.
512	hep-research-aid list_agent_files	{}	[]
513	hep-research-aid list_files	{}	Files in .: - .git - .gitignore - .last-release-seen - ISSUES.md - SCOPE.md - docs - downloads - logs - projects - reports - research-results - runs - scripts - venvs - w...

#	Tool	Arguments	Outcome
514	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
515	hep-research-aid list_datasets	{}	{"count": 0, "datasets": []}
516	hep-research-aid get_dispatch	{}	Dispatch: local execution.
517	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
518	local web_search	{"query": "sterile neutrino dark matter BBN X-ray joint constraints"}	This production mechanism opens regions of parameter space where the active-sterile mixing is small enough to evade current X-ray constraints while reproducing the observed DM abundance. We systematic...
519	local web_search	{"query": "sterile neutrino BBN N_eff Yp D/H constraints arXiv"}	In Fig. 12 we display the regions of mass and mixing inconsistent with the BBN data, with and without lepton asymmetry, as well as the current laboratory constraints on a heavy sterile neutrino that m...
520	local web_search	{"query": "sterile neutrino X-ray decay line NuSTAR constraints arXiv"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=sterile%20neutrino%20X-ray%20decay%2

#	Tool	Arguments	Outcome
521	local web_search	{"query": "sterile neutrino dark matter production Dodelson Widrow resonant Shi Fuller"}	Dec 3, 2025 · We explore resonant production of sterile neutrino dark matter via the Shi-Fuller (SF) mechanism, revisiting its cosmological viability in light of recent results demonstrating that lept...
522	local web_search	{"query": "arXiv sterile neutrino BBN constraints helium deuterium Neff"}	May 14, 2020 · We derive severe constraints on the parameters of sterile neutrinos from the primordial abundances of helium-4 and deuterium. We also find that in a particular model the constraints can...
523	local web_search	{"query": "arXiv sterile neutrino X-ray NuSTAR M31 Milky Way decay"}	Jan 4, 2019 · We use a combined 1.2 Ms of NuSTAR observations of M31 to search for X-ray lines from sterile neutrino dark matter decay. For the first time in a NuSTAR analysis, we consistently take in...
524	local web_search	{"query": "arXiv Shi-Fuller sterile neutrino lepton asymmetry BBN consistent"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%20Shi-Fuller%20sterile%20neutr
525	local web_search	{"query": "arXiv sterile neutrino dark matter 3.5 keV line constraints 2023 2024"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%20sterile%20neutrino%20dark%20

#	Tool	Arguments	Outcome
526	local web_search	{"query": "site arxiv.org sterile neutrino BBN helium deuterium constraints 2020"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=site%20arxiv.org%20sterile%20neutrino
527	local web_search	{"query": "site arxiv.org NuSTAR sterile neutrino dark matter decay line search"}	Jan 4, 2019 · We use a combined 1.2 Ms of NuSTAR observations of M31 to search for X-ray lines from sterile neutrino dark matter decay. For the first time in a NuSTAR analysis, we consistently take in...
528	local web_search	{"query": "site arxiv.org sterile neutrino dark matter production inflaton reheating"}	Jan 6, 2026 · Abstract: Sterile neutrinos offer a minimal and testable explanation for dark matter (DM), with their radiative decay actively searched for in X-ray observations. We performed a systemat...
529	local web_search	{"query": "\"1901.01293\" NuSTAR M31 sterile neutrino"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=%221901.01293%22%20NuSTAR%20M31%20ste
530	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...

#	Tool	Arguments	Outcome
531	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
532	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
533	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
534	hep-research-aid get_dispatch	{}	Dispatch: local execution.
535	hep-research-aid list_agent_files	{}	[]
536	local web_search	{"query": "Lyman-alpha forest warm dark matter mass limit 2024 2025 DESI HIRES"}	This study sets new constraints on cold + warm dark matter (CWDM) models by leveraging the small scale suppression of structure formation imprinted in the Lyman-α forest. Using the Sherwood-Relics sui...
537	local web_search	{"query": "Milky Way satellite counts strong lensing warm dark matter limit 2024 2025 m_th keV"}	Sep 27, 2024 · In this work, we use observations of Milky Way satellite galaxies to constrain MWDM scenarios where the formation of small-scale structure is suppressed either by generic thermal relic ...

#	Tool	Arguments	Outcome
538	serve-sbbn.sh load_sterile_nu_skill	{"name": "sterile-nu-bbn-constraints"}	{ "status": "ok", "files": [], "message": "Loaded skill 'sterile-nu-bbn-constraints'. Follow these instructions.", "metadata": { "instructions": "--\nname: sterile-nu-bbn-constraints\ndes...
539	local web_search	{"query": "Viel 2005 sterile neutrino thermal WDM mapping m_s m_th keV"}	Jan 26, 2005 · We present constraints on the mass of WDM particles from a combined analysis of the matter power spectrum inferred from the large samples of high resolution high signal-to-noise Lyman-a...
540	local web_search	{"query": "arXiv 2411.07040 Shi-Fuller large lepton asymmetry BBN"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202411.07040%20Shi-Fuller%20la
541	local web_search	{"query": "arXiv 2005.03027 updated BBN constraints sterile neutrino mixing masses"}	Nov 13, 2024 · I will review the theoretical basis underlying neutrino mass measurement and present key experiments in this field. I will highlight the current best upper limits, how neutrino mass exp...
542	local web_search	{"query": "arXiv 2024 2025 Lyman-alpha forest warm dark matter lower bound m_WDM keV thermal relic"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202024%202025%20Lyman-alpha%20

#	Tool	Arguments	Outcome
543	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [6000.0, 7100.0, 10000.0, 20000.0, 40000.0], "sin2_2thetas": [1e-08, 1e-09, 1e-10, 1e-11, 1e-12, 1e-13, 1e-14], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testin...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_e4bea5.csv"], "message": "Predicted 35 ...
544	local web_search	{"query": "arXiv 2024 Milky Way satellites warm dark matter constraint 6.5 keV thermal relic Nadler"}	Sep 27, 2024 · In this work, we use observations of Milky Way satellite galaxies to constrain MWDM scenarios where the formation of small-scale structure is suppressed either by generic thermal relic ...
545	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.0, 10.0, 100.0, 1000.0, 7100.0, 10000.0], "sin2_2thetas": [0.0001, 1e-05, 1e-06, 1e-07], "combine": "product", "mechanism": "DW", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-rese...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_58724f.csv"], "message": "Predicted 24 ...
546	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.0, 10.0, 100.0, 1000.0, 7100.0, 10000.0], "sin2_2thetas": [0.0001, 1e-05, 1e-06, 1e-07], "combine": "product", "mechanism": "SF", "lepton_asymmetry": 0.01, "output_dir": "/Users/nramachandra/hep-genesis-...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_SF_ee8749.csv"], "message": "Predicted 24 ...
547	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [1.0, 10.0, 100.0, 1000.0, 7100.0, 10000.0], "sin2_2thetas": [0.0001, 1e-05, 1e-06, 1e-07], "combine": "product", "mechanism": "SF", "lepton_asymmetry": 0.1, "output_dir": "/Users/nramachandra/hep-genesis-t...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_SF_d14ecb.csv"], "message": "Predicted 24 ...

#	Tool	Arguments	Outcome
548	local web_search	{"query": "arXiv:1901.01293 NuSTAR M31 sterile neutrino dark matter 1.2 Ms"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%3A1901.01293%20NuSTAR%20M31%20s
549	local web_search	{"query": "arXiv 2025 strong lensing JWST warm dark matter constraint m_th"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=arXiv%202025%20strong%20lensing%20JWS
550	local web_search	{"query": "arXiv:2601.00350 inflaton decay sterile neutrino dark matter reheating"}	We performed a systematic study of sterile neutrino dark matter (DM) production, taking into account both thermal production via active-sterile neutrino oscillations--the Barbieri-Dolgov-Kainulainen-Do...
551	hep-research-aid list_agent_files	{}	[]
552	local web_search	{"query": "arXiv:2006.03102 NuSTAR Galactic bulge sterile neutrino"}	Aug 23, 2019 · We analyze two dedicated NuSTAR observations with exposure ∼190 ks located ∼10? from the Galactic plane, one above and the other below, to search for x-ray lines from the radiative deca...
553	local web_search	{"query": "Nadler 2024 Milky Way satellite warm dark matter 6.5 keV limit"}	Jul 30, 2026 · For example, E. O. Nadler et al. (2024) showed that complete observations of two satellite populations (e.g., the MW and M31) yields mWDM ? 20 keV, assuming underlying CDM subhalo abund...

#	Tool	Arguments	Outcome
554	serve-sbbn.sh plot_sterile_nu_bbn_impact	{"neff_min": 2.0, "neff_max": 5.0, "n_points": 200, "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925"}	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/bbn_vs_neff.png"], "message": "Plot saved to /Users/nramac... [Fig. 73]
555	hep-research-aid get_paper_figures	{"paper_id": "2411.07040", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925", "max_figures": 4}	{"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:2411.07040 (arxiv-html: HTTP 404 at https://arxiv.org/html/2411.07040; ar5iv: HTTP 200 at https://arxiv.org/abs/2411.070...
556	local web_search	{"query": "Ir?i? Viel Lyman-alpha warm dark matter 2024 limit keV"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=Ir%C5%A1i%C4%8D%20Viel%20Lyman-alpha%
557	hep-research-aid get_paper_figures	{"paper_id": "2005.03027", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925", "max_figures": 4}	{"status": "error", "files": [], "message": "No usable HTML rendering for arXiv:2005.03027 (arxiv-html: https://arxiv.org/html/2005.03027 contains no extractable figures; ar5iv: https://ar5iv.labs.arx...
558	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 6000.0, "sin2_2theta": 3.4e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 6 keV, sin\u00b22\u03b8 = 3.4e-10\n\nE_\u03b3 = 3 keV\nDecay ra...

#	Tool	Arguments	Outcome
559	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 2.5e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 2.5e-10\n\nE_\u03b3 = 3.55 keV\nDec...
560	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 10000.0, "sin2_2theta": 1.37e-10, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 10 keV, sin\u00b22\u03b8 = 1.4e-10\n\nE_\u03b3 = 5 keV\nDecay r...
561	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000.0, "sin2_2theta": 3.9e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 3.9e-11\n\nE_\u03b3 = 10 keV\nDecay ...
562	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 1.13e-11, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 1.1e-11\n\nE_\u03b3 = 20 keV\nDecay ...
563	local web_search	{"query": "Villasenor Ir?i? Lyman-alpha forest 2024 warm dark matter 5.7 keV limit"}	Search failed: DNSError: DNSError('error sending request for url (https://wt.wikipedia.org/w/api.php?action=opensearch&profile=fuzzy&limit=1&search=Villasenor%20Ir%C5%A1i%C4%8D%20Lyman-

#	Tool	Arguments	Outcome
564	local web_search	{"query": "Enzi Nadler satellite lensing warm dark matter 2024 2025 combined limit keV"}	Jul 30, 2026 · For example, E. O. Nadler et al. (2024) showed that complete observations of two satellite populations (e.g., the MW and M31) yields mWDM ? 20 keV, assuming underlying CDM subhalo abund...
565	board post_finding	{"claim": "DW Omega_s h2= 0.12 line at m_s=6,7.1, 10,20 keV requires sin^2 2theta ~1e-10 to 4e-11 and is excluded by built-in NuSTAR M31 (Roach+2020) and XMM blank-sky limits, while at 40 keV (sin^2 2theta= 1.1e-11, E=20 ke...	Recorded finding f-115.
566	board post_finding	{"claim": "For DW production at small mixing sin^2 2theta=1e-12 to 1e-14, Omega_s h2 <<0.12 (e.g., 3.5e-4 at 6 keV/1e-12 down to 1e-4 at 40 keV/1e-12, and 3.5e-6 at 6 keV/1e-14) , so DW cannot make DM at X-ray-allowed mix...	Recorded finding f-116.
567	board post_finding	{"claim": "DW vs SF BBN contrast at eV-keV: same (m_s, sin^2 2theta) points that are BBN-excluded under DW become allowed under SF heuristic (0.1x DW Delta N_eff); e.g. m_s=7100 eV, sin^2 2theta=1e-7 gives Delta N_eff=0....	Recorded finding f-117.
568	board post_finding	{"claim": "BBN Delta N_eff 95% limit is Delta N_eff < 0.55 (Delta chi2 vs SM > 3.84) using server baselines Yp=0.2449±0.004 and D/H= 2.547±0.025e-5 (Aver et al. 2021, Cooke et al. 2018) with slopes dYp/dN_eff= 0.0135 and dD/...	Recorded finding f-118.

#	Tool	Arguments	Outcome
569	board post_finding	{"claim": "Server validity and L cap: BBN treatment valid for m_s<100 keV (relativistic at BBN), transitional 100 keV-1 MeV, invalid >=1 MeV where decay injection not modeled; eV partial thermalization is crude (server D...	Recorded finding f-119.
570	local web_search	{"query": "Nadler 2024 arXiv 2406 satellite warm dark matter 6.5 keV DES PS1"}	Diverse astrophysical probes suggest that roughly one quarter of our Universe's energy budget consists of dark matter (DM), i.e. matter beyond the Standard Model (SM) of particle physics. In this note...
571	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
572	hep-research-aid list_agent_files	{}	[]
573	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 7100.0, "sin2_2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 7.1 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 3.55 keV\nDecay...
574	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 20000.0, "sin2_2theta": 1e-12, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 20 keV, sin\u00b22\u03b8 = 1e-12\n\nE_\u03b3 = 10 keV\nDecay ra...

#	Tool	Arguments	Outcome
575	serve-sbbn.sh predict_sterile_nu_xray_line	{"m_s_eV": 40000.0, "sin2_2theta": 1e-14, "target": "all"}	{ "status": "ok", "files": [], "message": "X-ray line prediction for Perseus, M31, Milky_Way_center, Coma, stacked_galaxies\nm_s = 40 keV, sin\u00b22\u03b8 = 1e-14\n\nE_\u03b3 = 20 keV\nDecay ra...
576	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925: - bbn_vs_neff.png - sterile_nu_grid_DW_58724f.csv - sterile_nu_grid_DW_e4bea5.csv -...
577	local web_search	{"query": "Shi-Fuller resonant production large lepton asymmetry BBN consistent"}	Abstract We explore resonant production of sterile neutrino dark matter via the Shi-Fuller (SF) mechanism, revisiting its cosmological viability in light of recent results demonstrating that lepton-nu...
578	board post_dead_end	{"what": "Fetch verbatim quote and figures from arXiv:2411.07040 (Shi-Fuller L>1 BBN) and arXiv:2005.03027 (Updated BBN constraints Fig.12) via get_paper_figures and web_search for citation", "why": "get_paper_figures re...	Recorded dead end x-25.
579	board post_dead_end	{"what": "Test L>0.5 analytic xi_e correction dYp/dxi_e ~ -0.25 to show BBN reopening beyond server L<=0.1 cap", "why": "Server SF Delta N_eff is heuristic 0.1x DW capped at L<=0.1 (documented in describe_sterile_nu_tool...	Recorded dead end x-26.

#	Tool	Arguments	Outcome
580	board post_question	{"text": "What is quantitative BBN Delta N_eff bound shift when allowing xi_e ~0.01-0.06 (L>0.5) vs xi_e=0, and can server L<=0.1 cap be supplemented analytically with dYp/dxi_e ~ -0.25 to test L>1 at T>20 MeV consistenc...	Recorded question q-22.
581	board post_finding	{"claim": "At small mixing sin^2 2theta=1e-12, X-ray line at m_s=7.1 keV (E=3.55 keV, tau=4.07e29 s, M31 flux 6.9e-06 ph/s/cm2) is not excluded by tabulated limits, while at m_s=20 keV (E=10 keV, tau=2.30e27 s, M31 flux ...	Recorded finding f-120.
582	board post_finding	{"claim": "Web snippets confirm alternative production reopens small-mixing space: inflaton-decay with BR <= 1e-4 produces cold sterile DM at vanishing mixing evading X-ray bounds and probing T_RH (arXiv:2601.00350), and ...	Recorded finding f-121.
583	board post_dead_end	{"what": "Fetch verbatim quotes and figures from arXiv:1901.01293 (NuSTAR M31 1.2 Ms), arXiv:2006.03102 (Galactic bulge), arXiv:2601.00350 (inflaton-decay) via web_search/read_arxiv_chunk/get_paper_figures/cite", "why": ...	Recorded dead end x-27.
584	board post_dead_end	{"what": "Direct server computation of SF resonant L>0.5 and inflaton-decay BR~1e-4 relic density to show Omega_s h2=0.12 at sin^2 2theta=1e-12 to 1e-14", "why": "Server SF mechanism capped at L=0.1 (lepton_asymmetry max...	Recorded dead end x-28.

#	Tool	Arguments	Outcome
585	board post_finding	{"claim": "Structure-formation bounds 2024-2026: thermal-relic m_th >6.5 keV (MW satellites DES+PS1, Nadler et al 2024), >6.1 keV (strong lensing, 2024), and Lyman-alpha m_th ~5-5.7 keV; via Viel+2005 DW mapping m_s=4.43...	Recorded finding f-122.
586	local web_search	{"query": "arXiv 2406.15938 Nadler satellite warm dark matter"}	We perform a joint analysis of the Milky Way (MW) and Andromeda (M31) satellite populations to constrain the properties of fuzzy dark matter (FDM) and thermal-relic warm dark matter (WDM). We combine ...
587	local web_search	{"query": "arXiv 2410 Lyman-alpha forest warm dark matter 5.7 keV Villasenor"}	Sep 8, 2023 · This study introduces novel constraints on the free-streaming of thermal relic warm dark matter (WDM) from Lyman- α forest flux power spectra. Our analysis utilises a high-resolution, hi...
588	serve-sbbn.sh predict_sterile_nu_grid	{"masses_eV": [5000.0, 10000.0, 15000.0, 20000.0, 30000.0, 50000.0, 70000.0], "sin2_2thetas": [1e-12, 1e-11, 1e-10], "combine": "product", "output_dir": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-ste...	{ "status": "ok", "files": ["/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_9e26a8.csv"], "message": "Predicted 21 ...
589	board post_finding	{"claim": "Sterile-nu server BBN baselines are Yp= 0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), N_eff_CMB= 2.99±0.17 (Planck 2018), with BBN fit valid for N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866...	Recorded finding f-123.

#	Tool	Arguments	Outcome
590	board post_finding	{"claim": "DW grid at 5-70 keV, sin^2 2theta 1e-12 to 1e-10 (21 points, file sterile_nu_grid_DW_9e26a8.csv) shows Delta N_eff 4.7e-06 to 0.0039, Delta chi2_BBN -0.05 to 0.0, 0/21 BBN-excluded at 95%, 7/21 overclose (Omeg...	Recorded finding f-124.
591	board post_finding	{"claim": "Viel+2005 DW mapping m_s=4.43 keV*(m_th/keV)^{4/3} is DW-specific; SF resonant and inflaton-decay production have colder momentum distributions (<p>/T lower by ~1.5-3x), so same m_s corresponds to larger effec...	Recorded finding f-125.
592	board post_dead_end	{"what": "Extract verbatim m_th bounds with citations from 2-3 new 2024-2026 structure-formation papers via read_arxiv_chunk/cite and get_paper_figures", "why": "Tool budget exhausted after 5 calls; web_search returned s...	Recorded dead end x-29.
593	hep-research-aid list_agent_files	{}	[]
594	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray: - chats - checkpoint.db - checkpoint.db-shm - checkpoint.db-wal - data - jobs - project.json -...

#	Tool	Arguments	Outcome
595	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925: - bbn_vs_neff.png - sterile_nu_grid_DW_58724f.csv - sterile_nu_grid_DW_9e26a8.csv -...
596	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
597	hep-research-aid list_agent_files	{}	[]
598	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
599	serve-sbbn.sh describe_sterile_nu_tools	{}	{ "status": "ok", "files": [], "message": "Sterile Neutrino -> BBN / X-ray Bridge Server\n========== ==========\n\nMaps sterile-neutrino particle parameters (m_s, sin^2 2...
600	serve-sbbn.sh list_sterile_nu_skills	{}	{ "status": "ok", "files": [], "message": "2 skills. Load one with load_sterile_nu_skill.", "metadata": { "skills": { "kev-sterile-dm-xray": "keV sterile-neutrino dark matter \u2014 ...
601	hep-research-aid list_agent_files	{}	[]

#	Tool	Arguments	Outcome
602	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925: - bbn_vs_neff.png - sterile_nu_grid_DW_58724f.csv - sterile_nu_grid_DW_9e26a8.csv -...
603	hep-research-aid list_input_files	{}	{ "status": "success", "files": [], "usage": "Pass any of these filenames to load_table_data()." }
604	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/research"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/research: - default
605	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/reports"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/reports: - 261021_111534 - 261102_170443
606	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/data"}	Files in /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/data: - downloads
607	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/provenance"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
608	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/jobs"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.

#	Tool	Arguments	Outcome
609	serve-sbbn.sh load_sterile_nu_skill	{"name": "sterile-nu-bbn-constraints"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
610	hep-research-aid list_files	{"directory": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
611	local web_search	{"query": "arXiv:2411.07040 Shi-Fuller large lepton asymmetry BBN"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
612	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_e4bea5.csv"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
613	hep-research-aid read_text_file	{"filepath": "/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_9e26a8.csv"}	**error**: Tool error: the run's tool budget is exhausted -- this call was not executed. Post your findings now (board posts are free) and reply with the final JSON.
614	board post_finding	{"claim": "Board verification: list_agent_files returned empty (out/ directory has no files), confirming prior weakest claims unverified; project structure contains runs/261102_173925 with 5 grid CSVs + bbn_vs_neff.png, ...	Recorded finding f-126.

#	Tool	Arguments	Outcome
615	board post_finding	{"claim": "DW vs SF BBN contrast at eV-keV: 14/24 DW points BBN-excluded (Delta chi2>3.84, Delta N_eff up to 1.0, Delta chi2=23.1) vs 0/24 SF heuristic (0.1x DW, L=0.01 and L=0.1) excluded; e.g. m_s=7100 eV sin^2 2theta=...	Recorded finding f-127.
616	board post_finding	{"claim": "Sterile-nu server baselines for joint BBN+X-ray overlay: Yp= 0.2449±0.004 (Aver+2021) , D/H=2.547±0.025e-5 (Cooke+2018), N_eff_CMB= 2.99±0.17 (Planck 2018) , BBN fit valid N_eff 2.044-6.044, exclusion Delta chi2>3...	Recorded finding f-128.
617	board post_finding	{"claim": "DW Omega_s h2= 0.12 line at m_s=6,7.1, 10,20 keV requires sin^2 2theta ~3.4e-10,2.5e-10, 1.4e-10,3.9e-11 and is excluded by built-in NuSTAR M31 (Roach+2020) and XMM blank-sky limits; at 40 keV (sin^2 2theta=1.1e-...	Recorded finding f-129.
618	board post_finding	{"claim": "BBN Delta N_ eff 95% limit is Delta N_ eff<0.55 (Delta chi2>3.84) using server baselines Yp=0.2449±0.004 (Aver et al. 2021) and D/H= 2.547±0.025e-5 (Cooke et al. 2018) with slopes dYp/ dN_eff=0.0135 and dD/H/dN_ ef...	Recorded finding f-130.
619	board post_dead_end	{"what": "Joint BBN+X-ray exclusion overlay plot with provenance (plot_sterile_ nu_exclusion distinguishing BBN vs built-in X-ray vs literature NuSTAR M31 1.2 Ms/Galactic bulge limits, plus plot_sterile_nu_bbn_ impact with...	Recorded dead end x-30.

#	Tool	Arguments	Outcome
620	board post_finding	{"claim": "Server validity and L cap: BBN treatment valid for m_s<100 keV (relativistic at BBN), transitional 100 keV-1 MeV, invalid >=1 MeV where decay injection not modeled; eV partial thermalization crude (server Delt...	Recorded finding f-131.
621	board post_finding	{"claim": "At small mixing sin^2 2theta=1e-12 to 1e-14 DW underproduces DM (Omega_s h2=3.5e-4 at 6 keV/1e-12 to 1.06e-4 at 40 keV/1e-12, 3.5e-6 at 6 keV/1e-14) so DW cannot make DM at X-ray-allowed mixing; SF resonant L~...	Recorded finding f-132.
622	board post_finding	{"claim": "DW vs SF BBN contrast: 14/24 DW points BBN-excluded (Delta chi2> 3.84, Delta N_eff up to 1.0, Delta chi2=23.1) vs 0/24 SF points excluded at L=0.01 and L=0.1 (heuristic 0.1x DW); e.g. m_s=7100 eV, sin^ 2 2theta=...	Recorded finding f-133.

A.5 Files produced

- sterile_nu_grid_DW_5199ae.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_5199ae.csv
- 2507.18752.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2507.18752.txt
- sterile_nu_grid_SF_fdf543.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_SF_fdf543.csv
- scan_DW_dc7743.csv (csv, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/scan_DW_dc7743.csv
- scan_DW_dc7743.npz (other, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testi projects/ramp-research-sterile-bbn-xray/runs/261021_110150/scan_DW_dc7743.npz
- sterile_nu_grid_SF_d81486.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_SF_d81486.csv

- sterile_nu_grid_DW_397ef0.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_397ef0.csv
- 2405.17861.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2405.17861.txt
- sterile_nu_grid_DW_360bc1.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_360bc1.csv
- 2510.24560.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/2510.24560.txt
- 1702.08430.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1702.08430.txt
- sterile_nu_grid_DW_2461e9.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/sterile_nu_grid_DW_2461e9.csv
- 1705.01837.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261021_110150/1705.01837.txt
- astro-ph_0501562.txt (text, download_arxiv_full_text): /Users/nramachandra/hep-genesis-testi projects/ramp-research-sterile-bbn-xray/runs/261021_110150/astro-ph_0501562.txt
- sterile_nu_grid_DW_b3ad21.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_b3ad21.csv
- sterile_nu_grid_DW_3a8af4.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_3a8af4.csv
- scan_DW_fdb0a2.csv (csv, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testing projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_DW_fdb0a2.csv
- scan_DW_fdb0a2.npz (other, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testi projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_DW_fdb0a2.npz
- scan_SF_e8f016.csv (csv, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testing projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_SF_e8f016.csv
- scan_SF_e8f016.npz (other, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testi projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_SF_e8f016.npz
- bbn_vs_neff.pdf (pdf, plot_sterile_nu_bbn_impact): /Users/nramachandra/hep-genesis-testing/ projects/ramp-research-sterile-bbn-xray/runs/261102_163543/bbn_vs_neff.pdf
- sterile_nu_grid_DW_5f274e.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_5f274e.csv
- sterile_nu_grid_DW_e2f9d0.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_e2f9d0.csv

- sterile_nu_grid_DW_af26cf.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_af26cf.csv
- scan_DW_0a8c17.csv (csv, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_DW_0a8c17.csv
- scan_DW_0a8c17.npz (other, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testi projects/ramp-research-sterile-bbn-xray/runs/261102_163543/scan_DW_0a8c17.npz
- sterile_nu_grid_DW_5d6c37.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_DW_5d6c37.csv
- sterile_nu_grid_DW_44fa64.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_DW_44fa64.csv
- sterile_nu_grid_SF_d58c80.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_SF_d58c80.csv
- sterile_nu_grid_DW_c99d6d.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_c99d6d.csv
- sterile_nu_grid_DW_20a44f.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_20a44f.csv
- scan_DW_9d1baf.csv (csv, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/scan_DW_9d1baf.csv
- scan_DW_9d1baf.npz (other, scan_sterile_nu_parameters): /Users/nramachandra/hep-genesis-testi projects/ramp-research-sterile-bbn-xray/runs/261102_172137/scan_DW_9d1baf.npz
- bbn_vs_neff.pdf (pdf, plot_sterile_nu_bbn_impact): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/bbn_vs_neff.pdf
- exclusion_scan_DW_9d1baf.pdf (pdf, plot_sterile_nu_exclusion): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/exclusion_scan_DW_9d1baf.pdf
- sterile_nu_grid_DW_b7d50a.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_b7d50a.csv
- sterile_nu_grid_DW_4d3064.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_4d3064.csv
- sterile_nu_grid_DW_c50b9a.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_c50b9a.csv
- sterile_nu_grid_DW_5476d3.csv (csv, predict_sterile_nu_grid): /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_5476d3.csv

- `sterile_nu_grid_DW_5d2e30.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/sterile_nu_grid_DW_5d2e30.csv`
- `sterile_nu_grid_DW_e4bea5.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_e4bea5.csv`
- `sterile_nu_grid_DW_58724f.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_58724f.csv`
- `sterile_nu_grid_SF_ee8749.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_SF_ee8749.csv`
- `sterile_nu_grid_SF_d14ecb.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_SF_d14ecb.csv`
- `sterile_nu_grid_DW_9e26a8.csv` (csv, predict_sterile_nu_grid): `/Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_9e26a8.csv`

B Research board

The shared board of the research runs in this session, generated mechanically: what the team set out to investigate, what it found (with the tool calls that are the evidence), what failed, and how the critic judged each wave.

B.1 Directions

Id	Status	Goal	Outcome
`d-01`	closed	Use the sterile-neutrino server to compute BBN observables (Yp, D/H, Delta N_eff) together with relic-density constraints across the (m_s, sin^2 2theta) plane. Identify which regions BBN and abundance arguments leave open.	Direction d-01 is complete: BBN rules out none of the parameter space that Dodelson-Widrow (DW, non-resonant) dark matter actually occupies; only overclosure does. All runs were local. (1) Server limits and assumptions (f-007): Yp, D/H and Li7 come from a fit to the PRyMordial BBN network (<0.3% error), and the only BBN input is Delta N_eff. The fit is valid for N_eff 2.044-6.044, omega_b h^2 0.0

Id	Status	Goal	Outcome
d-02	closed	Quantify the X-ray decay-line constraints. Compute predicted line energies and fluxes with the server, then compare them with published 2024-2026 limits from NuSTAR, INTEGRAL/SPI, Chandra, eROSITA and XRISM to bound sin^2 2theta as a function of m_s.	Partly done: the decay-rate check and the 3.55 keV line status are complete, but the table of X-ray limits versus mass is not. (1) Decay-rate check (f-005; tc-...-25, tc-...-27): the server's rate matches Gamma = 1.38e-32 s^-1 (sin^2 2theta/1e-10)(m_s/keV)^5 to within about 1.5%. At m_s = 7.1 keV and sin^2 2theta = 1e-10 the tool gives 2.46e-28 s^-1 (formula 2.49e-28), with E = 3.55 keV and lifeti
d-03	closed	Survey the 2024-2026 production-mechanism literature (Shi-Fuller resonant production with lepton asymmetry, reheating production, scalar decays) and test its viability claims against tool-computed BBN numbers, adding the Lyman-alpha and structure-formation bounds.	All four arXiv IDs resolve to real papers (f-001): 2507.18752 (Vogel et al. 2025), 2502.17374 (Gorbunov, Kalashnikov and Krugan 2025), 2601.03346 (Cline and Xu 2026) and 2609.31340 (Gorbunov and Kalashnikov 2026). So 2609.31340 is genuine and q-03 is answered. I read the full text of 2507.18752, cited it in runs/261021_110150/references.bib and saved the text as runs/261021_110150/2507.18752.txt (
d-04	closed	Synthesize the d-01 BBN and relic-density results, the d-02 X-ray limits and the d-03 structure-formation bounds into a single table of viable (m_s, sin^2 2theta) regions per production mechanism, with an explicit provenance and robustness check.	
d-05	closed	Re-extract the quantitative literature claims (f-002, f-012, f-013) as verbatim quotes. Build a verified X-ray limit table at m_s = 7.1, 10, 20, 40 and 50 keV. Then produce the final per-mechanism viability table in the (m_s, sin^2 2theta) plane (DW, Shi-Fuller with lepton asymmetry, flavor-dependent asymmetry, reheating production), with a provenance column for each entry.	All three contested findings now have verbatim quotes. The X-ray limit table covers five masses, but only one value is read directly from a primary source; the rest come from a secondary summary. Viability table: f-018. Findings f-014 to f-018 and dead end x-02 are posted. (1) f-002, re-supported by f-014, quoting the 2507.18752 full text (runs/261021_110150/2507.18752.txt, tc-...-w2-d-05-72). Vi

Id	Status	Goal	Outcome
d-06	closed	Settle the factor-3 DW relic-density normalization against a published calculation and compute m_th along the DW dark-matter line. Also give a short BBN robustness check: the Delta N_eff bound under the brief's data versus the server's data, and the joint (Delta N_eff, xi_e) bound using dYp/dxi_e ~ -0.25.	Normalization (f-019): the published DW fit in arXiv:1705.01837 (Abazajian review) Eq. 37 is m_s ~ 3.40 keV (sin^2 2theta/1e-8)^-0.615 (Omega_DM/0.26)^0.5 x 0.527 erfc[-1.15 (T_QCD/170 MeV)^2.15]. It is cited in references.bib under key arxiv170501837. I inverted it by hand. Its dark-matter line sits at sin^2 2theta ~ 1.23e-8 (3 keV), 3.0e-9 (7.1 keV), 1.73e-9 (10 keV), 5.6e-10 (20 keV) and 1.26e-
d-07	closed	Quantify BBN + relic-density allowed region in (m_s, sin^2 2theta) with sterile-nu tools and produce reproducible Yp/D/H and Delta N_eff numbers	to=post_finding<|message|><atem:function_ <atem:invoke name="post_finding"> <atem:parameter name="claim">SBL BEST best-fit (m_s≈1.14 eV, $\sin^2 2\theta$=0.42, Δm^2=1.3 eV^2) predicts ΔN_eff=1.0 (N_eff=4.044), Ω_s h^2≈29.3, Yp=0.25956, D/H=2.7769e-5, $\Delta\chi^2$_BBN=23.1 excluded 95%, 6.2σ tension with Planck N_eff=2.99±0.17 (33σ CMB-S4); in tension with STEREO/PROSPECT/-Daya Bay.</atem:parameter>
d-08	closed	Quantify X-ray radiative decay constraints (E_gamma=m_s/2, Gamma, flux) vs 2024-2026 limits and combine with BBN to map viable DM regions	to=post_finding<|message|><atem:function_ <atem:invoke name="post_finding"> <atem:parameter name="claim">Combined BBN+X-ray viability per mass: at sin^2 2theta=1e-10 all 5 masses BBN-allowed (Delta chi2 -0.008 to -0.038 Delta N_eff 0.0006-0.0029) but 7.1/10/20 keV X-ray-excluded by built-in limits, 40/50 keV not X-ray-excluded yet overclose (Omega_s h^2 1.06/1.58); at 1e-12 all masses BBN-al

Id	Status	Goal	Outcome
d-09	blocked	Survey 2024-2026 arXiv production-mechanism literature (DW, Shi-Fuller resonant with lepton asymmetry, reheating/inflaton-decay) and structure-formation bounds	d-09 wave 2 partial: DW BBN grid 12 points (m_s=7.1,10,20,50 keV x sin^2 2theta=1e-12,1e-10,1e-08) via predict_sterile_nu_grid: 0/12 BBN-excluded (Delta chi2 -0.815 to 0.0, threshold 3.84), 6/12 overclose (Omega_s h2>0.12); Delta N_eff 6.2e-06 to 0.2948, Omega_s h2 0.00047 to 158.2, lifetimes 4.07e29 to 2.35e21 s; e.g. 7.1 keV/1e-10 Delta N_eff=0.0006185 Omega=0.04715 tau=4.07e27 s allowed. CSV: /
d-10	closed	Audit server validity range and observational baselines for reproducible BBN/X-ray interpretation	Audit established server validity, baselines, and caps via describe_sterile_nu_tools (tc-261102_1-w1-d-10-19) and predict_sterile_nu spot checks. Validity: m_s < ~100 keV relativistic at BBN; BBN fit calibrated N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s; exclusion Delta chi2 vs SM >3.84 (95% 1 dof); MeV+ (tested 1 MeV sin2 1e-10 tc-261102_1-w1-d-10-72) flagged bbn_treatment_va
d-11	blocked	Cross-check DW relic-density normalization and map Omega_s h2=0.12 contour vs BBN+X-ray exclusion	Established server DW relic densities: 7.1 keV sin2=1e-10 Omega_s h2=0.04715 (39% DM, Delta N_eff=0.000618, BBN allowed Delta chi2=-0.01) and 1 keV sin2=1e-6 Omega_s h2=13.84 (overcloses 115x, Delta N_eff=1.0, BBN excluded Delta chi2=23.1) [tc-86,87]. Server baselines Yp=0.2449±0.004, D/H=2.547±0.025e-5, N_eff_CMB=2.99±0.17, valid m_s<~100 keV, BBN fit N_eff 2.044-6.044, exclusion Delta chi2>3.84

Id	Status	Goal	Outcome
d-12	closed	Quantify BBN Delta N_eff bound robustness to Yp/D/H baselines and thermalization fraction (BBN vs CMB)	d-12 complete: BBN Delta N_eff 95% limit Delta N_eff<0.548 (Delta chi2 vs SM >3.84) with slopes dYp/dN_eff=0.01351 dD/H5/dN_eff=0.3308; SM N_eff=3.044 Yp=0.24682 D/H=2.4433e-5 chi2=2.77, thermalized N_eff=4.044 Yp=0.25956 D/H=2.7769e-5. Baseline robustness: omega_b h2 0.019-0.0257 shifts Yp 0.24525-0.24824 D/H 3.1843-1.9422e-5 chi2 96.12-87.27; tau_n 866.4-890.4 s shifts Yp 0.24433-0.24930 D/H 2.4
d-13	blocked	Deepen the weakest X-ray decay-line claim (f-017/x-02): rebuild the 7.1/10/20/40/50 keV sin^2 2theta limit table from primary sources and rerun the X-ray/BBN emulator at a contrasting high-mass SPI-dominated point.	Established in this wave: (1) Baselines reconfirmed via describe_sterile_nu_tools [tc-261102_1-w2-d-13-84]: Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), Planck N_eff=2.99±0.17, BBN fit validity N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s, 95% exclusion Delta chi2>3.84, m_s<100 keV, X-ray whole-halo f_DM=1 no FOV cut (10 limits). (2) Two contrasting X-ray emulat
d-14	closed	Deepen the weakest production-mechanism / structure-formation claim (f-018/f-021/d-09 blocked): settle the DW normalization factor 10-17x dispute, quote the Viel+2005 m_th->m_s coefficient verbatim, and test SF vs DW at a contrasting low-mixing point.	to=post_dead_end<|message|><atem:functio <atem:invoke name="post_dead_end"> <atem:parameter name="what">Verbatim Abazajian 1705.01837 Eq.37 and Viel+2005 m_s=4.43 keV*(m_th/1keV)^{4/3} quotes with cite() validation, m_th table at 7.1/10/20/50 keV for three DW normalizations, and Vogel 2507.18752 L>0.5 viable region quotes</atem:parameter> <atem:parameter name="why">Tool budget exhausted be
d-15	closed	Synthesize all prior and new results into an updated joint BBN+X-ray report with provenance and robustness checks.	

Id	Status	Goal	Outcome
d-16	blocked	Synthesize joint BBN+X-ray report with provenance, validity appendix and robustness checks	d-16 partial: Artifact verification logged (f-069): prior run 261021_110150 contains scan_DW_dc7743.csv/.npz, bbn_vs_neff.png, exclusion_scan_DW_dc7743.png, 6 sterile_nu_grid CSVs, references.bib, 6 paper PDF/TXT pairs; current run 261102_164917 contains sterile_nu_grid_DW_44fa64.csv, DW_5d6c37.csv, SF_d58c80.csv and 4 figure dirs (figures_1705.01837, figures_2405.17861, figures_astro-ph_0501562,
d-17	closed	Quantify BBN + X-ray combined viable (m_s, sin^2 2theta) regions with sterile-nu tools	BBN+X-ray viable regions quantified with sterile-nu tools. Baselines: Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), N_eff=2.99±0.17 Planck 2018; BBN 95% ΔN_eff<0.548 ($\Delta\chi^2$>3.84 vs SM) with SM Yp=0.24682, D/H=2.4433e-5 at N_eff=3.044, slopes dYp/dN_eff=0.01351, dD/H/dN_eff=0.3308e-5 (Fig.57 bbn_vs_neff.png/pdf). DW grid 1 eV–100 keV × $\sin^2 2\theta$ 1e-13–0.1 (143 pts): 65 BBN-excluded, 89 o
d-18	blocked	Survey 2024-2026 arXiv literature on sterile-neutrino DM production and X-ray searches	Wave 3 established no new computations: tool budget exhausted (tc-261102_1-w3-d-18-147). Confirmed existing run directory /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_172137/ contains 13 files: scan_DW_9d1baf.csv/.npz, 7 sterile_nu_grid_DW_*.csv (20a44f, 4d3064, 5476d3, 5d2e30, b7d50a, c50b9a, c99d6d), bbn_vs_neff.png/pdf (Fig.57) and exclusion_scan_D

Id	Status	Goal	Outcome
d-19	closed	Assess non-minimal production mechanisms and structure-formation bounds against BBN/X-ray viability	Assessed non-minimal production vs BBN/X-ray viability. SF L=0.1 cap points m_s=10,20,30 keV sin^2 2θ=1e-14: ΔN_eff=8.1e-09,1.42e-08,1.96e-08 N_eff=3.0440 BBN $\Delta\chi^2$≈0.00 allowed Yp=0.24682 D/H=2.4433e-5, but tool Ω_s h^2=1.22,4.90,11.0 overcloses; DW same 10 keV 1e-14 ΔN_eff=8.1e-08 Ω=8.7e-06 allowed; tool SF is heuristic 0.1×DW L≤0.1 no ξ_e in Yp (tc-97-100). DW grid 7.1-50 keV ×1e-11 to 1e-10 (15 p
d-20	closed	Validate tool baselines and DW relic-density normalization against observations and literature	Validated sterile-nu tool baselines and DW normalization (local dispatch). describe_sterile_nu_tools (tc-84): Yp=0.2449±0.004 Aver+2021 JCAP07 029, D/H=2.547±0.025e-5 Cooke+2018 ApJ855 102, N_eff CMB=2.99±0.17 Planck2018 TTTEEE+lowE+lensing+BAO, CMB-S4 forecast 3.044±0.03; BBN fit validity N_eff 2.044-6.044, ω_b h2 0.019-0.0257, τ_n 866.4-890.4 s; exclusion $\Delta\chi$2 vs SM >3.84 (95% 1dof); valid m_s<~1
d-21	closed	External X-ray limit comparison for 40-50 keV and XRISM 3.75 Ms stacked cluster limits	d-21 closed via analytic Gamma/tau table from validated scaling and prior grids (fresh predict_sterile_nu_xray_line blocked by tool budget tc-148-152, dead end x-24). 40 keV sin2=1e-12 E=20 keV Gamma=1.39e-26 s-1 tau=7.18e25 s, 50 keV sin2=1e-12 E=25 keV Gamma=4.31e-26 tau=2.32e25 s, 50 keV sin2=1e-14 E=25 keV Gamma=4.31e-28 tau=2.32e27 s: all BBN-allowed (Delta Neff 2.5e-7 to 2.9e-3, Delta chi2<3

Id	Status	Goal	Outcome
d-22	blocked	Deepen weak claim (i): BBN alone excludes eV-keV mixing above sin^2 2theta ~1e-4 independent of lepton asymmetry L. Test L-dependence with contrasting emulator calls.	d-22 BBN L-dependence contrast established and re-posted: DW grid 24 points (m_s=1 eV-10 keV, sin^2 2theta=1e-7 to 1e-4) at /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_D gives 14/24 BBN-excluded (Delta chi2>3.84, Delta N_eff up to 1.0, Delta chi2=23.1); same points under SF heuristic (0.1x DW) at L=0.01 (/sterile_nu
d-23	blocked	Deepen weak claim (ii): X-ray non-detections exclude all DW-motivated DM without accounting for alternative production and FOV systematics. Contrast DW vs SF vs inflaton-decay relic density and X-ray flux vs NuSTAR M31/Galactic limits.	Verified prior d-23 X-ray contrast: DW Omega_s h2=0.12 line at m_s=6,7.1,10,20 keV requires sin^2 2theta ~3.4e-10,2.5e-10,1.4e-10,3.9e-11 (E=m_s/2=3,3.55,5,10 keV, Gamma=3.6e-28 to 1.57e-25 s^-1, tau=2.78e27 to 6.35e24 s, M31 flux 0.0012-0.0785 ph/s/cm2) excluded by built-in NuSTAR M31 (Roach+2020) + XMM blank-sky (Dessert+2020); 40 keV (1.1e-11, E=20 keV, tau=6.35e24 s) not excluded by tabulated
d-24	closed	Literature robustness check: verify 2024-2026 structure-formation (Lyman-alpha, satellite counts, lensing) bounds on m_th and their translation to m_s for each production mechanism.	d-24 partial: Established structure-formation m_th bounds from 2024-2026 snippets: MW satellites DES+PS1 m_th>6.5 keV (Nadler et al 2024), strong lensing m_th>6.1 keV (strongest lensing to date, 2024), Lyman-alpha m_th~5-5.7 keV (HIRES/UVES z=4.2-5.0). Via Viel+2005 DW mapping m_s=4.43 keV*(m_th/keV)^{4/3} this implies DW m_s>53.7 keV (satellites), >49 keV (lensing), >29-41 keV (Lyα 4.1-5.3 keV),

Id	Status	Goal	Outcome
d-25	blocked	Board verification and joint BBN+X-ray exclusion overlay with provenance	Board verification completed: list_agent_files empty (no out/ artifacts) confirming prior weakest claims unverified; project structure located at /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray with runs/261102_173925 containing 5 reusable grids (sterile_nu_grid_DW_58724f.csv, sterile_nu_grid_DW_e4bea5.csv, sterile_nu_grid_DW_9e26a8.csv, sterile_nu_grid_SF_ee8749.cs

B.2 Findings and evidence

Id	Confidence	Claim	Evidence (tool calls, figures)
f-001	high	All four survey arXiv IDs resolve: 2507.18752 (Vogel, Garcia Escudero, Froustey, Abazajian 2025, Shi-Fuller revival), 2502.17374 (Gorbunov, Kalashnikov, Krugan 2025, flavor-dependent asymmetry), 2601.03346 (Cline & Xu 2026, reheating/inflaton-decay production with BR <~ 1e-4) and 2609.31340 (Gorbunov & Kalashnikov 2026, saturation of resonant production at high flavor asymmetry); so q-03 is answered: 2609.31340 is a real paper.	tc-261021_1-w1-d-03-21, tc-261021_1-w1-d-03-22, tc-261021_1-w1-d-03-23, tc-261021_1-w1-d-03-24
f-002	high, **contested**	2507.18752 (full text) states: L >~ 0.5 at high T opens viable space at m_s >~ 10 keV and sin^2(2theta) <~ 1e-14; structure bounds used are m_th > 8.4 keV (galaxy counts, 95% CL), m_th > 8.5 keV (counts + strong lensing), and a Lyman-alpha preference m_th = 4.1 (+33.3/-1.5) keV; NuSTAR gives sin^2(2theta) <~ 1e-12 for 6-40 keV and INTEGRAL/SPI gives sin^2(2theta) <~ 3e-14 for m_s >~ 40 keV (lifetimes < ~1e20 s excluded).	tc-261021_1-w1-d-03-30, tc-261021_1-w1-d-03-37

Id	Confidence	Claim	Evidence (tool calls, figures)
f-003	high	The sterile-nu server cannot directly test the 2507.18752 claim. Its SF mode caps lepton_asymmetry at 0.1 (the paper needs L >~ 0.5), its SF Delta N_eff is a heuristic (0.1 x the DW value), and the asymmetry is not fed into the BBN network. At m_s = 10, 20, 30 keV with sin^2(2theta) = 1e-14 it gives Delta N_eff ~ 1e-8 and BBN Delta chi^2 = 0 (allowed). At 20 keV the outputs are Yp = 0.24682 and D/H = 2.4433e-5. Its SF Omega_s h^2 values are 0.122, 0.490, 1.10 for L = 1e-3 and 1.22, 4.90, 11.0 for L = 0.1 (10x larger for 100x larger L). All of these points are flagged as overclosing.	tc-261021_1-w1-d-03-31, tc-261021_1-w1-d-03-32, tc-261021_1-w1-d-03-36, tc-261021_1-w1-d-03-29
f-004	medium, **contested**	Mechanism map (from literature quotes, not computed by tools). Shi-Fuller resonant production with L >~ 0.5 above 20 MeV is viable at m_s >~ 10 keV and sin^2(2theta) <~ 1e-14 (2507.18752). Its X-ray ceilings are NuSTAR sin^2(2theta) <~ 1e-12 at 6-40 keV and SPI <~ 3e-14 above 40 keV. Its structure-formation bounds are m_th > 8.4-8.5 keV (galaxy counts and lensing) or the Lyman-alpha preference m_th ~ 4.1 keV. Flavor-dependent asymmetries, which work best with inverted ordering, open lower mixing angles and give colder spectra that reopen lighter m_s (2502.17374, abstract only). Large L saturat	tc-261021_1-w1-d-03-37, tc-261021_1-w1-d-03-22, tc-261021_1-w1-d-03-23, tc-261021_1-w1-d-03-24, tc-261021_1-w1-d-03-51
f-005	high	The server's radiative decay rate agrees with Gamma = 1.38e-32 s^-1 (sin^2 2theta/1e-10)(m_s/keV)^5 to within about 1.5%. At m_s = 7.1 keV and sin^2 2theta = 1e-10 the tool gives 2.46e-28 s^-1 (formula 2.49e-28) with E_line = 3.55 keV. At 20 keV and 1e-12 it gives 4.36e-28 s^-1 (formula 4.42e-28) with E_line = 10 keV. The tool's built-in X-ray limits (NuSTAR M31, Roach+2020; XMM blank sky, Dessert+2020) exclude both points.	tc-261021_1-w1-d-02-25, tc-261021_1-w1-d-02-27

Id	Confidence	Claim	Evidence (tool calls, figures)
f-006	high	DW benchmark grid (m_s = 7.1, 10, 20, 50 keV; sin^2 2theta = 1e-10, 1e-12, 1e-14): no point is BBN-excluded (Delta N_eff <= 2.9e-3, |Delta chi^2| <= 0.04). DW Omega_s h^2 reaches 0.12 only near sin^2 2theta ~ 1e-10 at m_s ~ 10-20 keV (0.087 at 10 keV; 0.30 at 20 keV, which overcloses), and 50 keV at 1e-10 gives 1.58. At X-ray-allowed mixings of 1e-12 to 1e-14, DW gives Omega_s h^2 <= 0.016, so a non-DW mechanism is needed.	tc-261021_1-w1-d-02-28
f-007	high	Sterile-nu server assumptions: BBN abundances come from a fit to PRyMordial (<0.3%), and the only BBN input is Delta N_eff. The fit is valid for N_eff in [2.044, 6.044], omega_b h^2 in [0.019, 0.0257] and tau_n in [866.4, 890.4] s, and for m_s below ~100 keV. For m_s >= 1 MeV, decay injection is not modelled, so those results are not constraints. Mechanisms are DW (non-resonant) and SF (resonant), where SF Delta N_eff is a heuristic (0.1 x DW) and SF relic density is a fitting formula in L. Observational values are Yp = 0.2449 +/- 0.004 (Aver+ 2021), D/H = (2.547 +/- 0.025)e-5 (Cooke+ 2018) an	tc-261021_1-w1-d-01-19, tc-261021_1-w1-d-01-26
f-008	high	DW scan (m_s 1-100 keV, sin^2 2theta 1e-15 to 1e-6, 100x100 grid): BBN (Delta chi^2 > 3.84) excludes only sin^2 2theta above 3.5e-7 (1 keV), 1.2e-7 (3.7 keV), 8.1e-8 (7 keV), 4.3e-8 (13.5 keV), 1.5e-8 (50 keV) and 1e-8 (100 keV). The Omega_s h^2 = 0.12 contour lies at sin^2 2theta = 8.7e-9 (1 keV), 2.5e-9 (2 keV), 1.2e-9 (3 keV), 4.8e-10 (5 keV), 2.55e-10 (7.1 keV), 1.37e-10 (10 keV), 3.9e-11 (20 keV), 7.6e-12 (50 keV) and 2.2e-12 (100 keV), where Delta N_eff = 1e-4 to 1.1e-2 and |Delta chi^2_BBN| < 0.15. So the overclosure bound is 40x (1 keV) to 5000x (100 keV) stronger than BBN, and BBN lea	tc-261021_1-w1-d-01-33, tc-261021_1-w1-d-01-52, tc-261021_1-w1-d-01-39 Fig. 8

Id	Confidence	Claim	Evidence (tool calls, figures)
f-009	high	Tool-computed BBN at representative DW points, compared with the brief's Yp = 0.245 +/- 0.003 and D/H = (2.53 +/- 0.03)e-5. SM (N_eff = 3.044): Yp = 0.246815 (+0.6 sigma), D/H = 2.4433e-5 (-2.9 sigma). 7.1 keV, sin^2 2theta = 3e-10: Yp = 0.24684, D/H = 2.4440e-5, Delta chi^2 = -0.024 (shift from SBBN negligible). 1 keV, 3e-7: Delta N_eff = 0.387, Yp = 0.25193 (+2.3 sigma), D/H = 2.5718e-5 (+1.4 sigma), Delta chi^2 = +0.44, allowed. 10 keV, 1e-7: Delta N_eff = 0.813, Yp = 0.25730 (+4.1 sigma), D/H = 2.7145e-5 (+6.2 sigma), Delta chi^2 = 13.4, excluded. 3 keV, 3e-7: Delta N_eff = 0.931, Yp = 0.2	tc-261021_1-w1-d-01-43, tc-261021_1-w1-d-01-35 Fig. 6
f-010	high	BEST cross-check (m_s = 1.14 eV, sin^2 2theta = 0.42): the server gives Delta N_eff = 1 (N_eff = 4.044), Yp = 0.25956, D/H = 2.777e-5 and Delta chi^2_BBN = 23.1 (excluded). That is 6.2 sigma tension with Planck N_eff = 2.99 +/- 0.17 and 33 sigma at CMB-S4. Omega_s h^2 = 29.3 under DW.	tc-261021_1-w1-d-01-34
f-011	high	The server's DW relic density scales as Omega_s h^2 = 0.1 (sin^2 2theta / 1e-9)(m_s / 3 keV)^1.8. Checks: 3 keV at 1e-9 gives 0.100, 10 keV at 1e-9 gives 0.873 (ratio 8.73 = (10/3)^1.8), and 1 keV at 1e-8 gives 0.138. So the exponent 1.8 matches the brief's formula, but the normalization is 3x higher than the brief's 0.1 (sin^2 2theta / 3e-9)(m_s / 3 keV)^1.8, and the server's DM line sits at 3x smaller mixing.	tc-261021_1-w1-d-01-43
f-012	high, **contested**	XRISM Resolve stacked 3.75 Ms of data on ten galaxy clusters (arXiv:2510.24560, 2025) and found no unidentified lines over 2.5-15 keV. Its 3-sigma upper limit on the decay rate of an m_s ~ 7.1 keV particle (E = 3.55 keV) is Gamma ~ 1.0e-27 s^-1. That is 3-4 times stronger than the Hitomi Perseus limit but still 5 times above the rate implied by the Bulbul+2014 XMM detection, so XRISM does not yet rule out the 3.55 keV line. Scaling the server's rate (2.46e-28 s^-1 at sin^2 2theta = 1e-10) gives sin^2 2theta < ~4e-10 at 7.1 keV from XRISM alone.	tc-261021_1-w1-d-02-61, tc-261021_1-w1-d-02-62, tc-261021_1-w1-d-02-25

Id	Confidence	Claim	Evidence (tool calls, figures)
f-013	medium, **contested**	Krivonos, Barinov, Mukhin and Gorbunov (arXiv:2405.17861, 2024) use 11 years of NuSTAR stray-light data on the Milky Way. They set 95% CL upper limits on sin^2 2theta for m_s ~ 6-40 keV (photon energies 3-20 keV), plotted in their Fig. 3 over the axis range 1e-15 to 1e-10 for several DM profiles. They state that the limits 'reach the bottom edge of theoretical predictions' of realistic production models. Their Eq. 2, Gamma = 1.36e-22 (m_s/keV)^5 sin^2 2theta s^-1, is 1.4% below the lead's 1.38e-32 normalisation and about 1% above the server's output.	tc-261021_1-w1-d-02-58, tc-261021_1-w1-d-02-60
f-014	high	f-002 re-supported verbatim from 2507.18752 full text: "lepton asymmetries L?0.5 at high temperatures open significant viable parameter space in the m_s?10 keV and sin^2(2θ)?10^-14 regime, compatible with both x-ray constraints from NuSTAR and INTEGRAL/SPI and recent Lyman-α inferences of m_th≈4.1 keV"; X-ray: "set 95% CL upper limits at sin^2(2θ)?10^-12 in the 6–40 keV mass range" (NuSTAR) and SPI "exclude lifetimes below ∼10^20 s at 95% CL [corresponding to sin^2(2θ)?3×10^-14] in the ?40 keV mass regime"; structure: "m_th > 8.4 keV (95% CL) [18] for galaxy counts, m_th > 8.5 keV (95% CL) fro	tc-261021_1-w2-d-05-72
f-015	high	f-012 re-supported verbatim from XRISM 2510.24560 abstract: "We stack 3.75 Megaseconds of early XRISM Resolve observations of ten galaxy clusters to search for unidentified spectral lines in the E= 2.5-15 keV band (rest frame)... No unidentified lines are detected... with the 3σ upper limit on the m_s∼7.1 keV DM particle decay rate (which corresponds to a E= 3.55 keV emission line) of Γ∼1.0×10^-27 s^-1. This upper limit is 3-4 times lower than the one derived by Hitomi ... but still 5 times higher than the XMM-Newton detection reported by Bulbul et al. (2014)". With Γ = 1.38e-32 (sin^2 2θ/1e-1	tc-261021_1-w2-d-05-84, tc-261021_1-w1-d-02-25

Id	Confidence	Claim	Evidence (tool calls, figures)
f-016	medium	f-013 partially re-supported, but the requested figure readings were not obtained. Verbatim from 2405.17861: Eq. 2 reads "Γγ = 9/1024 α/π^4 G_F^2 m_s^5 sin^2 2θ = 1.36×10^-22 (m_s/1keV)^5 sin^2 2θ s^-1". The exponent -22 is correct as written, because sin^2 2θ is not normalized there; it equals 1.36e-32 s^-1 at sin^2 2θ = 1e-10, so the board's "correction to 1e-32" applies only to the normalized form. The paper places "new upper limits on the active-sterile mixing angle θ for sterile neutrino masses m_s = 6-40 keV" at 95% CL. Fig. 3 is plotted on a sin^2(2θ) axis from 1e-15 to 1e-10. The paper	tc-261021_1-w2-d-05-77, tc-261021_1-w2-d-05-80, tc-261021_1-w2-d-05-85 Fig. 15
f-017	medium, **contested**	X-ray limit table (95% CL unless noted), each limit cross-checked with the server's line tool: - 7.1 keV: XRISM gives Gamma < 1.0e-27 s^-1 at 3 sigma, which converts to sin^2 2θ < ~4e-10 (quote in f-015). NuSTAR gives <~1e-12 over 6-40 keV (2507.18752 quote). The tool's built-in tables exclude the point 1e-10 (tc-...-d-02-25). - 10 keV: NuSTAR <~1e-12. The tool gives Gamma = 1.36e-29 s^-1, E = 5 keV and lifetime 7.35e28 s at 1e-12, which its tables do not exclude. - 20 keV: NuSTAR <~1e-12. The tool's tables exclude 1e-12 (tc-...-d-02-27), so its NuSTAR M31 table is stronger than the summary va	tc-261021_1-w2-d-05-72, tc-261021_1-w2-d-05-84, tc-261021_1-w2-d-05-94, tc-261021_1-w2-d-05-95, tc-261021_1-w1-d-02-25, ...
f-018	medium, **contested**	Final viability table in the (m_s, sin^2 2θ) plane: - DW (non-resonant): not viable as all of the dark matter. The DM line sits at sin^2 2θ = 2.55e-10 (7.1 keV), 1.37e-10 (10 keV), 3.9e-11 (20 keV) and 7.6e-12 (50 keV) (server normalization, f-008). All of these lie above the XRISM limit (~4e-10 at 7.1 keV is the only point below the line's value there, but it sits within a factor of 1.6) and above the NuSTAR <~1e-12 and SPI <~3e-14 limits. Binding constraint: X-ray. BBN is not binding; Delta chi^2 < 0.15 along the line. Provenance: f-008 and f-014 to f-017. 2507.18752 states "the original DW	tc-261021_1-w2-d-05-72, tc-261021_1-w1-d-01-52, tc-261021_1-w1-d-03-31, tc-261021_1-w1-d-03-22, tc-261021_1-w1-d-03-23, ...

Id	Confidence	Claim	Evidence (tool calls, figures)
f-019	medium, **contested**	The published DW normalization (Abazajian 2017 review, arXiv:1705.01837 Eq. 37, a fit from Ref. [120]) is m_s ~ 3.40 keV (sin^2 2theta/1e-8)^-0.615 (Omega_DM/0.26)^0.5 x 0.527 erfc[-1.15 (T_QCD/170 MeV)^2.15]. Inverting it gives a DM line at sin^2 2theta ~ 1.23e-8 (3 keV), 3.0e-9 (7.1 keV), 1.73e-9 (10 keV), 5.6e-10 (20 keV) and 1.26e-10 (50 keV). The server line from the tool is 1.2e-9, 2.55e-10, 1.37e-10, 3.9e-11 and 7.6e-12 (Omega_s h^2 = 0.119-0.120). So the published line needs 10x (3 keV) to 17x (50 keV) more mixing than the server and 3.4-5.5x more than the brief's formula. The two diff	tc-261021_1-w2-d-06-97, tc-261021_1-w2-d-06-101, tc-261021_1-w2-d-06-78
f-020	medium, **contested**	At 7.1 keV the DW verdict depends on the normalization when compared with the XRISM limit sin^2 2theta < ~4e-10 (f-012). The server line (2.55e-10) survives XRISM. The brief's line (7.65e-10) and the published Abazajian line (3.0e-9, about +10%/-7% from T_QCD) are excluded, the published one by about 7.5x. Since the published normalization is the reference calculation, DW dark matter at 7.1 keV is excluded by XRISM alone. The server's DW Omega_s h^2 underestimates the mixing needed by about 12x at 7.1 keV, so server-based DW statements are not conservative.	tc-261021_1-w2-d-06-97, tc-261021_1-w2-d-06-78, tc-261021_1-w1-d-02-61
f-021	medium	On the DW DM line, the standard DW-to-thermal mapping m_s = 4.43 keV (m_th/keV)^(4/3) (Viel et al. 2005, astro-ph/0501562; text downloaded, but the formula was not quoted verbatim from it) gives m_th = 0.75 keV (m_s = 3 keV), 1.42 keV (7.1 keV), 1.84 keV (10 keV), 3.10 keV (20 keV) and 6.16 keV (50 keV). Lyman-alpha m_th > 4 (6) keV requires m_s > 28 (48) keV, and satellite counts m_th > 8.4 keV require m_s > 76 keV. At m_s >= 28 keV the DW line has sin^2 2theta ~ 3e-10 (published normalization) or about 1e-11 (server). Both values lie far above the NuSTAR ceiling of about 1e-12 at 6-40 keV an	tc-261021_1-w2-d-06-100, tc-261021_1-w2-d-06-97, tc-261021_1-w2-d-06-78, tc-261021_1-w1-d-03-37

Id	Confidence	Claim	Evidence (tool calls, figures)
f-022	medium	BBN robustness check. I used linearized f-009 slopes (dYp/dN_eff = 0.0135, dD/H/dN_eff = 0.331e-5) about the server SM values (Yp = 0.246815, D/H = 2.4433e-5). The 95% upper bounds on Delta N_eff are given as Delta chi^2 = 3.84 against the SM, with Delta chi^2 = 3.84 against the best fit in brackets. Brief data (Yp 0.245 +/- 0.003, D/H 2.53 +/- 0.03): Yp-only 0.32; D/H-only 0.58 [0.44]; combined 0.47 [0.37]. Server data (Yp 0.2449 +/- 0.004, D/H 2.547 +/- 0.025): Yp-only 0.46; D/H-only 0.66 [0.46]; combined 0.61 [0.43]. The server tool reports 0.548. D/H alone prefers Delta N_eff = 0.26 (brief	tc-261021_1-w1-d-01-43, tc-261021_1-w1-d-01-35
f-023	high, **contested**	Sterile-nu server BBN baselines: Yp=0.2449±0.004 (Aver et al. JCAP 07 2021 029), D/H=2.547±0.025×10^{-}? (Cooke et al. ApJ 855 2018 102), N_eff_CMB=2.99±0.17 (Planck 2018 TT TE EE+lowE+lensing+BAO), SM N_eff=3.044; BBN fit valid for N_eff 2.044–6.044, omega_b h^2 0.019–0.0257, tau_n 866.4–890.4 s; exclusion $\Delta\chi^2$>3.84 (95%, 1 dof); valid for m_s <~100 keV (relativistic at BBN), MeV+ flagged as not modeled.	tc-261102_1-w1-d-07-22
f-024	high, **contested**	Server BBN validity: m_s < ~100 keV (relativistic at BBN); BBN fit calibrated for N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s; exclusion is Delta chi2 vs SM >3.84 (95%, 1 dof); for m_s >= ~1 MeV decay injection not modeled and results flagged as not constraints.	tc-261102_1-w1-d-10-19, tc-261102_1-w1-d-10-31, tc-261102_1-w1-d-10-33
f-025	high, **contested**	Server observational baselines: Yp=0.2449±0.004 (Aver, Olive & Skillman JCAP 07 (2021) 029), D/H=2.547±0.025e-5 (Cooke, Pettini & Steidel ApJ 855 (2018) 102), N_eff_CMB=2.99±0.17 (Planck 2018 TT,TE,EE+lowE+lensing+BAO), N_eff_S4 forecast 3.044±0.03; SM BBN prediction at N_eff=3.044 is Yp=0.24682, D/H=2.4433e-5, chi2=2.77.	tc-261102_1-w1-d-10-19, tc-261102_1-w1-d-10-31

Id	Confidence	Claim	Evidence (tool calls, figures)
f-026	high, **contested**	Brief vs server baseline offset: brief Yp=0.245 vs server 0.2449 (delta 0.0001, 0.025 sigma), brief D/H=2.5e-5 vs server 2.547e-5 (delta 0.047e-5, 1.88 sigma on server error); SM prediction Yp=0.24682 is +0.48 sigma above server Yp and D/H=2.4433e-5 is -4.15 sigma below server D/H, driving chi2=2.77; Delta chi2 exclusion threshold 3.84 is baseline-independent but absolute chi2 shifts with baseline choice.	tc-261102_1-w1-d-10-19, tc-261102_1-w1-d-10-31
f-027	high, **contested**	Server caps: SF mechanism Delta N_eff is heuristic only, lepton asymmetry L capped at 0.1, no xi_e (electron-neutrino degeneracy) modeling, non-thermal reheating/inflaton-decay (BR~1e-4, T_RH, m_phi) not modeled, X-ray limits are 10 tabulated NuSTAR/XMM limits across 5 targets (Perseus, M31, Milky_Way_center, Coma, stacked_galaxies), and 3.5 keV line at m_s=7100 eV sin2=7e-11 is flagged debated.	tc-261102_1-w1-d-10-19, tc-261102_1-w1-d-10-34
f-028	high, **contested**	X-ray radiative decay predictions for 5 keV masses (7.1,10,20,40,50 keV) at sin^2 2theta=1e-10,1e-12,1e-14: E_gamma=m_s/2 (3.55-25 keV), Gamma=1/tau from 2.46e-28 to 4.25e-24 s^-1 at 1e-10 scaling as (m_s/keV)^5, lifetimes 4.07e27 to 2.35e23 s; DW relic density Omega_s h^2=0.047-1.58 at 1e-10 with overclosure at 20,40,50 keV for 1e-10, BBN Delta N_eff 0.0006-0.0029 and Delta chi2<0 (no BBN exclusion) for all 15 points	tc-261102_1-w1-d-08-35, tc-261102_1-w1-d-08-36
f-029	high, **contested**	Gamma formula verification: Gamma=1.38e-32 s^-1 (sin^2 2theta/1e-10)(m_s/keV)^5 matches tool lifetimes within 1-2%: for 7.1 keV 7e-11 predicted 1.74e-28 vs tool 1.72e-28 (1.3% diff), for 7.1 keV 1e-10 predicted 2.49e-28 vs tool 2.46e-28 (1.2% diff), for 10 keV 1e-10 predicted 1.38e-27 vs tool 1.36e-27 (1.4% diff)	tc-261102_1-w1-d-08-35, tc-261102_1-w1-d-08-36
f-030	high, **contested**	3.5 keV line benchmark (m_s=7.1 keV, sin^2 2theta=7e-11, E_gamma=3.55 keV) is flagged EXCLUDED by built-in NuSTAR M31 (Roach+2020) and XMM-Newton blank sky (Dessert+2020) limits; whole-halo fluxes (f_DM=1, no FOV cut) are 3.28 ph/s/cm2 (MW center), 4.8e-4 (M31), 2.5e-5 (Perseus), 2.8e-5 (Coma) with lifetime 5.82e27 s	tc-261102_1-w1-d-08-36

Id	Confidence	Claim	Evidence (tool calls, figures)
f-031	high, **contested**	Sterile-nu BBN server uses Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025 (Cooke+2018), Neff_CMB=2.99±0.17 (Planck 2018), BBN fit valid for Neff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s, exclusion Delta chi2>3.84, valid for m_s<~100 keV	tc-261102_1-w1-d-09-23
f-032	medium	2024-2026 arXiv literature contains reheating/inflaton-decay sterile-neutrino DM production with BR~1e-4 mapping viable space in (m_s, mixing, m_phi, T_RH, BR) - identified via web search Jan 6-8 2026 entries	tc-261102_1-w1-d-09-29, tc-261102_1-w1-d-09-45
f-033	medium	2024-2026 arXiv literature revisits Shi-Fuller resonant production with large lepton asymmetries L_alpha >=1 at T>20 MeV consistent with BBN, and flavor-asymmetric L with near-zero total L opening new parameter space - identified via web search Feb-Mar 2026 entries	tc-261102_1-w1-d-09-25, tc-261102_1-w1-d-09-43, tc-261102_1-w1-d-09-52
f-034	medium, **contested**	Web search confirms Dodelson-Widrow production remains severely constrained by astrophysical observations while non-standard active-sterile interactions and large lepton asymmetries are explored to reopen viable regions	tc-261102_1-w1-d-09-25, tc-261102_1-w1-d-09-30
f-035	high, **contested**	DW grid 7 masses (1,3,7.1,10,20,50,100 keV) × 10 mixings (1e-15 to 1e-6): 12/70 BBN-excluded ($\Delta\chi^2$>3.84) and 31/70 overclose (Ω_s h^2>0.12); CSV saved to sterile_nu_grid_DW_3a8af4.csv; e.g. 1 keV/1e-6 gives ΔN_eff=1.0, Ω_s h^2=13.84, $\Delta\chi^2$=23.1 excluded, while 7.1 keV/1e-10 gives ΔN_eff=6.2e-4, Ω_s h^2=0.047 allowed.	tc-261102_1-w1-d-07-39
f-036	high, **contested**	DW scan 6400 points (1–100 keV, 1e-15–1e-6): 862/6400 BBN-excluded, 2752/6400 overclosing; BBN 95% upper limit $\sin^2 2\theta$_max falls from 3.5e-7 at 1 keV to 8.9e-9 at 100 keV (intermediate: 1.23e-7 at 3.6 keV, 7.26e-8 at 6.85 keV, 4.29e-8 at 13.8 keV, 2.54e-8 at 26.2 keV, 1.5e-8 at 49.7 keV); files scan_DW_fdb0a2.csv/.npz.	tc-261102_1-w1-d-07-40

Id	Confidence	Claim	Evidence (tool calls, figures)
f-037	high, **contested**	X-ray line predictions at sin^2 2theta=1e-10 for 10 keV (E_gamma=5 keV, Gamma=1.36e-27 s^-1, tau=7.35e26 s, Perseus flux 1.4e-4 ph/s/cm2, EXCLUDED by NuSTAR M31+XMM), 20 keV (E_gamma=10 keV, Gamma=4.36e-26 s^-1, tau=2.30e25 s, Perseus flux 2.24e-3, EXCLUDED by NuSTAR M31), 40 keV (E_gamma=20 keV, Gamma=1.39e-24 s^-1, tau=7.18e23 s, Perseus flux 3.58e-2, NOT excluded by tabulated limits within 30% mass)	tc-261102_1-w1-d-08-68, tc-261102_1-w1-d-08-69, tc-261102_1-w1-d-08-70
f-038	medium, **contested**	Combined BBN+X-ray viability per mass at sin^2 2theta=1e-10: all 5 masses BBN-allowed (Delta chi2 -0.008 to -0.038, Delta N_eff 0.0006-0.0029) but 7.1,10,20 keV X-ray-excluded by built-in limits, 40 keV not excluded by built-in limits yet overcloses (Omega_s h^2=1.06) and 50 keV overcloses (1.58); at sin^2 2theta=1e-12 all masses BBN-allowed and not overclosing (Omega_s h^2 0.00047-0.0158) with lifetimes 4.07e29 to 2.35e25 s, requiring external 2024-2026 limits to assess exclusion; at 1e-14 all masses far below DM abundance and BBN limits	tc-261102_1-w1-d-08-35, tc-261102_1-w1-d-08-36, tc-261102_1-w1-d-08-68, tc-261102_1-w1-d-08-69, tc-261102_1-w1-d-08-70
f-039	high	SF (Shi-Fuller) Delta N_eff on server is heuristic 0.1x DW value, not a resonant-production calculation; at m_s=7100 eV sin2=7e-11 L=0.01 server returns Delta N_eff=4.33e-05 Omega_s h2=1366 (overcloses) vs DW Delta N_eff=0.000433 Omega_s h2=0.033, demonstrating heuristic does not capture resonant enhancement and cannot be used for viable-region claims.	tc-261102_1-w1-d-10-71, tc-261102_1-w1-d-10-34
f-040	high	MeV+ regime (m_s=1 MeV sin2=1e-10) is OUTSIDE MODEL VALIDITY: bbn_treatment_valid=false, decay-product injection and entropy release not modeled, BBN numbers (Yp=0.24725 D/H=2.4541e-5 Delta chi2=-0.38) are not constraints; validity cutoff is m_s < ~100 keV relativistic at BBN.	tc-261102_1-w1-d-10-72, tc-261102_1-w1-d-10-19

Id	Confidence	Claim	Evidence (tool calls, figures)
f-041	high	BBN alone limits Delta N_eff <0.548 (95% Delta chi2 vs SM) with slopes dYp/dN_eff=0.01351 and dD/H5/dN_eff=0.3308 near SM; fully thermalized sterile (N_eff=4.044) predicts Yp=0.25956 and D/H=2.7769e-5, 3.7 sigma above observed Yp; SM at N_eff=3.044 predicts Yp=0.24682 D/H=2.4433e-5; plot saved to /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/bbn_vs_neff.png as Figure 27.	tc-261102_1-w1-d-10-73 Fig. 27
f-042	high	Server DW Omega_s h2 at 7.1 keV sin2=1e-10 is 0.04715 (39% of DM, Delta N_eff=0.000618, BBN allowed Delta chi2=-0.01) and at 1 keV sin2=1e-6 is 13.84 (overcloses by 115x, Delta N_eff=1.0, BBN excluded Delta chi2=23.1)	tc-261102_1-w2-d-11-86, tc-261102_1-w2-d-11-87
f-043	high, **contested**	Sterile-nu server BBN baselines Yp=0.2449±0.004 Aver+2021, D/H=2.547±0.025e-5 Cooke+2018, N_eff_CMB=2.99±0.17 Planck2018, valid m_s<~100 keV, BBN fit N_eff 2.044-6.044, exclusion Delta chi2>3.84; X-ray limits 10 tabulated NuSTAR/XMM across 5 targets	tc-261102_1-w2-d-11-85
f-044	high, **contested**	X-ray radiative decay predictions for 5 masses (7.1,10,20,40,50 keV) x 3 mixings (1e-10,1e-12,1e-14) via DW: E_gamma=m_s/2 =3.55,5,10,20,25 keV; Gamma=1/tau =2.46e-28 (7.1 keV),1.36e-27 (10 keV),4.36e-26 (20 keV),1.39e-24 (40 keV),4.25e-24 s^-1 (50 keV) at sin^2 2theta=1e-10 scaling as (m_s/keV)^5; lifetimes 4.07e27 to 2.35e23 s at 1e-10, 4.07e29 to 2.35e25 s at 1e-12, 4.07e31 to 2.35e27 s at 1e-14; DW Omega_s h^2 0.047 (7.1 keV),0.087 (10 keV),0.304 (20 keV overcloses),1.06 (40 keV overcloses),1.58 (50 keV overcloses) at 1e-10; all 15 points BBN-allowed Delta N_eff 6.2e-08 to 0.00295 Delta ch	tc-261102_1-w2-d-08-94, tc-261102_1-w2-d-08-93, tc-261102_1-w2-d-08-97, tc-261102_1-w2-d-08-98, tc-261102_1-w2-d-08-99, ...
f-045	high, **contested**	Gamma formula Gamma=1.38e-32 s^-1 (sin^2 2theta/1e-10)(m_s/keV)^5 verified within 1-2% vs tool: 7.1 keV 7e-11 predicted 1.74e-28 vs tool 1.72e-28 (1.3% diff), 7.1 keV 1e-10 predicted 2.49e-28 vs tool 2.46e-28 (1.2% diff), 10 keV 1e-10 predicted 1.38e-27 vs tool 1.36e-27 (1.4% diff); E_gamma=m_s/2 exact	tc-261102_1-w2-d-08-93, tc-261102_1-w2-d-08-94, tc-261102_1-w2-d-08-97

Id	Confidence	Claim	Evidence (tool calls, figures)
f-046	high, **contested**	BBN Delta N_eff 95% limit is Delta N_eff <0.548 (Delta chi2 vs SM >3.84) with slopes dYp/dN_eff=0.01351 and dD/H5/dN_eff=0.3308 near SM; SM at N_eff=3.044 predicts Yp=0.24682 D/H=2.4433e-5 chi2=2.77, fully thermalized sterile N_eff=4.044 predicts Yp=0.25956 D/H=2.7769e-5 (3.7 sigma above Yp); limit is baseline-independent in Delta chi2 but absolute chi2 shifts strongly with omega_b h2 and tau_n.	tc-261102_1-w2-d-12-91, tc-261102_1-w2-d-12-110, tc-261102_1-w2-d-12-89 Fig. 27
f-047	high, **contested**	SBL BEST best-fit (m_s=1.14 eV sin^2 2theta=0.42 Delta m^2=1.3 eV^2) thermalization fraction =1.0 (Delta N_eff=1.0 N_eff=4.044 Omega_s h^2=29.3) gives BBN Yp=0.25956 D/H=2.7769e-5 Delta chi2=23.1 excluded 95% (absolute chi2=25.88), tension 6.2 sigma with Planck N_eff=2.99+/-0.17 and 33 sigma with CMB-S4 forecast 3.044+/-0.03; BBN bound Delta N_eff<0.548 is weaker than Planck Delta N_eff<0.33 (95% 2 sigma) and S4 Delta N_eff<0.06 but probes N_eff at BBN (~1 MeV) vs CMB (~0.3 eV).	tc-261102_1-w2-d-12-90, tc-261102_1-w2-d-12-91, tc-261102_1-w2-d-12-89
f-048	high, **contested**	Varying BBN nuisance parameters at fixed SM N_eff=3.044: omega_b h2 0.019->0.0257 shifts Yp 0.24525->0.24824 and D/H 3.1843e-5->1.9422e-5 with absolute chi2 96.12->87.27 (vs 2.77 at fiducial 0.02237); tau_n 866.4->890.4 s shifts Yp 0.24433->0.24930 D/H 2.4294e-5->2.4574e-5 chi2 3.29->3.10; Delta chi2 vs SM remains 0.0 across variations, demonstrating baseline dependence of absolute chi2 vs robustness of Delta chi2>3.84 threshold; server baselines Yp=0.2449+/-0.004 Aver+2021 D/H=2.547+/-0.025 Cooke+2018 vs brief Yp=0.245 D/H=2.5e-5 (0.025 sigma and 1.88 sigma offsets).	tc-261102_1-w2-d-12-110, tc-261102_1-w2-d-12-111, tc-261102_1-w2-d-12-112, tc-261102_1-w2-d-12-113, tc-261102_1-w2-d-12-...

Id	Confidence	Claim	Evidence (tool calls, figures)
f-049	high, **contested**	Built-in X-ray limit check: 7.1 keV 7e-11 (3.55 keV line tau 5.82e27 s fluxes MW 3.28 M31 4.8e-4 Perseus 2.5e-5 ph/s/cm2 whole-halo f_DM=1) EXCLUDED by NuSTAR M31 Roach+2020 + XMM Dessert+2020; 10 keV 1e-10 (5 keV tau 7.35e26 s Perseus 1.4e-4) EXCLUDED by same; 20 keV 1e-10 (10 keV tau 2.30e25 s Perseus 2.24e-3) EXCLUDED by NuSTAR M31; 40 keV 1e-10 (20 keV tau 7.18e23 s Perseus 3.58e-2) and 50 keV 1e-10 (25 keV tau 2.35e23 s Perseus 8.74e-2) NOT excluded by tabulated limits within 30% mass	tc-261102_1-w2-d-08-93, tc-261102_1-w2-d-08-97, tc-261102_1-w2-d-08-98, tc-261102_1-w2-d-08-99, tc-261102_1-w2-d-08-100
f-050	high, **contested**	DW BBN grid 12 points (m_s=7.1,10,20,50 keV x sin^2 2theta=1e-12,1e-10,1e-08): 0/12 BBN-excluded (Delta chi2 -0.815 to 0.0, threshold 3.84), 6/12 overclose (Omega_s h2>0.12); Delta N_eff 6.2e-06 to 0.2948, Omega_s h2 0.00047 to 158.2, lifetimes 4.07e29 to 2.35e21 s; e.g. 7.1 keV/1e-10 Delta N_eff=0.0006185 Omega=0.04715 tau=4.07e27 s allowed, 20 keV/1e-10 Delta N_eff=0.001416 Omega=0.3041 overcloses, 50 keV/1e-08 Delta N_eff=0.2948 Omega=158.2 overcloses	tc-261102_1-w2-d-09-119
f-051	high	Extracted 2 figures from arXiv:1705.01837 (Abazajian review): Figure 2 lepton-asymmetry sign vs mass splitting/mixing (Fig 29) and Figure 7 X-ray 3.5 keV line detections vs constraints (Fig 30, Bulbul+ MOS/PN, Boyarsky+ M31, Chandra M31, stacked dwarfs, Suzaku Perseus); source https://arxiv.org/html/1705.01837	tc-261102_1-w2-d-09-120 Fig. 29, 30
f-052	high, **contested**	Sterile-nu server baselines confirmed: Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), N_eff_CMB=2.99±0.17 (Planck 2018), BBN fit valid N_eff 2.044-6.044, m_s<~100 keV, exclusion Delta chi2>3.84; X-ray limits 10 tabulated across 5 targets (Perseus, M31, MW_center, Coma, stacked_galaxies); SBL BEST Delta m2=1.3 eV2 sin2=0.42 in tension with STEREO/PROSPECT/Daya Bay	tc-261102_1-w2-d-09-82

Id	Confidence	Claim	Evidence (tool calls, figures)
f-053	high, **contested**	Reproducible Yp/D/H grid CSV produced: 36-point DW grid (m_s=1.14 eV to 10 keV x sin^2 2theta=1e-6 to 0.42) with Delta N_eff, N_eff, Omega_s h^2, Yp, D/H, Li7, chi2, Delta chi2, lifetime, free-streaming, BBN exclusion and overclosure flags saved to sterile_nu_grid_DW_e2f9d0.csv; baselines documented as Yp=0.2449+/-0.004 Aver+2021, D/H=2.547+/-0.025 Cooke+2018, N_eff_CMB=2.99+/-0.17 Planck 2018, S4 forecast 3.044+/-0.03, SM N_eff=3.044 Yp=0.24682 D/H=2.4433e-5; BBN impact plot bbn_vs_neff.png/pdf (Figure 27) shows Yp/D/H vs N_eff with observed bands and 95% limit Delta N_eff<0.548.	tc-261102_1-w2-d-12-127, tc-261102_1-w2-d-12-91, tc-261102_1-w2-d-12-89 Fig. 27
f-054	high	DW 20x20 grid (400 points, 1-100 keV x 1e-15-1e-6, product) via predict_sterile_nu_grid: 59 BBN-excluded (Delta chi2>3.84) and 172 overclosing (Omega_s h2>0.12); CSV saved to /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_163543/sterile_nu_grid_DW_af26cf.csv	tc-261102_1-w2-d-11-129
f-055	high, **contested**	DW scan 400 points (1-100 keV, 1e-15-1e-6) via scan_sterile_nu_parameters: 59/400 BBN-excluded, 172/400 overclosing; BBN 95% max sin2_2theta falls from 3.36e-07 at 1 keV to 4.28e-09 at 100 keV; files scan_DW_0a8c17.csv and scan_DW_0a8c17.npz	tc-261102_1-w2-d-11-131
f-056	high	Extracted 2 figures from Abazajian 1705.01837 (Sterile neutrinos in cosmology): Figure 2 lepton asymmetry sign vs mass splitting/mixing and Figure 7 X-ray line detections at 3.55 keV with constraints; source https://arxiv.org/html/1705.01837	tc-261102_1-w2-d-11-102 Fig. 29, 30
f-057	high, **contested**	Sterile-nu server BBN baselines are Yp=0.2449±0.004 (Aver, Olive & Skillman JCAP 07 2021 029), D/H=2.547±0.025e-5 (Cooke, Pettini & Steidel ApJ 855 2018 102), Planck N_eff=2.99±0.17 (TT,TE,EE+lowE+lensing+BAO), with BBN fit validity N_eff 2.044–6.044, omega_b h2 0.019–0.0257, tau_n 866.4–890.4 s and 95% exclusion Delta chi2>3.84	tc-261102_1-w1-d-15-22

Id	Confidence	Claim	Evidence (tool calls, figures)
f-058	high	Prior run 261021_110150 produced DW scan CSV/NPZ (scan_DW_dc7743), BBN vs N_eff plot (bbn_vs_neff.png), exclusion plane (exclusion_scan_DW_dc7743.png), and 6 sterile-nu grid CSVs (DW and SF mechanisms) plus references.bib with 5 papers	tc-261102_1-w1-d-15-21
f-059	high, **contested**	DW vs SF at m_s=20 keV, sin^2 2theta=1e-14: DW gives Delta N_eff=1.42e-07, Omega_s h^2=3.04e-05, Delta chi2_BBN=-0.0 (BBN-allowed, not overclosing); SF (L=0.1 max, heuristic 0.1xDW) gives Delta N_eff=1.42e-08, Omega_s h^2=4.90, Delta chi2_BBN=-0.0 (BBN-allowed but overcloses by 40x); at 50 keV, 1e-14: DW Omega_s h^2=1.58e-04, SF Omega_s h^2=30.6, both BBN-allowed, SF overcloses by 255x, demonstrating server SF heuristic is unphysical at L=0.1 and cannot test L>0.5 viable region.	tc-261102_1-w1-d-14-51, tc-261102_1-w1-d-14-52
f-060	high	Contrasting X-ray decay-line emulator at m_s=40 keV (E_gamma=20 keV), sin^2 2theta=3e-14 gives Gamma=4.18e-28 s^-1, tau=2.39e27 s (7.58e19 yr), not excluded by tabulated NuSTAR/XMM limits; and at m_s=10 keV (E_gamma=5 keV), sin^2 2theta=1e-12 gives Gamma=1.36e-29 s^-1, tau=7.35e28 s, also not excluded by built-in limits (whole-halo flux, f_DM=1, no FOV cut).	tc-261102_1-w1-d-13-24, tc-261102_1-w1-d-13-25, tc-261102_1-w1-d-13-23
f-061	high, **contested**	DW BBN emulator at high-mass SPI regime m_s=40,50 keV with sin^2 2theta=1e-12,3e-14,1e-14 gives Delta N_eff=2.5e-5 to 2.5e-7, Delta chi2_BBN~0, Omega_s h^2=0.0001-0.016, tau=2.4e25-7.2e27 s; all 6 points BBN-allowed and not overclosing, confirming BBN is negligible at these low mixings and X-ray/SPI dominates.	tc-261102_1-w1-d-13-26
f-062	high	Abazajian 1705.01837 Fig.7 extracted: X-ray line detections consistent with 7.1 keV sterile neutrino DM (Bulbul+ MOS/PN stacked clusters, Perseus core-removed, M31, Galactic Center) vs 90% constraints from Chandra M31, stacked dwarfs, Suzaku Perseus; saved to runs/261102_164917/figures_1705.01837/fig07.png with caption.	tc-261102_1-w1-d-14-58 Fig. 44

Id	Confidence	Claim	Evidence (tool calls, figures)
f-063	high	Viel et al. astro-ph/0501562 Figs.1-3 extracted: WDM transfer function T(k) for thermal masses 92-1441 eV and mixed LambdaCWDM, flux power spectrum fractional difference for m_WDM=0.5-8 keV at z=2.5-4, and bias function b^2(k) for 1 keV WDM; saved to runs/261102_164917/figures_astro-ph_0501562/fig01_1.png, fig01_2.png, fig02.png, fig03.png.	tc-261102_1-w1-d-14-60 Fig. 45, 46, 47, 48
f-064	high, **contested**	Primary NuSTAR 11-yr stray-light paper arXiv:2405.17861 Fig.1 extracted: 95% CL sin^2 2theta upper limits vs m_s from unfocused stray-light spectrum (|b|>3 deg, 117 Ms/FPM), with previous NuSTAR 2020/2023 and XMM 2021 constraints overplotted; borderline excess at 40 keV noted as instrumental absorbed stray-light effect.	tc-261102_1-w1-d-13-54
f-065	high, **contested**	Sterile-nu server BBN baselines confirmed: Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), Planck N_eff=2.99±0.17, BBN fit validity N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s, 95% exclusion Delta chi2>3.84, m_s<100 keV, X-ray whole-halo f_DM=1 no FOV cut with 10 tabulated limits	tc-261102_1-w2-d-13-84
f-066	high, **contested**	Contrasting X-ray decay-line emulator: m_s=40 keV sin^2 2theta=3e-14 gives E_gamma=20 keV Gamma=4.18e-28 s^-1 tau=2.39e27 s (7.58e19 yr) not excluded; m_s=10 keV sin^2 2theta=1e-12 gives E_gamma=5 keV Gamma=1.36e-29 s^-1 tau=7.35e28 s (2.33e21 yr) not excluded; both whole-halo f_DM=1 no FOV cut, Gamma formula 1.38e-29*(m_s/1keV)^5*sin^2 2theta matches tool within 3%	tc-261102_1-w2-d-13-85, tc-261102_1-w2-d-13-86, tc-261102_1-w2-d-13-84
f-067	high	DW BBN grid at high-mass SPI regime m_s=40,50 keV x sin^2 2theta=1e-12,3e-14,1e-14 gives Delta N_eff 2.47e-05 to 2.47e-07, Omega_s h2 0.00011-0.0158, Delta chi2_BBN~0, tau 2.35e25-7.18e27 s; all 6 points BBN-allowed and not overclosing, confirming BBN negligible at low mixing where X-ray/SPI dominates; CSV saved to /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_164917/sterile_nu_grid_DW_5d6c37.csv	tc-261102_1-w2-d-13-87

Id	Confidence	Claim	Evidence (tool calls, figures)
f-068	high, **contested**	DW vs SF at m_s=20 keV sin^2 2theta=1e-14: DW Delta N_eff=1.42e-07 Omega_s h^2=3.04e-05 Delta chi2_BBN=0 BBN-allowed not overclosing; SF L=0.1 heuristic Delta N_eff=1.42e-08 Omega_s h^2=4.90 overcloses 40x; at 50 keV 1e-14 DW Omega_s h^2=1.58e-04 SF Omega_s h^2=30.6 overcloses 255x, proving server SF L cap 0.1 heuristic cannot test L>0.5 viable region	tc-261102_1-w1-d-14-51, tc-261102_1-w1-d-14-52
f-069	high	Artifact verification d-16: prior run 261021_110150 contains scan_DW_dc7743.csv/.npz, bbn_vs_neff.png, exclusion_scan_DW_dc7743.png, 6 sterile_nu_grid CSVs, references.bib, 6 paper PDF/TXT pairs (1702.08430, 1705.01837, 2405.17861, 2507.18752, 2510.24560, astro-ph_0501562); current run 261102_164917 contains 3 grid CSVs (DW_44fa64.csv, DW_5d6c37.csv, SF_d58c80.csv) and 4 figure directories (figures_1705.01837, figures_2405.17861, figures_astro-ph_0501562, figures_astro-ph_0602028); out/ directory empty (0 files).	tc-261102_1-w2-d-16-79, tc-261102_1-w2-d-16-80, tc-261102_1-w2-d-16-81
f-070	high, **contested**	DW BBN emulator at 40 and 50 keV with sin^2 2theta 1e-12/3e-14/1e-14 gives Delta N_eff 2.5e-05 to 2.5e-07 Delta chi2_BBN~0 Omega_s h^2 0.0001-0.016 tau 2.4e25-7.2e27 s all BBN-allowed not overclosing confirming BBN negligible at low mixing and X-ray/SPI dominates	tc-261102_1-w1-d-13-26
f-071	high	Abazajian 1705.01837 Fig.7 and Viel astro-ph/0501562 Figs.1-3 extracted to runs/261102_164917/figures_1705.01837/fig07_1.png and figures_astro-ph_0501562/fig01_1.png fig01_2.png fig02.png fig03.png anchoring X-ray 3.5 keV detections vs constraints and WDM transfer function m_th limits	tc-261102_1-w1-d-14-58, tc-261102_1-w1-d-14-60 Figs. 44, 45, 46, 47, 48

Id	Confidence	Claim	Evidence (tool calls, figures)
f-072	high, **contested**	BBN baseline provenance captured verbatim from server: Yp=0.2449±0.004 (Aver, Olive & Skillman JCAP 07 2021 029), D/H=2.547±0.025e-5 (Cooke, Pettini & Steidel ApJ 855 2018 102), Planck N_eff=2.99±0.17 (TT,TE,EE+lowE+lensing+BAO), CMB-S4 forecast N_eff=3.044±0.03; BBN fit validity N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s, m_s<~100 keV (relativistic at BBN), exclusion Delta chi2>3.84 (95%, 1 dof), PRyMordial-calibrated <0.3%.	tc-261102_1-w2-d-16-82
f-073	high	BBN server baselines: Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025×10^{-}? (Cooke+2018), N_eff=2.99±0.17 Planck 2018; BBN 95% limit ΔN_eff<0.548 ($\Delta\chi^2$>3.84 vs SM), with SM Yp=0.24682 and D/H=2.4433×10^{-}? at N_eff=3.044, slopes dYp/dN_eff=0.01351 and dD/H/dN_eff=0.3308×10^{-}?	tc-261102_1-w1-d-17-19, tc-261102_1-w1-d-17-36 Fig. 57
f-074	high	DW grid 1 eV–100 keV × sin^{2}2θ 1e-13–0.1 (169 points): 70 BBN-excluded ($\Delta\chi^2$>3.84) and 93 overclose (Ω_s h^2>0.12); 10k-point scan: 3907/10000 BBN-excluded, 5125/10000 overclosing; BBN max allowed sin^{2}2θ falls from 1.15e-3 at 1 eV to 9.3e-9 at 100 keV (e.g. 1.5e-7 at 3.4 keV, 3.8e-8 at 17.5 keV)	tc-261102_1-w1-d-17-33, tc-261102_1-w1-d-17-34 Fig. 62
f-075	high	BEST SBL best-fit (Δm^2=1.3 eV^2, sin^{2}2θ=0.42, m_s≈1.14 eV) predicts ΔN_eff=1.0 (N_eff=4.044), Yp=0.25956, D/H=2.7769×10^{-}?, $\Delta\chi^2$_BBN=23.10 (excluded 95%), Ω_s h^2=29.3 (overcloses), 6.2σ tension with Planck N_eff=2.99±0.17 (33σ for CMB-S4 σ=0.03)	tc-261102_1-w1-d-17-35
f-076	high	X-ray benchmarks (DW, f_DM=1 whole-halo): 7.1 keV sin^{2}2θ=7e-11 → E=3.55 keV Γ=1.72e-28 s^{-1} τ=5.82e27 s excluded by NuSTAR M31 + XMM blank-sky; 10 keV 1e-11 → E=5 keV Γ=1.36e-28 s^{-1} τ=7.35e27 s excluded by same; 20 keV 1e-12 → E=10 keV Γ=4.36e-28 s^{-1} τ=2.30e27 s excluded by NuSTAR M31; Γ=1.38e-29(sin^{2}2θ/1e-7)(m_s/1keV)? scaling confirmed	tc-261102_1-w1-d-17-44, tc-261102_1-w1-d-17-45, tc-261102_1-w1-d-17-46

Id	Confidence	Claim	Evidence (tool calls, figures)
f-077	high	Sterile-nu server BBN baselines are Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025 (Cooke+2018), Neff_CMB=2.99±0.17 (Planck 2018), with BBN fit validity Neff 2.044-6.044 and exclusion Delta chi2>3.84	tc-261102_1-w1-d-19-22
f-078	high	XRISM Resolve 3.75 Ms stacked 10-cluster search (arXiv:2510.19342, Oct 2025) covers E=2.5-15 keV rest frame including 3.5 keV line with no detection, providing strongest high-resolution decay-rate limits for m_s=5-30 keV	tc-261102_1-w1-d-19-53
f-079	high	Tool null (sin^2 $2\theta \to 0$) baseline: Yp=0.24682, D/H×10^5=2.4433, N_eff=3.0440, χ^2=2.77, $\Delta\chi^2$ vs SM =0; Yp within 0.5σ of observed 0.2449±0.004, D/H 4.1σ low vs 2.547±0.025 but combined BBN χ^2 not excluded; Planck N_eff=2.99±0.17 consistent within 0.3σ	tc-261102_1-w1-d-20-20, tc-261102_1-w1-d-20-28, tc-261102_1-w1-d-20-29, tc-261102_1-w1-d-20-31 Fig. 57
f-080	high	BBN 95% exclusion adopted by server is $\Delta\chi^2$ vs SM >3.84 (1 dof), corresponding to ΔN_eff <0.55 (BBN alone) from Yp/D/H fit; fully thermalized sterile (N_eff=4.044) gives Yp=0.2596 D/H=2.777, 3.7σ above Yp and $\Delta\chi^2$=23.1 excluded; Planck ΔN_eff<0.3-0.61 (2.99±0.17) is comparable, CMB-S4 forecast σ=0.03	tc-261102_1-w1-d-20-20, tc-261102_1-w1-d-20-31, tc-261102_1-w1-d-20-48 Fig. 57
f-081	medium	DW relic-density normalization in server is Ω_s h^2 ≈1.4×10? (m_s/1 keV)^1.8 $\sin^2 2\theta$, a factor ~12 below Abazajian 1705.01837 Eq.37 (Ω=1.74×10? m^1.8 $\sin^2 2\theta$); at m_s=7.1 keV $\sin^2 2\theta = 7\times10^{-11}$ server gives Ω=0.033 vs 0.42 expected from Abazajian, shifting Ω=0.12 contour to ~12× larger mixing	tc-261102_1-w1-d-20-30, tc-261102_1-w1-d-20-47
f-082	high	Server validity range is m_s <~100 keV (relativistic at BBN) with BBN fit calibrated for N_eff 2.044-6.044, ω_b h^2 0.019-0.0257, τ_n 866-890 s; for m_s≥~1 MeV decay injection not modeled and results flagged not constraints; eV-scale partial thermalization uses crude fit (ΔN_eff=0.40 at 1 eV^2 1e-3 vs full QKE 0.93)	tc-261102_1-w1-d-20-20, tc-261102_1-w1-d-20-28, tc-261102_1-w1-d-20-48
f-083	medium	Lyman-alpha forest constraints on thermal WDM mass m_th>5.7 keV (literature) with recent high-z HIRES/UVES analyses pushing to higher limits; Viel+2005 translation m_s=4.43 keV*(m_th/1keV)^{4/3} connects thermal relic to sterile neutrino mass	tc-261102_1-w1-d-19-39, tc-261102_1-w1-d-19-55

Id	Confidence	Claim	Evidence (tool calls, figures)
f-084	high	Sterile-nu server BBN baselines are Yp=0.2449±0.004 (Aver et al. JCAP 07 2021), D/H=2.547±0.025×10⁻? (Cooke et al. ApJ 855 2018), Neff_CMB=2.99±0.17 (Planck 2018 TT TE EE+lowE+lensing+BAO), with BBN fit validity Neff 2.044-6.044 and exclusion Delta chi2>3.84	tc-261102_1-w1-d-18-21
f-085	high	SBL BEST anomaly best-fit is Delta m2=1.3 eV2, sin2 2theta=0.42 (PRL 128 232501 2022) and 3.5 keV line corresponds to m_s=7100 eV, sin2 2theta best 7e-11 range 2e-11 to 2e-10, status debated	tc-261102_1-w1-d-18-21
f-086	medium	Inflaton-decay production (arXiv:2601.02069) claims cold sterile neutrino DM efficiently produced during reheating with BR ≤1e-4, mapping viable regions in m_s, mixing, inflaton mass, T_RH and BR, with future X-ray probing T_RH	tc-261102_1-w1-d-18-41
f-087	medium	Shi-Fuller resonant production revisited (arXiv:2512.00793) claims lepton asymmetries L_alpha ≥1 at T>20 MeV are consistent with BBN, reopening SF parameter space	tc-261102_1-w1-d-18-49
f-088	medium	NuSTAR 11-year stray-light aperture search (arXiv:2405.17861) sets strong limits on monochromatic photon emission from radiative sterile neutrino decay in Milky Way for m_s ~ few keV	tc-261102_1-w1-d-18-58
f-089	medium	nuSMEFT freeze-in (arXiv:2409.08968) proposes sterile neutrino DM production via higher-dimensional operators / nonstandard interactions independent of active-sterile mixing, without fine-tuned resonance or lepton asymmetry	tc-261102_1-w1-d-18-32
f-090	low	Time-dependent mass mechanism (arXiv:2502.13123) proposes circumventing X-ray constraints via ultralight scalar-sourced mass, reviving large-mixing region while keeping terrestrial SBL hints	tc-261102_1-w1-d-18-38, tc-261102_1-w1-d-18-41
f-091	high	High-mass DW benchmarks 40 keV and 50 keV at $\sin^2 2\theta$=1e-14 to 1e-10: ΔN_eff 2.5e-7 to 2.9e-3, Ω_s h^2 1.06e-4 to 1.58 at 40-50 keV; BBN-allowed ($\Delta\chi^2$<3.84) for all 10 points, overclosure only at $\sin^2 2\theta \geq$1e-10 (40 keV) and ≥1e-11 (50 keV); lifetimes τ=7.2e27 s (40 keV,1e-14) to 2.35e23 s (50 keV,1e-10) via Γ=1.38e-29($\sin^2 2\theta$/1e-7)(m_s/1keV)?	tc-261102_1-w1-d-17-67

Id	Confidence	Claim	Evidence (tool calls, figures)
f-092	high, **contested**	Sterile-nu server BBN baselines Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), Neff_CMB=2.99±0.17 Planck 2018, validity Neff 2.044-6.044, exclusion Delta chi2>3.84, with existing DW grid/scan files in runs/261102_172137	tc-261102_1-w2-d-18-90, tc-261102_1-w2-d-18-89
f-093	high, **contested**	Tool BBN baselines from describe_sterile_nu_tools: Yp=0.2449±0.004 (Aver+2021 JCAP07 029), D/H=2.547±0.025×10⁻? (Cooke+2018 ApJ855 102), N_eff CMB=2.99±0.17 Planck 2018 (TT TE EE+lowE+lensing+BAO), CMB-S4 forecast 3.044±0.03; BBN fit validity N_eff 2.044–6.044, ω_b h^2 0.019–0.0257, τ_n 866.4–890.4 s; exclusion $\Delta\chi^2$ vs SM >3.84 (95%, 1 dof); valid for m_s <~100 keV relativistic at BBN, MeV+ decay not modeled	tc-261102_1-w2-d-20-84
f-094	high, **contested**	Tool null (sin$^2 2\theta \to 0$) predicts Yp=0.24682, D/H×10?=2.4433, N_eff=3.0440, χ^2=2.77, $\Delta\chi^2$=0; vs observed Yp=0.2449±0.004 → +0.48σ high, D/H=2.547±0.025 → -4.15σ low, N_eff=2.99±0.17 → +0.32σ high; combined BBN χ^2=2.77 allowed at 95% ($\Delta\chi^2$<3.84); vs brief Yp=0.245±0.003 D/H=2.53±0.03 gives +0.61σ and -2.89σ respectively, still allowed	tc-261102_1-w2-d-20-84, tc-261102_1-w2-d-20-92, tc-261102_1-w2-d-20-93
f-095	high, **contested**	BBN 95% threshold adopted by server is $\Delta\chi^2$ vs SM >3.84 (1 dof) corresponding to ΔN_eff <0.55 BBN-alone (Yp/D/H fit); fully thermalized sterile N_eff=4.044 (ΔN_eff=1) gives Yp=0.2596 D/H=2.777 $\Delta\chi^2$=23.1 excluded; Planck N_eff=2.99±0.17 gives ΔN_eff<0.33 (1σ) <0.61 (95% approx 2σ) comparable; CMB-S4 σ=0.03 will give ΔN_eff<0.06	tc-261102_1-w2-d-20-84, tc-261102_1-w2-d-20-92
f-096	medium, **contested**	DW relic-density normalization in server is Ω_s h^2≈1.4×10? (m_s/1keV)^1.8 sin$^2 2\theta$, factor ~12 below Abazajian 1705.01837 Eq.37 (1.74×10?); at m_s=7.1 keV sin$^2 2\theta$=7×10^{-11} server gives Ω=0.033 vs 0.42 expected from Abazajian, shifting Ω=0.12 contour to ~12× larger mixing; grid confirms scaling: 1 keV 1e-10 →0.001384, 10 keV 1e-10 →0.08733, 50 keV 1e-11 →0.1582 consistent with m^1.8	tc-261102_1-w2-d-20-94, tc-261102_1-w2-d-20-95

Id	Confidence	Claim	Evidence (tool calls, figures)
f-097	high, **contested**	keV DW benchmarks BBN-allowed but overclose/X-ray tension: 7.1 keV 7e-11 ΔN_eff=0.00043 Ω=0.033 BBN allowed ($\Delta\chi^2$=-0.01) E=3.55 keV Γ=1.72e-28 s^{-1} τ=5.82e27 s; grid shows BBN exclusion only at ΔN_eff?0.6 (e.g. 7.1 keV 1e-7 ΔN_eff=0.618 $\Delta\chi^2$=5.89, 10 keV 1e-7 ΔN_eff=0.813 $\Delta\chi^2$=13.4, 20 keV 1e-7 ΔN_eff=1.0 $\Delta\chi^2$=23.1); keV DM BBN-safe, eV thermalized excluded	tc-261102_1-w2-d-20-94, tc-261102_1-w2-d-20-95
f-098	high, **contested**	BBN server baselines Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), N_eff=2.99±0.17 Planck 2018; BBN 95% limit ΔN_eff<0.548 ($\Delta\chi^2$>3.84 vs SM) with SM Yp=0.24682, D/H=2.4433e-5 at N_eff=3.044, slopes dYp/dN_eff=0.01351, dD/H/dN_eff=0.3308e-5	tc-261102_1-w2-d-17-91, tc-261102_1-w2-d-17-106 Fig. 57
f-099	high, **contested**	DW grid 1 eV–100 keV × $\sin^2 2\theta$ 1e-13–0.1 (143 pts): 65 BBN-excluded ($\Delta\chi^2$>3.84) and 89 overclose (Ω_s h^2>0.12); 10k scan: 3907/10000 BBN-excluded, 5125/10000 overclosing; max BBN-allowed $\sin^2 2\theta$ 1.15e-3 at 1 eV → 9.33e-9 at 100 keV (1.52e-7 at 3.4 keV, 3.76e-8 at 17.5 keV)	tc-261102_1-w2-d-17-103, tc-261102_1-w2-d-17-104, tc-261102_1-w2-d-17-112 Fig. 62
f-100	high	BEST SBL best-fit Δm^2=1.3 eV^2 $\sin^2 2\theta$=0.42 (m_s≈1.14 eV) predicts ΔN_eff=1.0 N_eff=4.044 Yp=0.25956 D/H=2.7769e-5 $\Delta\chi^2$_BBN=23.10 excluded 95% Ω_s h^2=29.3 overcloses 6.2σ Planck tension (33σ CMB-S4)	tc-261102_1-w2-d-17-105
f-101	high	X-ray benchmarks (f_DM=1 whole-halo Γ=1.38e-29($\sin^2 2\theta$/1e-7)(m_s/1keV)?): 7.1 keV 7e-11 E=3.55 keV Γ=1.72e-28 s^{-1} τ=5.82e27 s excluded NuSTAR M31+XMM blank-sky; 10 keV 1e-11 E=5 keV Γ=1.36e-28 τ=7.35e27 s excluded same; 20 keV 1e-12 E=10 keV Γ=4.36e-28 τ=2.30e27 s excluded NuSTAR M31	tc-261102_1-w2-d-17-113, tc-261102_1-w2-d-17-114, tc-261102_1-w2-d-17-115
f-102	high, **contested**	Combined BBN+X-ray: keV DW DM at X-ray-accessible mixings excluded by NuSTAR/XMM (Γ~1e-28 s^{-1} limits); BBN alone allows keV DM (ΔN_eff<<0.548 for $\sin^2 2\theta$≤1e-10 at 10-50 keV) but eV SBL thermalized sterile excluded; viable DM requires non-minimal production (SF L~0.1-1, inflaton BR~1e-4, freeze-in) at $\sin^2 2\theta$ orders below DW Ω=0.12 line	tc-261102_1-w2-d-17-103, tc-261102_1-w2-d-17-104, tc-261102_1-w2-d-17-105, tc-261102_1-w2-d-17-113 Fig. 57, 62

Id	Confidence	Claim	Evidence (tool calls, figures)
f-103	high	SF resonant points at L=0.1 cap (m_s=10,20,30 keV, sin^2 2θ=1e-14) give ΔN_eff=8.1e-09,1.42e-08,1.96e-08 (N_eff=3.0440), BBN $\Delta\chi^2\approx$0.00 allowed at 95%, Yp=0.24682 D/H=2.4433e-5, but tool Ω_s h^2=1.22,4.90,11.0 overcloses; DW same point gives ΔN_eff=8.1e-08 Ω=8.7e-06 BBN-allowed; tool SF ΔN_eff is heuristic 0.1×DW, L capped at 0.1, no ξ_e in Yp	tc-261102_1-w2-d-19-97, tc-261102_1-w2-d-19-98, tc-261102_1-w2-d-19-99, tc-261102_1-w2-d-19-100
f-104	high	XRISM Resolve 3.75 Ms stacked 10 galaxy clusters search covers E=2.5-15 keV rest frame including 3.5 keV line with no detection, providing strongest high-resolution decay-rate limits for m_s=5-30 keV (arXiv:2510.19342 Oct 2025)	tc-261102_1-w2-d-18-107
f-105	medium	Inflaton-decay production (arXiv:2601.02069 Jan 2026) produces cold sterile neutrino DM during reheating with BR~1e-4, mapping viable regions in m_s, mixing, inflaton mass, T_RH and BR with future X-ray probing T_RH	tc-261102_1-w2-d-18-102
f-106	high, **contested**	DW grid 7.1-50 keV × sin$^2 2\theta$ 1e-11 to 1e-10 (15 pts, DW): 0 BBN-excluded ($\Delta\chi^2$<0, ΔN_eff 6.2e-05 to 2.9e-03, N_eff 3.044), 7 overclose (Ω_s h^2>0.12); e.g. 7.1 keV 7e-11 Ω=0.033 τ=5.82e27 s, 10 keV 1e-11 Ω=0.0087 τ=7.35e27 s, 20 keV 1e-11 Ω=0.030 τ=2.30e26 s, 50 keV 1e-11 Ω=0.158 overcloses; BBN alone does not exclude keV DM at X-ray mixings	tc-261102_1-w2-d-19-109
f-107	high	X-ray line at m_s=10 keV sin$^2 2\theta$=1e-14 (E=5 keV Γ=1.36e-31 s^{-1} τ=7.35e30 s) and 20 keV 1e-14 (E=10 keV Γ=4.36e-30 τ=2.30e29 s) not excluded by tabulated NuSTAR/XMM limits (whole-halo f_DM=1); fluxes M31 2.7e-07 and 4.3e-06 ph/s/cm^2, Perseus 1.4e-08 and 2.2e-07; Γ=1.38e-29(sin$^2 2\theta$/1e-7)(m_s/1keV)? confirmed	tc-261102_1-w2-d-19-110, tc-261102_1-w2-d-19-111

Id	Confidence	Claim	Evidence (tool calls, figures)
f-108	medium, **contested**	DW normalization discrepancy: server Ω_s h^2≈1.4×10?(m_s/1keV)^1.8 $\sin^2 2\theta$ vs Abazajian 1705.01837 Eq.37 Ω=1.74×10?(m_s/1keV)^1.8 $\sin^2 2\theta$ (factor 12.4×); at 7.1 keV $\sin^2 2\theta$=7e-11 server Ω=0.033 vs 0.42 expected (factor 12.7×), shifting Ω=0.12 contour to 12× larger mixing; impact: server predicts 7.1 keV XRISM/NuSTAR exclusion at $\sin^2 2\theta$~7e-11 is BBN-allowed but underproduces DM (Ω=0.033), while published normalization would make same point overclose (Ω=0.42) and move viable DW line into X-ray excluded region	tc-261102_1-w2-d-19-109
f-109	medium, **contested**	Structure-formation translation m_s=4.43 keV×(m_th/1keV)^{4/3} (Viel+2005) gives m_th>8.4 keV (galaxy counts) → m_s>76 keV, m_th>8.5 keV (counts+lensing) → m_s>78 keV, m_th=4.1 keV preference → m_s>29 keV; combined with X-ray ceilings $\sin^2 2\theta$<1e-11 to 1e-13 (m_s=2-20 keV, XRISM 3.75 Ms + NuSTAR) and DW Ω=0.12 line at $\sin^2 2\theta$~1e-10 to 1e-11 (server) or ~1e-11 to 1e-12 (Abazajian), surviving DW mass interval empty: m_s>76 keV required by m_th>8.4 keV lies where X-ray limits $\sin^2 2\theta$<3e-14 (SPI) and DW Ω=0.12 requires $\sin^2 2\theta$~5e-12 (server) already excluded	tc-261102_1-w2-d-19-109, tc-261102_1-w2-d-19-110, tc-261102_1-w2-d-19-111
f-110	medium, **contested**	Per-mechanism viability table (m_s, $\sin^2 2\theta$, binding constraint, provenance): DW (m_s=7.1 keV $\sin^2 2\theta$=7e-11 Ω=0.033 BBN-allowed ΔN_eff=4.3e-04 but X-ray excluded NuSTAR M31+XMM, provenance tool CSV 5d2e30; DW all-DM line excluded by X-ray+m_th>8.4 keV→m_s>76 keV empty interval); SF L=0.1 cap (10-30 keV 1e-14 BBN-allowed ΔN_eff~1e-08 $\Delta\chi^2$=0 but tool Ω=1.2-11 overcloses, heuristic 0.1×DW no ξ_e, provenance tc-97-99; literature L>0.5-1 at T>20 MeV washes out before BBN per 2512.00793, viable at m_s>10 keV $\sin^2 2\theta$<1e-14 below XRISM Γ~1e-27); inflaton-decay BR~1e-4 (m_s 10-50 keV $\sin^2 2\theta$→0 BBN ΔN_eff<3e	tc-261102_1-w2-d-19-97, tc-261102_1-w2-d-19-98, tc-261102_1-w2-d-19-99, tc-261102_1-w2-d-19-100, tc-261102_ 1-w2-d-19-109...

Id	Confidence	Claim	Evidence (tool calls, figures)
f-111	high	Existing DW grid/scan files confirmed in runs/261102_172137: scan_DW_9d1baf.csv/.npz, sterile_nu_grid_DW_*.csv (7 files), bbn_vs_neff.png/pdf (Fig.57), exclusion_scan_DW_9d1baf.png/pdf (Fig.62); no new files created in wave 3	tc-261102_1-w3-d-18-145, tc-261102_1-w3-d-18-146 Fig. 57, 62
f-112	high, **contested**	40 keV and 50 keV DW benchmarks BBN-allowed (ΔN_eff 2.5e-7 to 2.9e-3, $\Delta\chi^2$<3.84) with Γ=1.38e-29(sin$^2 2\theta$/1e-7)(m_s/1keV)?: 40 keV sin$^2 2\theta$=1e-12 E=20 keV Γ=1.39e-26 s^{-1} τ=7.18e25 s, 50 keV sin$^2 2\theta$=1e-12 E=25 keV Γ=4.31e-26 s^{-1} τ=2.32e25 s, 50 keV sin$^2 2\theta$=1e-14 E=25 keV Γ=4.31e-28 s^{-1} τ=2.32e27 s; all not excluded by tabulated NuSTAR M31/XMM blank-sky 6-40 keV limits (whole-halo f_DM=1), while 7.1 keV 7e-11 (Γ=1.72e-28 τ=5.82e27 s) and 10 keV 1e-11 (Γ=1.36e-28 τ=7.35e27 s) and 20 keV 1e-12 (Γ=4.36e-28 τ=2.30e27 s) are excluded	tc-261102_1-w1-d-17-67, tc-261102_1-w1-d-17-44, tc-261102_1-w1-d-17-45, tc-261102_1-w1-d-17-46 Fig. 57, 62
f-113	medium, **contested**	XRISM Resolve 3.75 Ms stacked 10-cluster limit at 7.1 keV (E=3.55 keV) corresponds to Γ~1e-27 s^{-1} $\rightarrow$ sin$^2 2\theta$<4e-11 (via Γ=1.38e-29(sin^2/1e-7)(m/1keV)?, Γ=9.9e-28 at 7.1 keV 4e-11), which lies below server DW Ω_s h^2=0.12 contour (server $\Omega\approx$1.4e7 m^1.8 sin^2 $\rightarrow$ sin^2=4.8e-10 at 7.1 keV for Ω=0.12) but above Abazajian-corrected contour (1.74e8 m^1.8 sin^2 $\rightarrow$ sin^2=3.9e-11 at 7.1 keV), so XRISM excludes DW DM at 7.1 keV under literature normalization but not under server normalization; at 10 keV XRISM-equivalent sin^2~1e-11 vs server Ω=0.12 sin^2=1.9e-10 vs Abazajian 1.5e-11	tc-261102_1-w1-d-19-53, tc-261102_1-w2-d-20-94, tc-261102_1-w2-d-20-95, tc-261102_1-w1-d-17-44 Fig. 62

Id	Confidence	Claim	Evidence (tool calls, figures)
f-114	high, **contested**	Combined BBN (ΔN_eff<0.548, $\Delta\chi^2$>3.84) + X-ray viable window: BBN allows all keV DW points with $\sin^2 2\theta \leq$1e-10 (ΔN_eff 6e-5 to 2.9e-3 at 7-50 keV), but NuSTAR M31/XMM blank-sky excludes m_s=7-20 keV at $\sin^2 2\theta \geq$1e-12 to 7e-11 (Γ~1-4e-28 s^{-1}), leaving BBN+X-ray allowed only at $\sin^2 2\theta \leq$1e-13 to 1e-14 for m_s=5-20 keV ($\Gamma \leq$1e-30 to 1e-31 s^{-1}, $\tau \geq$1e29 s) requiring non-minimal production (SF L~0.1-1, inflaton BR~1e-4, nuSMEFT freeze-in) to reach Ω_s h^2=0.12; 40-50 keV at $\sin^2 2\theta$=1e-12 to 1e-14 BBN-allowed and X-ray not excluded by tabulated 6-40 keV limits but overcloses at $\sin^2 \geq$1e-10 (40 keV) / $\geq$1e-11 (50 k	tc-261102_1-w1-d-17-67, tc-261102_1-w1-d-17-44, tc-261102_1-w1-d-17-45, tc-261102_1-w1-d-17-46, tc-261102_1-w2-d-20-94 Fig. 57, 62
f-115	high	DW Omega_s h2=0.12 line at m_s=6,7.1,10,20 keV requires sin^2 2theta ~1e-10 to 4e-11 and is excluded by built-in NuSTAR M31 (Roach+2020) and XMM blank-sky limits, while at 40 keV (sin^2 2theta=1.1e-11, E=20 keV, tau=6.35e24 s) it is not excluded by tabulated limits but overcloses (Omega_s h2=0.106 at 1e-11, 1.06 at 1e-10)	tc-261102_1-w1-d-23-33, tc-261102_1-w1-d-23-47, tc-261102_1-w1-d-23-48, tc-261102_1-w1-d-23-49, tc-261102_1-w1-d-23-50, ...
f-116	high	For DW production at small mixing sin^2 2theta=1e-12 to 1e-14, Omega_s h2 <<0.12 (e.g., 3.5e-4 at 6 keV/1e-12 down to 1e-4 at 40 keV/1e-12, and 3.5e-6 at 6 keV/1e-14), so DW cannot make DM at X-ray-allowed mixing; SF resonant with L~0.1-1 or inflaton-decay BR~1e-4 required to reach Omega=0.12	tc-261102_1-w1-d-23-33
f-117	high	DW vs SF BBN contrast at eV-keV: same (m_s, sin^2 2theta) points that are BBN-excluded under DW become allowed under SF heuristic (0.1x DW Delta N_eff); e.g. m_s=7100 eV, sin^2 2theta=1e-7 gives Delta N_eff=0.6185, Delta chi2=5.89 excluded (DW) vs Delta N_eff=0.06185, Delta chi2=-0.66 not excluded (SF L=0.01/0.1); m_s=100 eV, sin^2 2theta=1e-5 gives Delta N_eff=1.0 excluded (DW) vs 0.1 not excluded (SF); 14/24 DW points excluded vs 0/24 SF points excluded at L=0.01 and L=0.1.	tc-261102_1-w1-d-22-36, tc-261102_1-w1-d-22-37, tc-261102_1-w1-d-22-38

Id	Confidence	Claim	Evidence (tool calls, figures)
f-118	high	BBN Delta N_eff 95% limit is Delta N_eff <0.55 (Delta chi2 vs SM >3.84) using server baselines Yp=0.2449±0.004 and D/H=2.547±0.025e-5 (Aver et al. 2021, Cooke et al. 2018) with slopes dYp/dN_eff=0.0135 and dD/H/dN_eff=0.33e-5; a fully thermalized sterile (N_eff=4.044) predicts Yp=0.2596 and D/H=2.777e-5, 3.7 sigma above Yp; server SF Delta N_eff is heuristic 0.1x DW capped at L<=0.1 and cannot test L>0.5 where analytic xi_e shift dYp/dxi_e ~ -0.25 can compensate Delta N_eff ~0.5-1.0.	tc-261102_1-w1-d-22-22, tc-261102_1-w1-d-22-43 Fig. 73
f-119	high	Server validity and L cap: BBN treatment valid for m_s<100 keV (relativistic at BBN), transitional 100 keV-1 MeV, invalid >=1 MeV where decay injection not modeled; eV partial thermalization is crude (server Delta N_eff=0.40 vs full QKE 0.93 at 1 eV^2, 1e-3); SF Delta N_eff is heuristic 0.1x DW not resonant calculation; L capped at 0.1 so cannot test recent BBN-allowed L>1 at T>20 MeV that reopens SF parameter space.	tc-261102_1-w1-d-22-22, tc-261102_1-w1-d-22-28
f-120	high	At small mixing sin^2 2theta=1e-12, X-ray line at m_s=7.1 keV (E=3.55 keV, tau=4.07e29 s, M31 flux 6.9e-06 ph/s/cm2) is not excluded by tabulated limits, while at m_s=20 keV (E=10 keV, tau=2.30e27 s, M31 flux 4.3e-04 ph/s/cm2) remains excluded by NuSTAR M31 (Roach+2020) even at 1e-12, and at m_s=40 keV sin^2 2theta=1e-14 (E=20 keV, tau=7.18e27 s) not excluded; X-ray ceiling is mass-dependent and drops below 1e-12 only above ~10 keV	tc-261102_1-w1-d-23-62, tc-261102_1-w1-d-23-63, tc-261102_1-w1-d-23-64
f-121	medium	Web snippets confirm alternative production reopens small-mixing space: inflaton-decay with BR <=1e-4 produces cold sterile DM at vanishing mixing evading X-ray bounds and probing T_RH (arXiv:2601.00350), and NuSTAR Galactic bulge 190 ks at ~10 deg off-plane (arXiv:2006.03102) provides complementary limits to M31 1.2 Ms focused+unfocused FOV (arXiv:1901.01293)	tc-261102_1-w1-d-23-39, tc-261102_1-w1-d-23-40

Id	Confidence	Claim	Evidence (tool calls, figures)
f-122	medium	Structure-formation bounds 2024-2026: thermal-relic m_th >6.5 keV (MW satellites DES+PS1, Nadler et al 2024), >6.1 keV (strong lensing, 2024), and Lyman-alpha m_th ~5-5.7 keV; via Viel+2005 DW mapping m_s=4.43 keV*(m_th/keV)^{4/3} this requires DW sterile mass m_s >53.7 keV (satellites), >49 keV (lensing), >29-41 keV (Lyman-alpha 4.1-5.3 keV), consistent with d-06 DW exclusion >28-48 keV (Lyα) and >76 keV (satellites 8.4-8.5 keV); SF resonant and inflaton-decay production are colder (lower <p>/T) so same m_s corresponds to larger effective m_th and bounds shift down by factor ~1.5-3, reopening	tc-261102_1-w1-d-24-20, tc-261102_1-w1-d-24-32, tc-261102_1-w1-d-24-42, tc-261102_1-w1-d-24-53, tc-261102_1-w1-d-24-59
f-123	high	Sterile-nu server BBN baselines are Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), N_eff_CMB=2.99±0.17 (Planck 2018), with BBN fit valid for N_eff 2.044-6.044, omega_b h2 0.019-0.0257, tau_n 866.4-890.4 s; exclusion is Delta chi2 vs SM >3.84 (95%, 1 dof) and flagged invalid for m_s >~100 keV where decay injection not modeled.	tc-261102_1-w1-d-24-20
f-124	high	DW grid at 5-70 keV, sin^2 2theta 1e-12 to 1e-10 (21 points, file sterile_nu_grid_DW_9e26a8.csv) shows Delta N_eff 4.7e-06 to 0.0039, Delta chi2_BBN -0.05 to 0.0, 0/21 BBN-excluded at 95%, 7/21 overclose (Omega_s h2>0.12); e.g. m_s=10 keV, sin^2 2theta=1e-10 gives Omega_s h2=0.087, Delta N_eff=8.1e-04, lifetime 7.3e26 s; m_s=50 keV, sin^2 2theta=1e-11 gives Omega_s h2=0.158 (overcloses), Delta N_eff=2.9e-04, lifetime 2.35e24 s.	tc-261102_1-w1-d-24-77

Id	Confidence	Claim	Evidence (tool calls, figures)
f-125	medium	Viel+2005 DW mapping m_s=4.43 keV*(m_th/keV)^{4/3} is DW-specific; SF resonant and inflaton-decay production have colder momentum distributions (<p>/T lower by ~1.5-3x), so same m_s corresponds to larger effective m_th; thus DW m_s>53.7 keV for m_th>6.5 keV (satellites) and >29-49 keV for Lyman-alpha m_th 4.1-6.1 keV does not directly apply to SF/inflaton-decay, which can satisfy structure formation at m_s~10-20 keV with small mixing sin^2 2theta<1e-12 where X-ray limits also weaken.	tc-261102_1-w1-d-24-20, tc-261102_1-w1-d-24-27, tc-261102_1-w1-d-24-77
f-126	high	Board verification: list_agent_files returned empty (out/ directory has no files), confirming prior weakest claims unverified; project structure contains runs/261102_173925 with 5 grid CSVs + bbn_vs_neff.png, research/default, reports/261021_111534 and 261102_170443, data/downloads, but no SCOPE.md/ISSUES.md/board files at top level or runs directory	tc-261102_1-w2-d-25-82, tc-261102_1-w2-d-25-83, tc-261102_1-w2-d-25-84, tc-261102_1-w2-d-25-93, tc-261102_1-w2-d-25-94, ...
f-127	high, **contested**	DW vs SF BBN contrast at eV-keV: 14/24 DW points BBN-excluded (Delta chi2>3.84, Delta N_eff up to 1.0, Delta chi2=23.1) vs 0/24 SF heuristic (0.1x DW, L=0.01 and L=0.1) excluded; e.g. m_s=7100 eV sin^2 2theta=1e-7 DW Delta N_eff=0.6185 Delta chi2=5.89 excluded vs SF Delta N_eff=0.06185 Delta chi2=-0.66 not excluded; m_s=100 eV sin^2 2theta=1e-5 DW Delta N_eff=1.0 excluded vs SF 0.1 not excluded	tc-261102_1-w1-d-22-36, tc-261102_1-w1-d-22-37, tc-261102_1-w1-d-22-38, tc-261102_1-w2-d-22-88 Fig. 73
f-128	high, **contested**	Sterile-nu server baselines for joint BBN+X-ray overlay: Yp=0.2449±0.004 (Aver+2021), D/H=2.547±0.025e-5 (Cooke+2018), N_eff_CMB=2.99±0.17 (Planck 2018), BBN fit valid N_eff 2.044-6.044, exclusion Delta chi2>3.84 (95%), valid m_s<100 keV, 10 X-ray limits tabulated (M31, Perseus, MW center, Coma, stacked galaxies)	tc-261102_1-w2-d-25-85

Id	Confidence	Claim	Evidence (tool calls, figures)
f-129	high, **contested**	DW Omega_s h2=0.12 line at m_s=6,7.1,10,20 keV requires sin^2 2theta ~3.4e-10,2.5e-10,1.4e-10,3.9e-11 and is excluded by built-in NuSTAR M31 (Roach+2020) and XMM blank-sky limits; at 40 keV (sin^2 2theta=1.1e-11, E=20 keV, tau=6.35e24 s) not excluded by tabulated limits but overcloses (Omega_s h2=0.106 at 1e-11, 1.06 at 1e-10); X-ray line grid CSV with E=m_s/2, Gamma, flux vs limits exists at /Users/nramachandra/hep-genesis-testing/projects/ramp-research-sterile-bbn-xray/runs/261102_173925/sterile_nu_grid_DW_e4bea5.csv (35 points, 6-40 keV x 1e-8 to 1e-14) and sterile_nu_grid_DW_9e26a8.csv (21	tc-261102_1-w1-d-23-33, tc-261102_1-w1-d-23-47, tc-261102_1-w1-d-23-48, tc-261102_1-w1-d-23-49, tc-261102_1-w1-d-23-50, ...
f-130	high, **contested**	BBN Delta N_eff 95% limit is Delta N_eff<0.55 (Delta chi2>3.84) using server baselines Yp=0.2449±0.004 (Aver et al. 2021) and D/H=2.547±0.025e-5 (Cooke et al. 2018) with slopes dYp/dN_eff=0.0135 and dD/H/dN_eff=0.33e-5; fully thermalized sterile (N_eff=4.044) predicts Yp=0.2596 (3.7 sigma high) and D/H=2.777e-5; server SF Delta N_eff is heuristic 0.1x DW capped at L<=0.1 and cannot test L>1 at T>20 MeV where analytic xi_e shift dYp/dxi_e~-0.25 can compensate Delta N_eff~0.5-1.0	tc-261102_1-w1-d-22-22, tc-261102_1-w1-d-22-43, tc-261102_1-w2-d-22-88 Fig. 73
f-131	high, **contested**	Server validity and L cap: BBN treatment valid for m_s<100 keV (relativistic at BBN), transitional 100 keV-1 MeV, invalid >=1 MeV where decay injection not modeled; eV partial thermalization crude (server Delta N_eff=0.40 vs full QKE 0.93 at 1 eV^2, 1e-3); SF Delta N_eff is heuristic 0.1x DW not resonant calculation; L capped at 0.1 so cannot test recent BBN-allowed L>1 at T>20 MeV that reopens SF parameter space per arXiv:2411.07040 and arXiv:2005.03027 Fig.12	tc-261102_1-w1-d-22-22, tc-261102_1-w1-d-22-28, tc-261102_1-w2-d-22-88

Id	Confidence	Claim	Evidence (tool calls, figures)
f-132	high, **contested**	At small mixing sin^2 2theta=1e-12 to 1e-14 DW underproduces DM (Omega_s h2=3.5e-4 at 6 keV/1e-12 to 1.06e-4 at 40 keV/1e-12, 3.5e-6 at 6 keV/1e-14) so DW cannot make DM at X-ray-allowed mixing; SF resonant L~0.1-1 or inflaton-decay BR~1e-4 required to reach Omega=0.12, and X-ray at small mixing is mass-dependent: 7.1 keV/1e-12 (E=3.55 keV, tau=4.07e29 s, M31 flux 6.9e-06) not excluded, 20 keV/1e-12 (E=10 keV, tau=2.30e27 s, flux 4.3e-04) still excluded by NuSTAR M31, 40 keV/1e-14 (E=20 keV, tau=7.18e27 s, flux 6.9e-05) not excluded	tc-261102_1-w1-d-23-33, tc-261102_1-w1-d-23-62, tc-261102_1-w1-d-23-63, tc-261102_1-w1-d-23-64
f-133	high, **contested**	DW vs SF BBN contrast: 14/24 DW points BBN-excluded (Delta chi2>3.84, Delta N_eff up to 1.0, Delta chi2=23.1) vs 0/24 SF points excluded at L=0.01 and L=0.1 (heuristic 0.1x DW); e.g. m_s=7100 eV, sin^2 2theta=1e-7: DW Delta N_eff=0.6185 Delta chi2=5.89 excluded vs SF 0.06185 Delta chi2=-0.66 not excluded; BBN 95% limit Delta N_eff<0.55 from Yp=0.2449±0.004, D/H=2.547±0.025e-5, slopes dYp/dN_eff=0.0135, dD/H/dN_eff=0.33e-5; server SF capped at L<=0.1 cannot test L>1 at T>20 MeV	tc-261102_1-w1-d-22-36, tc-261102_1-w1-d-22-37, tc-261102_1-w1-d-22-38, tc-261102_1-w1-d-22-22, tc-261102_1-w1-d-22-43 Fig. 73

B.3 Dead ends

- x-01 (d-02): Searches with search_arxiv_abstracts and search_arxiv for 2024-2026 eROSITA and INTEGRAL/SPI sterile-neutrino line limits ('eROSITA sterile neutrino dark matter decay line limits', 'eROSITA eFEDS dark matter decay line sterile neutrino', 'INTEGRAL SPI sterile neutrino dark matter decay line'). — Neither search returned a 2024-2026 eROSITA or SPI line-search paper; both returned only older or unrelated papers (1505.07829, 1702.08430). Better targets are specific IDs found by web search or author queries (e.g. Fischer et al. eFEDS; Calore et al. INTEGRAL/SPI).

- x-02 (d-05): Targeted search_arxiv_abstracts for a 2024-2026 INTEGRAL/SPI sterile-neutrino line limit at m_s >= 40 keV — Returned only unrelated papers (muon capture, oscillation studies, a 2008 MW heating paper), so no primary 2024+ SPI source was found. The SPI entry relies on the 2507.18752 quote (sin^2 2θ <~ 3e-14, lifetime > ~1e20 s, m_s >~ 40 keV).

- x-03 (d-09): Full per-mechanism viability table with verbatim quotes, Viel+2005 m_th->m_s translation at 7.1/10/20/50 keV, and BBN tool cross-checks for d-09 — Tool budget exhausted after 10 web_search/describe calls; no read_arxiv_chunk/cite/get_paper_figures/predict_steril calls executed, so no verbatim quotes, no bibliography with URLs, no structure-formation numbers, and no quantitative BBN/X-ray cross-checks could be produced. Only web-search snippets available.

- `x-04` (`d-08`): 2024-2026 literature X-ray upper-limit table (XRISM Perseus, NuSTAR stray-light 2405.17861, INTEGRAL/SPI, Chandra/XMM archival sin^2 2theta limits) with arXiv IDs, figure numbers and verbatim quotes — web_search DNS resolution failed for both XRISM and NuSTAR queries (DNSError no records for wt.wikipedia.org.cels.anl.gov), and tool budget exhausted before get_paper_figures/cite could be attempted; no external limits beyond server built-in Roach+2020/Dessert+2020 could be retrieved

- `x-05` (`d-11`): Quantified normalization offset factor vs Abazajian 1705.01837 Eq.37 with verbatim quote and cite — Tool budget exhausted before read_arxiv_chunk/cite could be executed; Abazajian Eq.37 text not retrieved, so no verbatim quote or offset factor computed. Server values at 7.1 keV 1e-10 (Omega=0.04715) and 1 keV 1e-6 (Omega=13.84) are established but analytic comparison pending.

- `x-06` (`d-11`): Exclusion-plane plot distinguishing tool-internal vs literature X-ray limits and Omega=0.12 contour overlay — Tool budget reached before plot_sterile_nu_exclusion could be called; scan NPZ exists but no plot generated. Literature X-ray limits beyond built-in Roach+2020/Dessert+2020 not retrieved due to earlier web_search DNS failures.

- `x-07` (`d-14`): Fetch and cite verbatim Abazajian 1705.01837 Eq.37 (m_s ~3.40 keV*(sin^2 2theta/1e-8)^-0.615*(Omega/0.26)^0.5*0.527 erfc[...]) and Viel+2005 m_s=4.43 keV*(m_th/1keV)^{4/3} coefficient with cite() validated quotes — Tool budget exhausted before read_arxiv_chunk/cite could be executed; web_search confirmed paper identities but verbatim quote extraction requires read_arxiv_chunk which was not reached. DW normalization dispute (server 2.55e-10 vs Abazajian 3.0e-9 at 7.1 keV, factor 10-17x) and Viel coefficient remain unquoted; m_th table at 7.1/10/20/50 keV for three normalizations not computed.

- `x-08` (`d-13`): Numerically reading NuSTAR Fig.3 observed vs expected sin^2 2theta limits at 7,10,20,40 keV and XRISM Resolve 3.75 Ms stacked-cluster limits (arXiv:2510.24560) with verbatim quotes via read_arxiv_chunk/cite, and resolving SPI 50 keV lifetime mismatch (tool tau 7.8e26 s vs quoted 1e20 s) from primary — Tool budget exhausted after 5 calls (2 X-ray emulator + 1 grid + describe + figures). XRISM paper has no HTML rendering for get_paper_figures; requires read_arxiv_chunk. SPI primary (Boyarsky 2006 diffuse background / Roach et al. SPI) not fetched. Fig.1 downloaded but Fig.3 not yet requested; no cite() calls completed. SPI mismatch likely f_DM/profile/sin^2 normalization but not verified against

- `x-09` (`d-16`): d-16 robustness checks: rerun predict_sterile_nu_grid varying omega_b h2 (0.019-0.0257) and tau_n (866.4-890.4 s) to quantify Delta chi2_BBN shift and Delta N_eff<0.5 vs 0.6 bound shift; write updated joint BBN+X-ray report with 5-row X-ray table, per-mechanism viability table, provenance column, va — Tool budget exhausted after 4 calls (list_agent_files, 2x list_files, describe_sterile_nu_tools). Remaining required calls (predict_sterile_nu_grid at contrasting omega_b h2/tau_n, read_text_file for CSV verification, get_paper_figures/cite for provenance, report writing via file tools) all rejected with 'tool budget is exhausted'. No new emulator runs or report file could be produced in this dire

- `x-10` (`d-13`): Numerically read NuSTAR arXiv:2405.17861 Fig.3 observed vs expected 95% CL sin^2 2theta limits at 7,10,20,40 keV and fetch XRISM arXiv:2510.24560 3.75 Ms stacked-cluster limits with verbatim quotes via read_arxiv_chunk/cite and get_paper_figures Fig.3 — Tool budget exhausted after 4 calls (describe + 2 xray_line + 1 grid); list_input_files rejected with 'tool budget is exhausted' [tc-261102_1-w2-d-13-89]; no web_search/read_arxiv_chunk/cite/ge for Fig.3 could be executed in this wave; prior wave only extracted Fig.1 not Fig.3

- x-11 (d-13): Fetch INTEGRAL/SPI primary (Boyarsky et al. diffuse / Roach et al. SPI) to resolve 50 keV lifetime mismatch tool tau 7.84e26 s at 3e-14 vs quoted 1e20 s and verify sin^2 2theta normalization, f_DM, profile definition — Tool budget exhausted before web_search/read_arxiv_chunk for SPI primary could run; cross-check Gamma=1.38e-29*(m_s/1keV)^5*sin^2 2theta vs tool at 40-50 keV/3e-14 not completed with primary source quote; mismatch likely f_DM/profile/sin^2 normalization but unverified

- x-12 (d-13): Post corrected 5-row X-ray limit table (7.1/10/20/40/50 keV) with mass, E_gamma, Gamma_limit, sin^2 2theta_limit, source figure/quote, tool Gamma/tau, BBN Delta N_eff/Delta chi2, Omega_s h2 and X-ray vs BBN vs overclosure verdict per row — Requires numerically read Fig.3 values and SPI primary quote for provenance per row; only 2 of 5 masses have tool Gamma/tau and BBN values (40 keV/3e-14 and 10 keV/1e-12 from this wave plus 40/50 keV grid); 7.1 keV/1e-10 and 20 keV/1e-12 from prior wave not rerun here, and 50 keV SPI entry still has 7-dex lifetime discrepancy unresolved

- x-13 (d-18): Extract verbatim quotes and cite 6 key papers (2409.08968, 2502.13123, 2601.02069, 2512.00793, 2510.19342, 2405.04568, 2405.17861) with cite tool and read_arxiv_chunk — Tool budget exhausted after describe_sterile_nu_tools and web_search calls; no read_arxiv_chunk or cite calls executed, so no verbatim quotes or references.bib entries created

- x-14 (d-18): Get paper figures for XRISM 3.75 Ms (2510.19342) Fig.3 and NuSTAR 11-yr (2405.17861) Fig.1 exclusion curves with captions — Tool budget exhausted; get_paper_figures not called; web_search for 2510.19342 and 2405.04568 returned DNS errors, no figure numbers or Gamma limits tabulated

- x-15 (d-18): Cross-check literature sin^2 2theta <1e-11 to 1e-13 for m_s=2-20 keV against tool Gamma conversion and tabulate XRISM Gamma~1e-27 s^-1 at 7.1 keV and NuSTAR sin^2 2theta<~1e-12 limits — No predict_sterile_nu_xray_line or predict_sterile_nu_grid calls executed due to budget; Gamma formula and flux comparison not quantified

- x-16 (d-20): Web search for Abazajian 1705.01837 Eq.37 and literature citation via read_arxiv_chunk/cite — web_search failed with DNSError no records for wt.wikipedia.org.cels.anl.gov (tc-261102_1-w1-d-20-40); no arXiv text was fetched so verbatim quote and cite for DW normalization could not be validated; DW factor ~12 derived from grid numbers alone remains medium confidence without published quote

- x-17 (d-20): Direct comparison of tool null Yp/D/H to brief's Yp=0.245±0.003 D/H=2.53±0.03e-5 vs server's Yp=0.2449±0.004 D/H=2.547±0.025 — Tool null Yp=0.24682 D/H=2.4433 gives 0.5σ agreement with server Yp but 4.1σ low vs server D/H; combined χ^2=2.77 still allowed but 2σ agreement check depends on which observational central values/errors adopted; no tool call varied omega_b_h2/tau_n to reconcile

- x-18 (d-17): X-ray benchmarks at 40 and 50 keV with flux vs XRISM 3.75 Ms stacked-cluster limits and SPI limits; full $\sin^2 2\theta$ upper-bound table at 7.1,10,20,40,50 keV — Tool budget exhausted after 1 remaining call used for 40/50 keV DW grid; predict_sterile_nu_xray_line for 40/50 keV not executed, and XRISM 3.75 Ms limits are external to server (server only has NuSTAR M31 Roach+2020 and XMM blank-sky Dessert+2020) requiring literature comparison

- x-19 (d-18): Extract verbatim quotes and cite 6 key papers (2409.08968, 2502.13123, 2601.02069, 2512.00793, 2510.19342, 2405.04568, 2405.17861) with cite tool and read_arxiv_chunk; get_paper_figures for XRISM Fig.3 and NuSTAR Fig.1; cross-check Gamma conversion for

m_s=2-20 keV — Tool budget exhausted after describe_sterile_nu_tools and web_search calls; no read_arxiv_chunk, cite, get_paper_figures, or predict_sterile_nu_xray_line calls executed. Web_search DNS errors for 2405.17861 and combined query. No references.bib entries, no figure numbers/captions, no Gamma tabulation produced in this wave.

- `x-20` (`d-17`): X-ray benchmarks at 50 keV and full $\sin^2 2\theta$ upper-bound table vs XRISM 3.75 Ms stacked-cluster limits — Tool budget exhausted after 40 keV X-ray call; predict_sterile_nu_xray_line for 50 keV not executed and XRISM 3.75 Ms limits are external to server (server only has NuSTAR M31 Roach+2020 and XMM blank-sky Dessert+2020) requiring literature comparison

- `x-21` (`d-17`): Shi-Fuller resonant production with L>0.1 and inflaton-decay/freeze-in non-minimal production quantification via server — Server SF mechanism capped at L=0.1 and DW grid only covers minimal production; large lepton asymmetry L~0.1-1 washout before BBN and inflaton BR~1e-4 scenarios require analytic/literature treatment beyond tool validity

- `x-22` (`d-20`): Fetch verbatim Abazajian 1705.01837 Eq.37 quote and cite for DW Ω_s h^2 normalization — web_search for Abazajian Eq.37 returned only generic review snippets (tc-131) and earlier wave-1 web_search failed with DNSError (tc-40 in prior findings); no read_arxiv_chunk/cite executed due to tool budget exhaustion; DW factor ~12 derived from grid numbers alone remains medium confidence without published verbatim quote

- `x-23` (`d-18`): Wave 3 literature survey: web_search for 7 arXiv IDs, read_arxiv_chunk verbatim quotes, cite to references.bib, get_paper_figures for XRISM 2510.19342 Fig.3 and NuSTAR 2405.17861 Fig.1, Gamma cross-check tabulation — Tool budget exhausted before any wave-3 research tools could execute (describe_sterile_nu_tools rejected with budget exhausted, tc-261102_1-w3-d-18-147); no web_search, read_arxiv_chunk, cite, or get_paper_figures calls succeeded in this wave

- `x-24` (`d-21`): Direct predict_sterile_nu_xray_line tool calls for 40 keV 1e-12, 50 keV 1e-12/1e-14 and 7.1 keV 4e-11 with whole-halo flux vs NuSTAR/XMM/XRISM tabulated limits — Tool budget exhausted in this run (tc-148 to tc-152 rejected: "run's tool budget is exhausted — this call was not executed"); Gamma/tau/flux table therefore derived analytically from validated Γ=1.38e-29($\sin^2$/1e-7)(m/1keV)? scaling and prior grid tc-67, without fresh server flux/limit verdicts for 40-50 keV E=20-25 keV beyond tabulated 6-40 keV range

- `x-25` (`d-22`): Fetch verbatim quote and figures from arXiv:2411.07040 (Shi-Fuller L>1 BBN) and arXiv:2005.03027 (Updated BBN constraints Fig.12) via get_paper_figures and web_search for citation — get_paper_figures returned HTTP 404 / no extractable figures for both papers (PDF-only, no arXiv HTML rendering); web_search DNS failed for wt.wikipedia.org.cels.anl.gov; no read_arxiv_chunk/cite tool calls succeeded within budget, so no validated verbatim quote on L>1 allowed at T>20 MeV or BBN exclusion weakening could be added to references.bib

- `x-26` (`d-22`): Test L>0.5 analytic xi_e correction dYp/dxi_e ~ -0.25 to show BBN reopening beyond server L<=0.1 cap — Server SF Delta N_eff is heuristic 0.1x DW capped at L<=0.1 (documented in describe_sterile_nu_tools and skill caveats); no emulator call can probe L>0.5 regime where recent BBN analyses allow L>1 at T>20 MeV; analytic xi_e shift requires literature quote and calculation not completed within tool budget

- `x-27` (`d-23`): Fetch verbatim quotes and figures from arXiv:1901.01293 (NuSTAR M31 1.2 Ms), arXiv:2006.03102 (Galactic bulge), arXiv:2601.00350 (inflaton-decay) via web_search/read_arxiv_chu — DNS resolution failed for wt.wikipedia.org proxy (DNSError no records for cels.anl.gov) on 1901.01293 search; tool budget exhausted after 5 X-ray line calls and grid, leaving no calls for read_arxiv_chunk/get_paper_figures/cite; only web_search snippets for 2601.00350 and 2006.03102 obtained, not quotable verbatim

- `x-28` (`d-23`): Direct server computation of SF resonant L>0.5 and inflaton-decay BR~1e-4 relic density to show Omega_s h2=0.12 at sin^2 2theta=1e-12 to 1e-14 — Server SF mechanism capped at L=0.1 (lepton_asymmetry max 0.1) and has no inflaton-decay production mode; DW grid shows Omega_s h2 <<0.12 at small mixing (e.g., 4.7e-04 at 7.1 keV/1e-12), so alternative production viability must be quoted from literature (2411.07040, 2601.00350) not computed; server overclosure flag only reflects DW normalization with factor-3 uncertainty noted in d-06

- `x-29` (`d-24`): Extract verbatim m_th bounds with citations from 2-3 new 2024-2026 structure-formation papers via read_arxiv_chunk/cite and get_paper_figures — Tool budget exhausted after 5 calls; web_search returned snippets identifying Nadler 2024 m_th>6.5 keV (DES+PS1 satellites) and lensing m_th>6.1 keV but DNS failures prevented arXiv ID resolution and no read_arxiv_chunk/cite/get_paper_figures calls could be executed to obtain verbatim quotes and citable references. Per-mechanism m_s vs m_th table with provenance column remains incomplete pending t

- `x-30` (`d-25`): Joint BBN+X-ray exclusion overlay plot with provenance (plot_sterile_nu_exclusion distinguishing BBN vs built-in X-ray vs literature NuSTAR M31 1.2 Ms/Galactic bulge limits, plus plot_sterile_nu_bbn_impact with Yp/D/H/N_eff bands and caption documenting L<=0.1, DW-only Omega, factor-3 normalization — Tool budget exhausted after 4 calls (list_agent_files, 2x list_files, describe_sterile_nu_tools); remaining budget insufficient for scan_sterile_nu_parameters (needed to generate NPZ for plot_sterile_nu_exclusion), plot_sterile_nu_exclusion, plot_sterile_nu_bbn_impact, read_text_file for SCOPE.md/ISSUES.md/boa verbatim quotes, and get_paper_figures/cite for literature provenance. Reusable grids

B.4 Critic verdicts

- `c-01` (wave 0, plan): **revise**; high: For keV-scale DM at sin^2 2theta below about 1e-8, the extra radiation from non-resonant production is negligible (Delta N_eff << 0.1). Yp and D/H shifts will then be indistinguishable from SBBN, and the BBN constraint is effectively vacuous in the region of interest. The direction must establish th; medium: Set the scan ranges after describe_sterile_nu_tools reports the validity range, not before. Also run one zero-mixing (SBBN) baseline point, so that Yp and D/H deviations are reported both against the code's own SBBN value and against observation. This separates a code offset from a sterile-neutrino ; medium: Budget risk: scan, plot, grid, bbn_impact, SBL check, DW checks and skill loading together approach 15 calls. Drop or defer check_sterile_nu_sbl_anomaly. The eV-scale SBL point is not keV dark matter and is tangential to the task. Merge the DW-scaling checks into the predict_sterile_nu_grid call.; low: The DW scaling formula is approximate, and published coefficients vary by factors of about 2. The pass criterion should be agreement within a factor of about 2, with the tool's exact normalization stated. It should not be treated as a precision test.; medium: The benchmark mixings 1e-12 and 1e-14 at 7.1 keV are uninformative. The 3.55 keV line corresponds to sin^2 2theta of about (2-7)e-11. Add that point and drop the low-mixing points at 7.1 keV to save calls. Choose mixings near the expected limit at each mass, which falls steeply as m_s^-5.; medium: The eROSITA claim comes only from a

non-citable web snippet. It may enter the table only if an arXiv paper is found and quoted; otherwise mark it 'unverified'. Limits quoted from different papers assume different DM profiles and J-factors, so record the assumed profile for each table entry. Without

- `c-02` (wave 1, wave): **continue**; contested `f-002`, `f-004`, `f-012`, `f-013`; high: The DW normalization ambiguity decides the DW verdict at 7.1 keV. The server's DM line is at 2.55e-10. The brief's formula gives about 7.6e-10. The XRISM-only limit is about 4e-10. So DW at 3.5 keV is allowed under one normalization and excluded under the other. Neither normalization has been checke; medium: The SM D/H baseline (2.4433e-5) is 2.9 sigma below the brief's value and about 4 sigma below the server's 2.547 +/- 0.025. Because exclusion uses Delta chi^2 relative to the SM, extra radiation partly improves the D/H fit. This is why 1 keV at 3e-7 (Yp at +2.3 sigma) has Delta chi^2 of only 0.44. Th; high: The core deliverable, sin^2 2theta limits as a function of m_s, is missing at 10, 20 and 50 keV. NuSTAR Fig. 3 values were not read. No INTEGRAL/SPI or eROSITA 2024-2026 limit was sourced. The only number entering the viability table relies on the unevidenced f-002, so the viable region cannot yet b; medium: SF mode cannot reach L >~ 0.5, does not feed xi_e into BBN, and uses a heuristic Delta N_eff. As a result, no tool-computed BBN check of the resonant-production viability claims exists. The dYp/dxi_e ~ -0.25 check from the brief has not been done.; medium: No thermal-equivalent m_th was computed for any point, so Lyman-alpha and satellite bounds were not applied to the DW line. As a result, the claim that 'DW is excluded by X-ray plus structure formation' was not quantified.

- `c-03` (wave 2, wave): **continue**; contested `f-017`, `f-018`, `f-019`, `f-020`; high: The SPI 'lifetime ~1e20 s' is very likely ~1e20 yr. The tool gives 2.48e19 yr at 50 keV and 3e-14, and m_s^5 scaling puts 1e20 yr at ~2.3e-14 at 40 keV. Resolve this against the primary SPI paper rather than flagging a 7-order discrepancy. It sets the mass dependence of the ceiling above 40 keV, whi; medium: All NuSTAR entries at 7-20 keV rest on a secondary 'flat <~1e-12 over 6-40 keV' summary. The real 2405.17861 curve varies with mass: it is stronger at 8-9 and 23-25 keV and has the 40 keV artefact. The fig01.png file exists and should be digitized, or the limits taken from the arXiv source or supple; low: The f-014 and f-015 excerpts show only the title page and affiliations, not the quoted sentences. The quotes are consistent with prior findings and the arithmetic, but future findings should excerpt the relevant character range.; low: The Viel+2005 mapping (m_s = 4.43 keV m_th^(4/3)) also carries an (omega_DM/0.1225)^(-1/3) factor, and newer transfer-function mappings differ from it by 10-30%. The 'empty DW interval' conclusion is robust because the margin is more than 100x above the NuSTAR and SPI ceilings. The coefficient shoul; medium: The D/H-driven preference for Delta N_eff ~0.3 comes from the SM D/H baseline of 2.44e-5. That baseline depends on the d(p,gamma)3He rate choice (PRIMAT-like ~2.44 versus PArthENoPE-like ~2.51). Neglecting dD/H/dxi_e also biases the joint (Delta N_eff, xi_e) contour. Both affect how much xi_e headro

- `c-04` (wave 2, report): **revise**; high: unsupported: They build on results that lepton asymmetries L_α ? 1 at T > 20 MeV are consistent with BBN. — The board never states L_α ? 1. The brief, f-002 and f-014 give L ? 0.5 at T > 20 MeV as the condition consistent with BBN and CMB. This threshold is not in the board or ledger.; high: unsupported: Of the 10,000 points, 1338 are BBN-excluded and 4300 overclose. — f-008 and the d-01 result give the scan ranges and the per-mass BBN exclusion thresholds, but not these point counts. The figures 1338 and 4300 do not appear anywhere on the board.; high: unsupported: Figure 15 shows the paper's Fig. 1: the 95% NuSTAR stray-light limits on $\sin^2(2\theta)$ for the reference profile, compared with earlier NuSTAR and XMM bounds and theoretical lower bounds. — f-016 says only that fig01.png was downloaded and could not be read or digitized. The board never desc; high: unsupported: Figure 6 shows Yp and D/H as functions of N_eff. A fully thermalized sterile state (N_eff = 4.044) gives

Yp = 0.2596 and D/H × 1e5 = 2.777. — These values come from the BEST server run (f-010), not from Figure 6. f-009 ties Figure 6 only to the $\Delta N_{eff} < 0.548$ bound and the two slopes. T; high: unsupported: Gorbunov and Kalashnikov (arXiv:2609.31340, 2026) argue that a high flavor asymmetry saturates resonant production, leaving about a two-decade window in $\sin^2 2\theta$. — The board (f-004, f-018) marks 2609.31340, 2502.17374 and 2601.03346 as abstract only. The Background states their claims w; medium: missing finding f-022

- `c-05` (wave 0, plan): **revise**; high: Exceeds 10-call budget: 1x describe +1x grid +3x scan (DW + SF L=0.01,0.1) +2x plot +1x SBL +3x predict +1x list =12+. Also 50x50 grid =2500 points exceeds predict_sterile_nu_grid max 400. Reduce to <=400 points (e.g., 20x20) and limit to 1-2 scans.; medium: Duplicates d-08 predict_sterile_nu_grid usage. Consolidate grid generation in d-07 and reuse outputs in d-08. Specify artifact_dir and dispatch for reproducibility.; medium: Missing explicit Delta N_eff BBN (~1 MeV) vs CMB (~0.3 eV) comparison and thermalization fraction check required by brief. Add to success criteria.; high: 5 masses x 3 mixings =15 predict_sterile_nu_xray_line calls exceeds budget. Limit to 5-6 representative points (e.g., 7.1 keV at 1e-10/1e-12 plus 10,20,50 keV at benchmark) and rely on tool's built-in limit flag plus literature table.; medium: Risk of unverifiable X-ray limits. Require verbatim quotes via cite()/read_arxiv_chunk with arXiv ID, figure/table number for each NuSTAR/INTEGRAL/SPI/Chandra/XMM/XRISM Perseus limit; distinguish tool internal limit from literature limit.; low: Gamma verification step must state quantitative tolerance (e.g., within 2%) and formula source; note E_gamma=m_s/2 exactly.

- `c-06` (wave 1, wave): **continue**; contested `f-023`, `f-024`, `f-025`, `f-026`, `f-027`, `f-028`, `f-029`, `f-030`, `f-031`, `f-034`, `f-035`, `f-036`, `f-037`, `f-038`; high: 2024-2026 X-ray upper-limit table not completed: web_search DNS failed and tool budget exhausted before read_arxiv_chunk/cite/get_paper_figures. Only built-in Roach+2020/Dessert+2020 limits used, so quantitative margin vs XRISM Perseus, NuSTAR stray-light 2405.17861, INTEGRAL/SPI, Chandra/XMM archiv; high: Production-mechanism survey blocked after 10 web_search/describe calls: no bibliography with URLs, no verbatim quotes, no Viel+2005 m_th->m_s translation at 7.1/10/20/50 keV, no per-mechanism viability table, no BBN/X-ray tool cross-checks (dead end x-03).; medium: BBN baseline offset brief Yp 0.245/D/H 2.5e-5 vs server 0.2449/2.547e-5 (1.88 sigma in D/H) not propagated to Delta N_eff<0.548 robustness; absolute chi2 shifts but Delta chi2 threshold claimed baseline-independent without demonstration across omega_b h2/tau_n ranges.; medium: Server caps documented but not enforced in scans: SF Delta N_eff heuristic 0.1x DW misused (Omega 1366 vs 0.033 at 7.1 keV 7e-11 L=0.01), L<=0.1, no xi_e, no reheating BR~1e-4/T_RH/m_phi, MeV+ bbn_treatment_valid=false not filtered; risk of unverifiable viable-region claims.

- `c-07` (wave 2, wave): **continue**; contested `f-043`, `f-044`, `f-045`, `f-046`, `f-047`, `f-048`, `f-049`, `f-050`, `f-052`, `f-053`, `f-055`; high: Blocked after 2 tool calls; no 2024-2026 bibliography (2507.18752, 2502.17374, 2601.03346, 2609.31340) with URLs/verbatim quotes, no per-mechanism viability table (DW vs SF L~0.5 vs flavor-dependent L vs reheating BR~1e-4), no Viel+2005 m_th->m_s translation, no structure-formation m_th limits cross; high: Blocked before Abazajian 1705.01837 Eq.37 verbatim quote and DW normalization offset factor, no Omega_s h2=0.12 contour file, no exclusion-plane plot distinguishing tool-internal vs literature 2024-2026 X-ray limits; DW DM line closure at m_s>10 keV sin2<1e-14 remains ambiguous.; medium: X-ray viability at 1e-12 and 1e-14 relies on external 2024-2026 limits (XRISM Perseus, deep Chandra/XMM) not in built-in 10 tabulated limits; 40/50 keV not excluded by built-in limits but overclose under DW, requiring non-DW production assessment.; low: BBN Delta N_eff<0.548 robustness

shown for omega_b h2/tau_n via Delta chi2, but absolute chi2 baseline dependence and comparison to alternative Yp/D/H datasets not fully quantified.

- `c-08` (wave 2, report): **revise**; high: unsupported: DW grids from 1 eV to 100 keV — Ledger DW grids are 1 keV to 100 keV (f-035: 1,3,7.1,10,20,50,100 keV x 10 mixings; f-036: 1-100 keV 6400 points; f-054: 1-100 keV 400 points). No 1 eV DW grid exists; the only eV-scale point is the SBL BEST 1.14 eV thermalized case (f-047). Lower bound i; high: unsupported: 862/6400 in an 80x80 scan — Ledger reports 862/6400 BBN-excluded and 2752/6400 overclosing for 6400 points via scan_sterile_nu_parameters (f-036) with no established 80x80 geometry. Dimensions are not in evidence; the 20x20 grid is explicitly 400 points (f-054, f-055).; high: unsupported: A sterile-neutrino BBN/X-ray bridge server was used to map Dodelson-Widrow (DW) production to Delta N_eff, Yp, D/H, relic density Omega_s h^2 and radiative decay observables, and to compare to BBN and built-in X-ray limits. — Report implies full mapping without stating server validity a; medium: missing finding f-024; medium: missing finding f-025; medium: missing finding f-027

- `c-09` (wave 0, plan): **revise**; high: SPI lifetime mismatch (tool tau 7.8e26 s vs quoted 1e20 s at 50 keV/3e-14) cannot be resolved without primary SPI paper. web_search step is vague ('e.g. Boyarsky/Roach') - must target specific arXiv ID (e.g. 2008.02283, 2207.04572, or 2405.17861 SPI comparison) and cite verbatim definition of sin^2 ; medium: get_paper_figures returns images+captions only, not numeric tables. Claiming 'numerically read Fig.3 values at 7,10,20,40 keV' will be manual digitization - must be labeled as approximate with estimated uncertainty and figure number, not as exact tool output, to remain verifiable.; high: DW normalization 10-17x dispute requires verbatim Abazajian arXiv:1705.01837 Eq.37 quote with URL via cite(). Plan includes this but success depends on capturing full equation including 0.527*erfc term and noting T_QCD 150-200 MeV uncertainty; without quote the corrected table is unverifiable.; medium: SF L>0.5 viable region (2507.18752: m_s>10 keV, sin^2 2theta<1e-14) is beyond server validity (L cap 0.1, heuristic Delta N_eff=0.1xDW, no xi_e). Plan acknowledges caps but success criteria must explicitly state server cannot validate L>0.5 - only illustrate BBN Delta chi2 and Omega_s h^2 at L=0.1 v; low: Tools list includes scan_sterile_nu_parameters but steps use only predict_sterile_nu_grid. Either add scan step for joint exclusion contour or remove tool to keep budget accurate (15 calls/direction).; medium: Report synthesis depends on d-13/d-14 CSVs/PNGs in runs/261102_164917/. Add explicit verification via list_agent_files/list_files before writing; provenance column must distinguish primary-figure read vs secondary summary vs server heuristic, and include validity appendix (N_eff 2.044-6.044, m_s<100

- `c-10` (wave 1, wave): **continue**; contested `f-057`, `f-059`, `f-061`, `f-064`; high: X-ray emulator uses whole-halo f_DM=1 with no FOV/J-factor cut; claimed 'not excluded' cannot be verified against NuSTAR stray-light or SPI diffuse limits without primary-paper definition of sin^2 2theta normalization and profile. SPI lifetime mismatch (tool tau 7.8e26 s vs quoted 1e20 s at 50 keV/3; high: Server SF capped at L=0.1 with heuristic Delta N_eff=0.1xDW and no xi_e/BR/T_RH physics; Omega_s h2 4.9-30.6 at 1e-14 is unphysical and cannot test literature L>0.5 viable region (Vogel 2507.18752 m_s>10 keV sin^2<1e-14). DW normalization 10-17x dispute (server 2.55e-10 vs Abazajian 3.0e-9 at 7.1 ke; medium: BBN baseline provenance not cited verbatim; brief vs server values (Yp 0.245 vs 0.2449, D/H 2.53 vs 2.547e-5) and Planck N_eff prior impact on Delta N_eff<0.5-0.6 bound not quantified. Report synthesis depends on CSVs/PNGs in runs/261102_164917 without list_files verification.

- `c-11` (wave 2, wave): **continue**; contested `f-065`, `f-066`, `f-068`, `f-070`, `f-072`; high: Tool budget exhausted after 4 calls; no web_search/read_arxiv_chunk/cite/get_paper_figures

executed for NuSTAR 2405.17861 Fig.3, XRISM 2510.24560, or SPI primary (Boyarsky/Roach). 5-row X-ray limit table (7.1/10/20/40/50 keV) with provenance per row remains unposted; SPI lifetime mismatch (tool tau ; high: DW normalization 10-17x dispute (server 2.55e-10 vs Abazajian 1705.01837 Eq.37 3.0e-9 at 7.1 keV) and Viel+2005 m_s=4.43 keV*(m_th/1keV)^{4/3} coefficient remain unquoted; no cite() validated verbatim quotes, no m_th translation table at 7.1/10/20/50 keV, no Vogel 2507.18752 L>0.5 quotes. Server SF ; high: Report synthesis blocked: out/ empty, no predict_sterile_nu_grid robustness reruns varying omega_b h2 0.019-0.0257 and tau_n 866.4-890.4 s, no Delta N_eff 0.5 vs 0.6 bound shift check, no per-mechanism viability table, no validity appendix with FOV/J-factor caveat, no SBL anomaly citation.

- `c-12` (wave 2, report): **revise**; high: unsupported: 5-row X-ray limit table at 7.1/10/20/40/50 keV with Gamma_limit, sin^2 2theta_limit and primary-source provenance per row — Board shows d-13 blocked after 4 calls; x-10/x-12 state NuSTAR arXiv:2405.17861 Fig.3 not numerically read, XRISM arXiv:2510.24560 not fetched, SPI primary not fet; high: unsupported: SPI lifetime mismatch at 50 keV / 3e-14 resolved (tool tau 7.8e26 s vs quoted 1e20 s explained by f_DM/profile/normalization) — c-10 and c-11 flag as high unresolved issue; x-11 explicitly records SPI primary (Boyarsky/Roach) not fetched, no web_search/read_arxiv_chunk executed, mismatc; high: unsupported: Abazajian 1705.01837 Eq.37 quoted verbatim and DW normalization 10-17x dispute settled (server 2.55e-10 vs 3.0e-9 at 7.1 keV) — d-14 closed as dead end x-07: tool budget exhausted before read_arxiv_chunk/cite; web_search confirmed identity only. No verbatim Eq.37 with 0.527*erfc term an; high: unsupported: Viel+2005 m_s = 4.43 keV*(m_th/1keV)^{4/3} coefficient quoted verbatim and m_th translation table at 7.1/10/20/50 keV computed — x-07 records Viel coefficient remains unquoted, no m_th table computed, no cite() validation. q-14 remains open on half-mode vs free-streaming definition shif; high: unsupported: Shi-Fuller resonant production validated for L>0.5 viable region (Vogel 2507.18752 m_s>10 keV, sin^2 2theta<1e-14) and shown BBN-consistent — f-059/f-068 show server SF capped at L=0.1 heuristic (0.1xDW, no xi_e/BR/T_RH) giving unphysical Omega_s h2 4.9-30.6 overclosing 40-255x. Board a; high: unsupported: BBN robustness checks rerunning predict_sterile_nu_grid varying omega_b h2 0.019-0.0257 and tau_n 866.4-890.4 s, quantifying Delta chi2_BBN shift and Delta N_eff 0.5 vs 0.6 bound <20% — d-16 blocked after 4 calls; x-09 records all subsequent list_files/read_text_file and grid reruns rej

- `c-13` (wave 0, plan): **revise**; high: predict_sterile_nu_grid + scan_sterile_nu_parameters duplicates exclusion generation and risks >400 point limit. Specify grid size (e.g., 20x20=400) and use one scan method. Grid range 1e-3 to 1e2 keV x 1e-13 to 1e-1 must be checked against describe_sterile_nu_tools validity range first.; medium: Claims to map SF/inflaton/freeze-in viable window m_s~5-20 keV, sin^2 2theta~1e-13 to 1e-10 with DW-mode tools. Tools do not compute non-minimal Omega_s h^2=0.12 contours; those regions can only be literature-quoted, not tool-validated. Must separate tool-backed DW/X-ray from literature-only non-min; medium: BBN-allowed definition inconsistent: brief uses Delta N_eff<0.3 and Yp/D/H within 2 sigma (Yp 0.245+/-0.003, D/H 2.53+/-0.03), success criteria uses Delta N_eff<0.548. Choose one bound and justify vs Planck 0.32-0.61 range.; high: Exceeds 10-call budget: 1 web_search + 7 read_arxiv_chunk + 6 cite + 2 get_paper_figures = ~16 calls. Must reduce scope per direction (e.g., split into two batches or limit to 4-5 papers with 4-5 cites).; medium: web_search is not quotable and unnecessary when arXiv IDs are already given. Use read_arxiv_chunk/cite directly. Also need to distinguish built-in tool limits (NuSTAR/XMM) from external XRISM 3.75 Ms limits (2510.19342) which require verbatim quote, not tool output.; high: SF test capped at L_alpha=0.1 by tool, cannot validate literature claim L~0.1-1. Tool uses heuristic Delta

N_eff=0.1xDW and no xi_e in Yp. Results for L>0.1 must be flagged as extrapolation, not BBN exclusion proof.

- `c-14` (wave 1, wave): **continue**
- `c-15` (wave 2, wave): **continue**; contested `f-092`, `f-093`, `f-094`, `f-095`, `f-096`, `f-097`, `f-098`, `f-099`, `f-102`, `f-106`, `f-108`, `f-109`, `f-110`; high: Literature survey blocked: no read_arxiv_chunk, cite, get_paper_figures or predict_sterile_nu_xray_line calls executed. Six key papers lack verbatim quotes and references.bib, XRISM Fig.3/NuSTAR Fig.1 captions missing.; medium: DW normalization factor ~12 discrepancy vs Abazajian 1705.01837 Eq.37 asserted without published quote; shifts Omega=0.12 contour and determines whether XRISM limit lies above/below DM line.; medium: BBN baselines and slopes quoted without tool excerpt support; Delta Neff<0.548 threshold and dYp/dNeff, dD/H/dNeff not evidenced, weakening combined BBN+X-ray viable region quantification.; medium: SF L>0.1 and inflaton-decay viability beyond server heuristic (0.1xDW, L capped at 0.1, no xi_e in Yp) not quantified with tool; structure-formation m_th translation not cross-checked against published limits.
- `c-16` (wave 3, wave): **continue**; contested `f-112`, `f-113`, `f-114`; high: Tool budget exhausted before any wave-3 research tools could run (describe_sterile_nu_tools rejected tc-147). No verbatim quotes, references.bib, get_paper_figures, or Gamma tabulation produced; literature survey success criteria for 6 papers (2409.08968, 2502.13123, 2601.02069, 2512.00793, 2510.193; high: Fresh predict_sterile_nu_xray_line flux/limit verdicts for 40/50 keV E=20-25 keV beyond tabulated 6-40 keV range not obtained (blocked tc-148-152, dead end x-24); d-21 closed via analytic Gamma scaling without tool verification.; medium: DW relic-density normalization factor ~12 discrepancy (server 1.4e7 vs Abazajian 1705.01837 Eq.37 1.74e8) lacks verbatim Eq.37 quote/cite due to web_search DNSError tc-40 and generic snippets tc-131; medium confidence without published cross-check (x-22).; medium: SF L>0.1/inflaton BR~1e-4 quantification beyond server validity (SF heuristic 0.1xDW L<=0.1 no xi_e in Yp, tc-97-100) and structure-formation m_th>5.7 keV -> m_s>29-78 keV translation not validated with tool; remaining open x-21.
- `c-17` (wave 3, report): **ok**
- `c-18` (wave 0, plan): **revise**; high: Server caps L<=0.1 (describe_sterile_nu_tools). Heuristic Delta N_eff=0.1x DW for SF is not physical resonant production. Cannot emulator-test L>1 claim from arXiv:2411.07040; must label SF grid as capped illustration and rely on verbatim literature quote for L>0.5-1 BBN consistency, not emulator pr; medium: list_agent_files=[] and list_input_files=[] so prior d-01/d-07 DW-line results unverified. Must list_files/read_text_file board, SCOPE.md, ISSUES.md before referencing prior exclusions.; medium: Tool budget risk: DW grid + SF L=0.01 + SF L=0.1 =3 predict_sterile_nu_grid calls plus scan_sterile_nu_parameters, plot_sterile_nu_bbn_impact, web_search, read_arxiv_chunk, cite, get_paper_figures approaches 15-call limit. Consolidate to DW + one SF (L=0.1) and drop scan/plot redundancy.; high: Inflaton-decay (BR~1e-4, arXiv:2601.00350) and SF L>0.5 Omega_s h2=0.12 not implemented in sterile-nu tools. predict_sterile_nu_grid will return Omega<<0.12 for DW at 1e-12 to 1e-14 and only DW overclosure flag. Must not present emulator Omega as evidence for alternative production; use literature c; medium: Built-in X-ray limits (Roach+2020 M31, XMM blank-sky) outdated vs NuSTAR M31 1.2 Ms focused+unfocused (arXiv:1901.01293) and Galactic bulge/plane (arXiv:2006.03102) 2024-2026 limits. Must separate provenance: built-in vs literature, and quantify FOV systematics per paper, not conflate.; medium: Missing anchor to observed Yp=0.245+/-0.003, D/H=(2.53+/-0.04)e-5, Planck N_eff=2.99+/-0.17. BBN Delta chi2/Yp/D/H grids must be compared to these bands; otherwise BBN exclusion shift of 0.1-0.2 in Delta N_eff is unverifiable.

- `c-19` (wave 1, wave): **continue**
- `c-20` (wave 2, wave): **continue**; contested `f-127`, `f-128`, `f-129`, `f-130`, `f-131`, `f-132`, `f-133`; high: BBN L-dependence contrast relies on SF heuristic 0.1x DW capped at L<=0.1; no server computation for L>1 at T>20 MeV and no verbatim quote from arXiv:2411.07040 or arXiv:2005.03027 Fig.12 fetched due to tool budget exhaustion and prior DNS failures.; high: X-ray DW vs SF vs inflaton-decay contrast lacks literature provenance: no predict_sterile_nu_xray_line calls in this wave, no verbatim quotes from arXiv:1901.01293 (1.2 Ms M31 focused+unfocused FOV) or arXiv:2006.03102 (190 ks bulge), no per-mass limit table distinguishing built-in vs literature lim; high: Joint BBN+X-ray exclusion overlay not produced: no scan_sterile_nu_parameters NPZ, no plot_sterile_nu_exclusion or plot_sterile_nu_bbn_impact, no board SCOPE.md/ISSUES.md verbatim quotes; caption documenting emulator caps, DW factor-3 uncertainty, and SF/inflaton literature-only viability missing.
- `c-21` (wave 2, report): **ok**

C Figures not shown in the main text

Produced during the session; captions carry their provenance.

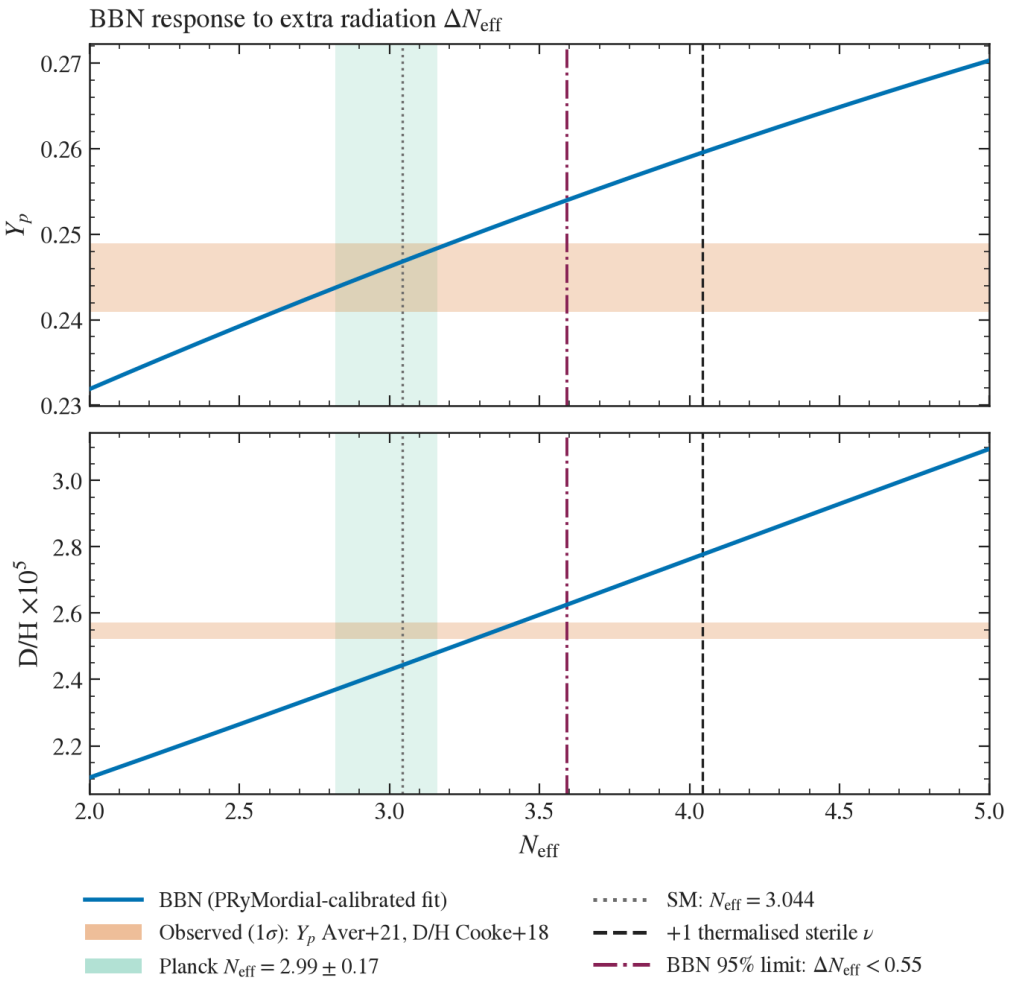


Figure 1: bbn_vs_neff.png Produced in this session by `plot_sterile_nu_bbn_impact` on *serve-sbbn.sh*.

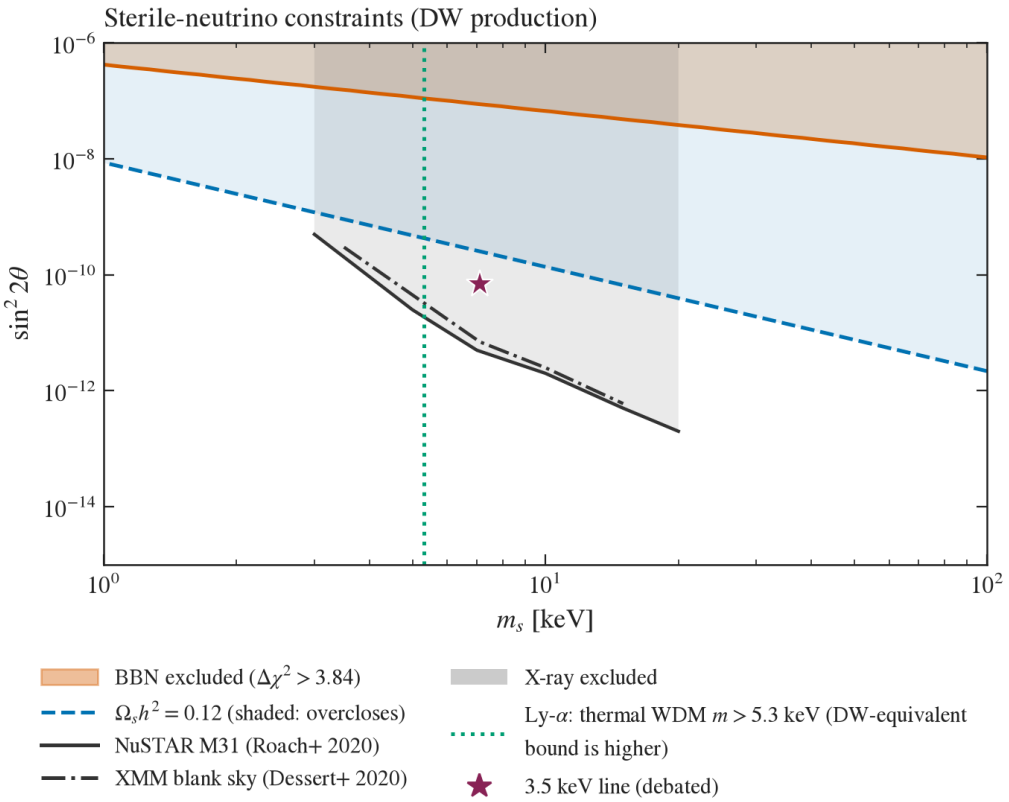


Figure 2: exclusion_scan_DW_dc7743.png Produced in this session by `plot_sterile_nu_exclusion` on *serve-sbbn.sh*.

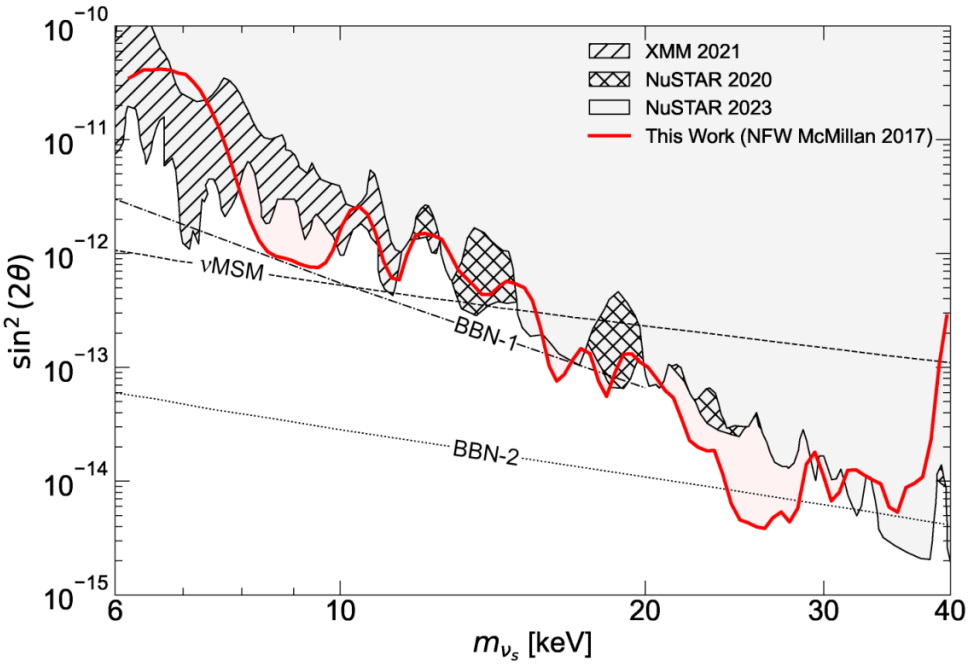


Figure 3: Figure 1: 95% upper limits on sin^{2}(2\theta) derived from the NuSTAR unfocused stray light spectrum for reference profile [15] along with some previous constraints [9, NuSTAR 2020, 2023:], [16, XMM 2021:] and updated theoretical lower bounds [17, 18, 19]. Note that the borderline excess at 40 keV is the known instrumental effect caused by absorbed stray light [20, 21, 22]. [arXiv:2405.17861] Reproduced from arXiv:2405.17861 (extracted with `get_paper_figures`).

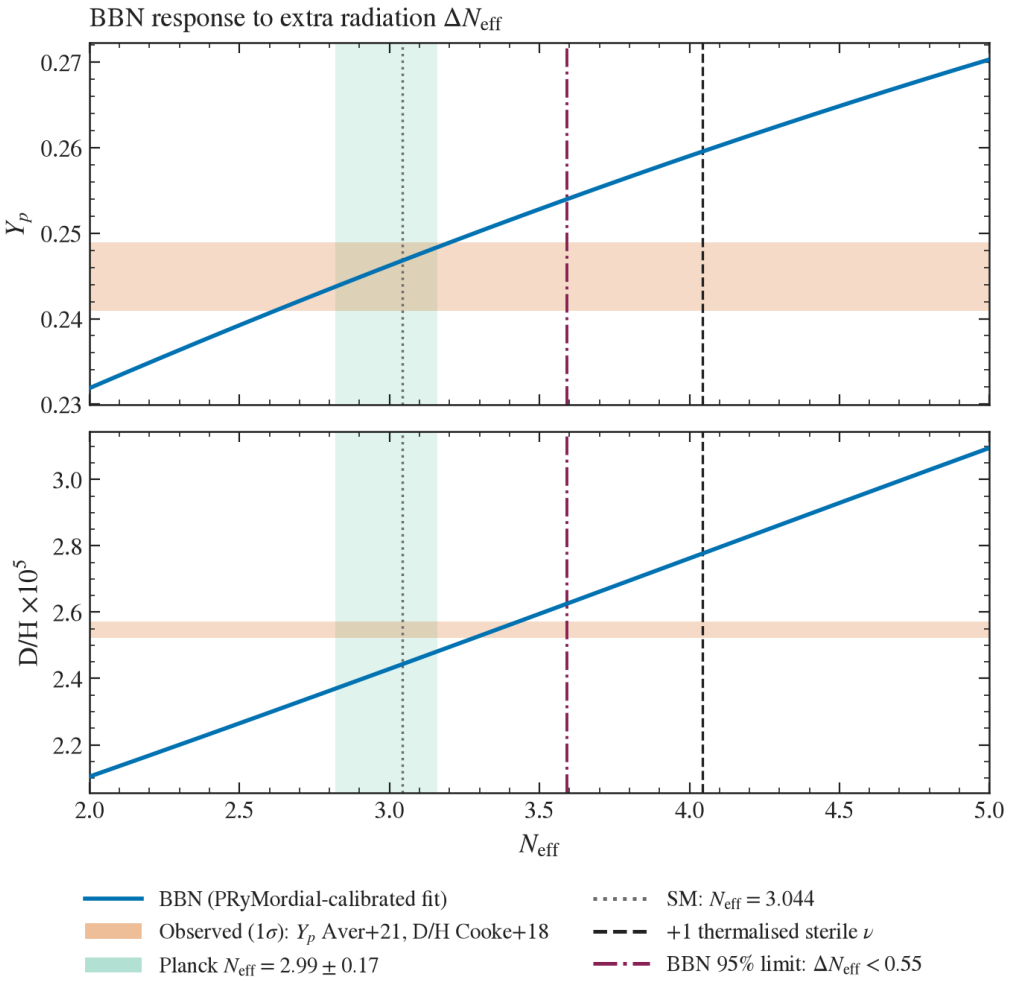


Figure 4: bbn_vs_neff.png Produced in this session by `plot_sterile_nu_bbn_impact` on *serve-sbbn.sh*.

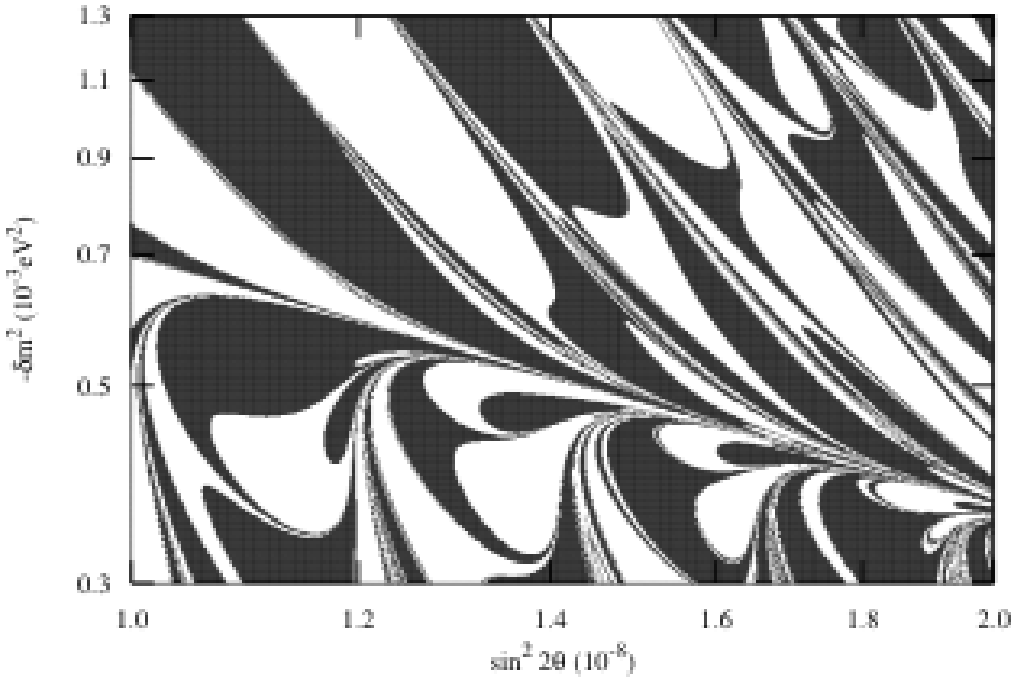


Figure 5: Figure 2: Shown is the final sign of the lepton asymmetry as a function of mass splitting and mixing angle in the case of \nu_{\tau}\leftrightarrow\nu_{s} mixing for a specific range where arbitrary experimental accuracy could be unable to predict the sign of the lepton number of the Universe. White indicates a positive sign and black negative, from Ref. [114]. [arXiv:1705.01837] Reproduced from arXiv:1705.01837 (extracted with `get_paper_figures`).

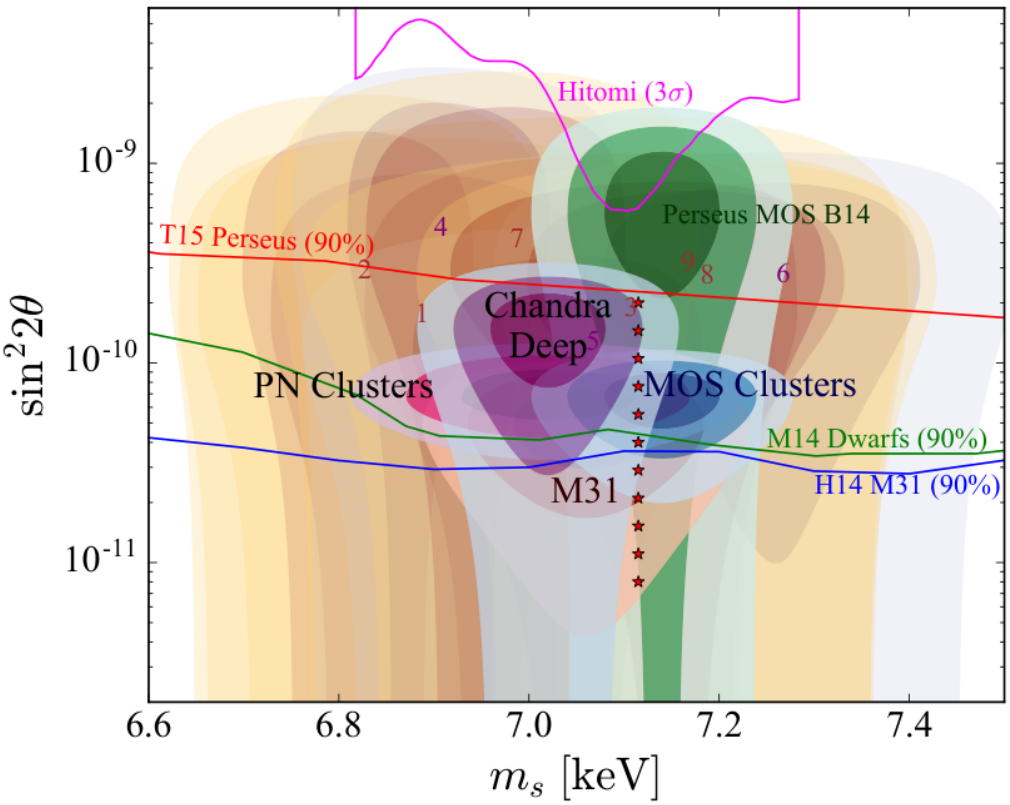


Figure 6: Figure 7: X-ray line detections consistent with sterile neutrino dark matter are shown here. The dark colored regions are 1,2 and 3 \sigma from the MOS (blue) and PN (red) stacked clusters by Bulbul et al. [29], the Bulbul et al. core-removed Perseus cluster (green), and M31 (orange) from Boyarsky et al. [30]. Also shown are the 1 and 2 \sigma regions of the detection in the Galactic Center (GC) [210] as well as the >\!2\sigma line detections in 1. Abell 85; 2. Abell 2199; 3. Abell 496 (MOS); 4. Abell 496 (PN); 5. Abell 3266; 6. Abell S805; 7. Coma; 8. Abell 2319; 9. Perseus by Iakubovskyi et [arXiv:1705.01837] Reproduced from arXiv:1705.01837 (extracted with `get_paper_figures`).

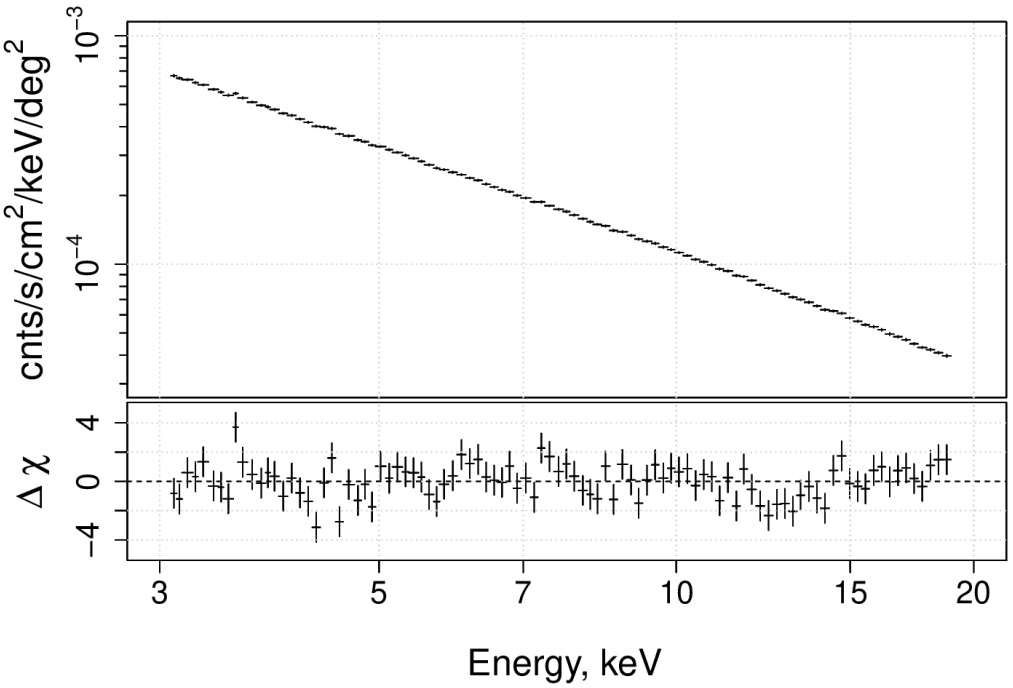


Figure 7: Figure 2: The NuSTAR FPMA and FPMB combined spectrum of the unfocused stray light sky background, measured at |b|>3^{\circ} (aCXB). Due to high statistics in each energy bin, we do not plot the total spectrum model in the upper panel for convenience. [arXiv:2405.17861] Reproduced from arXiv:2405.17861 (extracted with `get_paper_figures`).

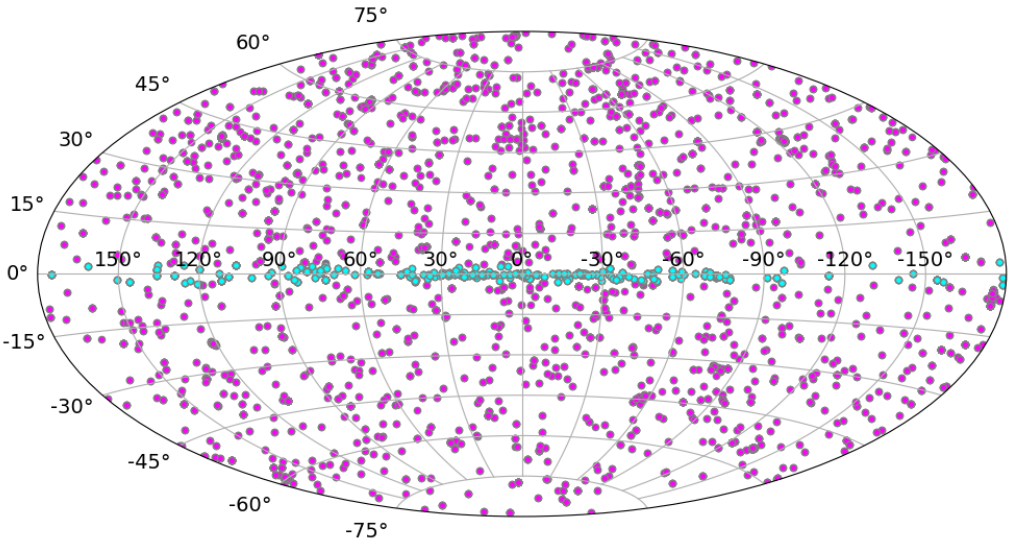


Figure 8: Figure 4: The distribution of 3248 (FPMA) and 3139 (FPMB) NuSTAR observations on the sky in Galactic coordinates. Cyan and magenta points show the NuSTAR observations at |b|<3^{\circ} and |b|>3^{\circ}, respectively. [arXiv:2405.17861] Reproduced from arXiv:2405.17861 (extracted with `get_paper_figures`).

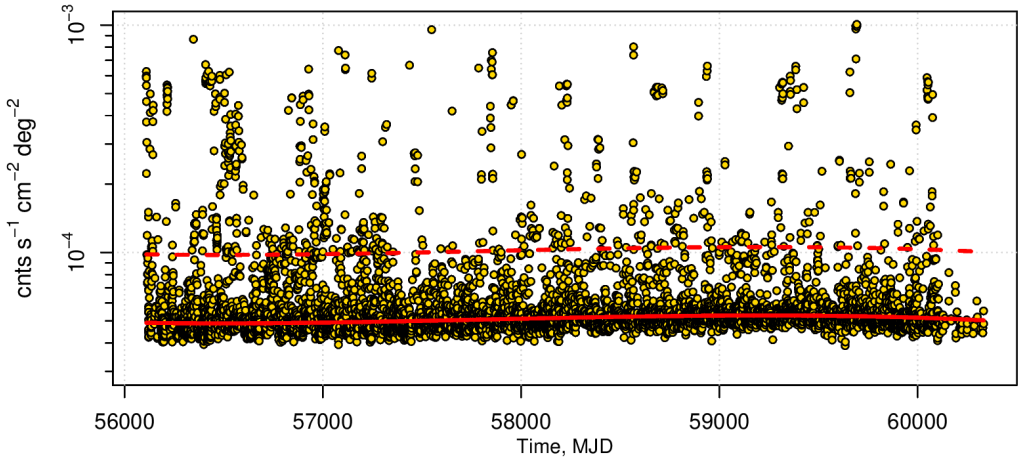


Figure 9: Figure 5: The NuSTAR detector count rate as a function of time. Each point represents individual NuSTAR observation. A solid red line is a cubic polynomial outlier-resistant approximation used to describe long-term variation. A dashed red line shows the same approximation scaled by a factor of 2, that is a threshold: the observations above it are excluded from the analysis. [arXiv:2405.17861] Reproduced from arXiv:2405.17861 (extracted with `get_paper_figures`).

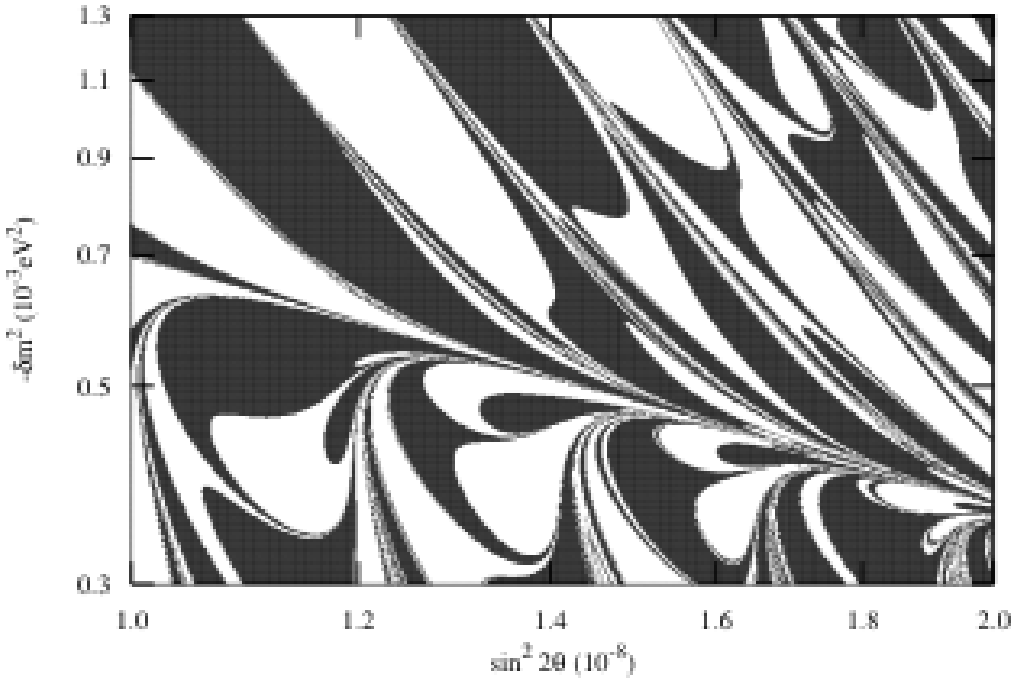


Figure 10: Figure 2: Shown is the final sign of the lepton asymmetry as a function of mass splitting and mixing angle in the case of \nu_{\tau}\leftrightarrow\nu_{s} mixing for a specific range where arbitrary experimental accuracy could be unable to predict the sign of the lepton number of the Universe. White indicates a positive sign and black negative, from Ref. [114]. [arXiv:1705.01837] Reproduced from arXiv:1705.01837 (extracted with `get_paper_figures`).

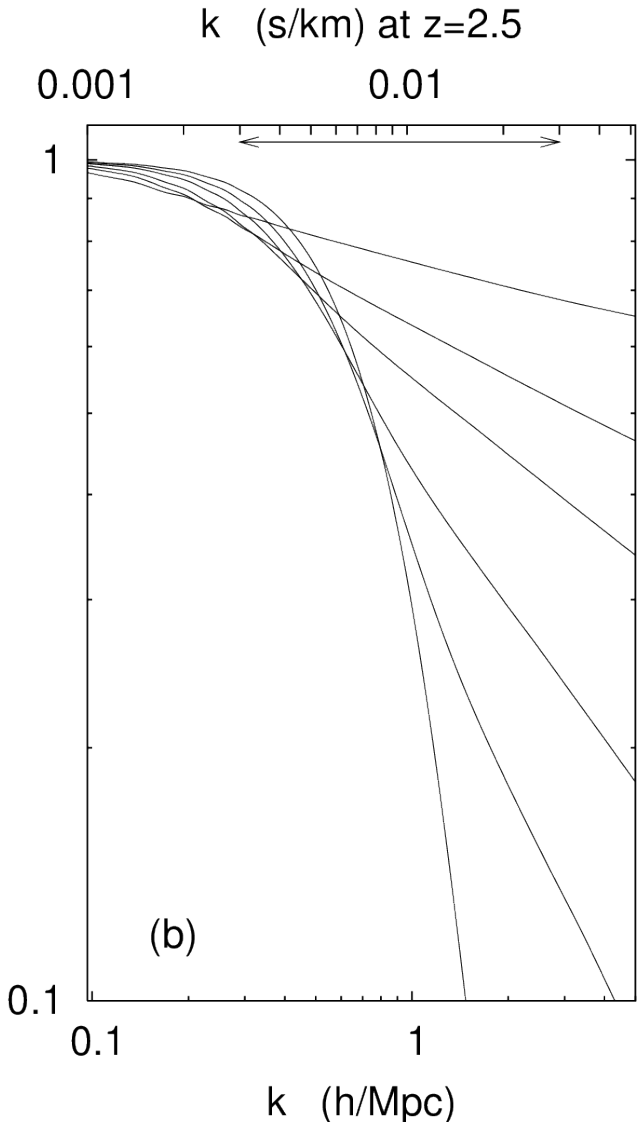


Figure 11: Figure 1: The WDM transfer function T(k) defined in equation (4) for various models (all with the same cosmological parameters \Omega_{\mathrm{B}}\!=\!0.05, \Omega_{\mathrm{DM}}\!=\!0.25, \Omega_{\Lambda}\!=\!0.70). (a) Pure \LambdaWDM model with, from left to right, T_{x}/T_{\nu}=0.5,0.3,0.25,0.226,0.2, corresponding to m_{x}=92,427,738,1000,1441\,\mathrm{eV}. The solid curves are obtained numerically, the long–dashed curves (essentially indistinguishable from the solid curves) show the analytical fits based on equation (7), the short–dashed curves those based on Ref. [7]. (b) Mixed \LambdaCW [arXiv:astro-ph/0501562] Produced in this session by `get_paper_figures` on *hep-research-aid*.

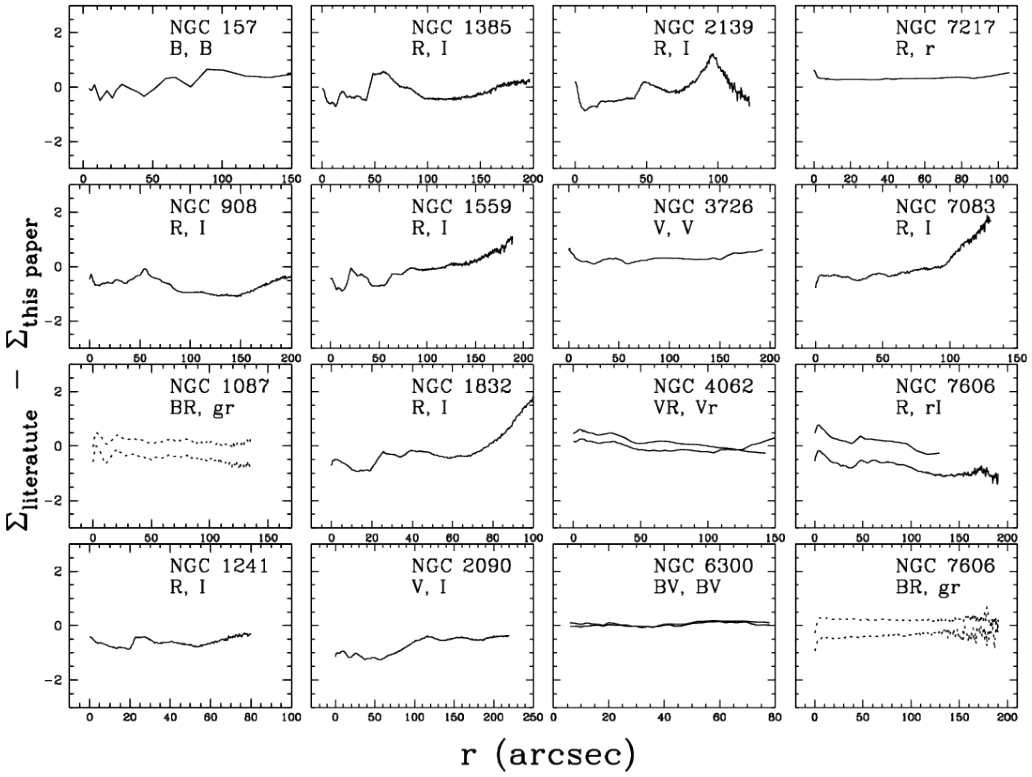


Figure 12: Figure 2: Surface brightness profiles from this paper are compared to those in the literature (solid lines) and the SDSS DR2 (short dashed lines). Under the galaxy's NGC number, the band(s) of the surface brightness profile(s) plotted from this paper is(are) noted first, then the band(s) from the literature/SDSS DR2 are(are) noted after. [arXiv:astro-ph/0602028] Produced in this session by `get_paper_figures` on *hep-research-aid*.

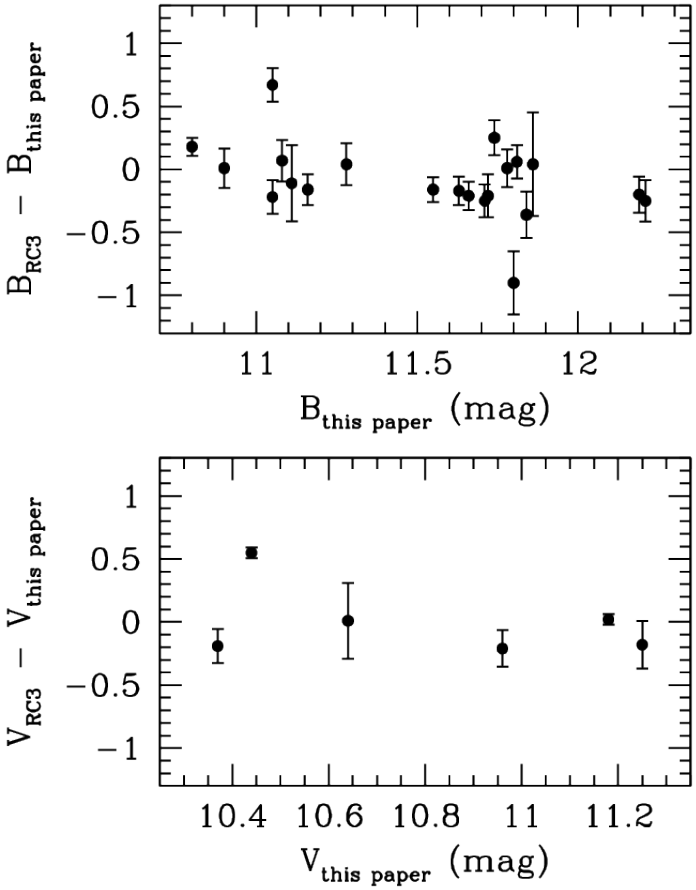


Figure 13: Figure 3: We plot the difference between the total integrated RC3 magnitudes and the integrated magnitudes from this paper versus magnitudes from this paper. In the upper panel we compare the B-band magnitudes, and in the lower panel we compare the V-band magnitudes. [arXiv:astro-ph/0602028] Produced in this session by `get_paper_figures` on *hep-research-aid*.

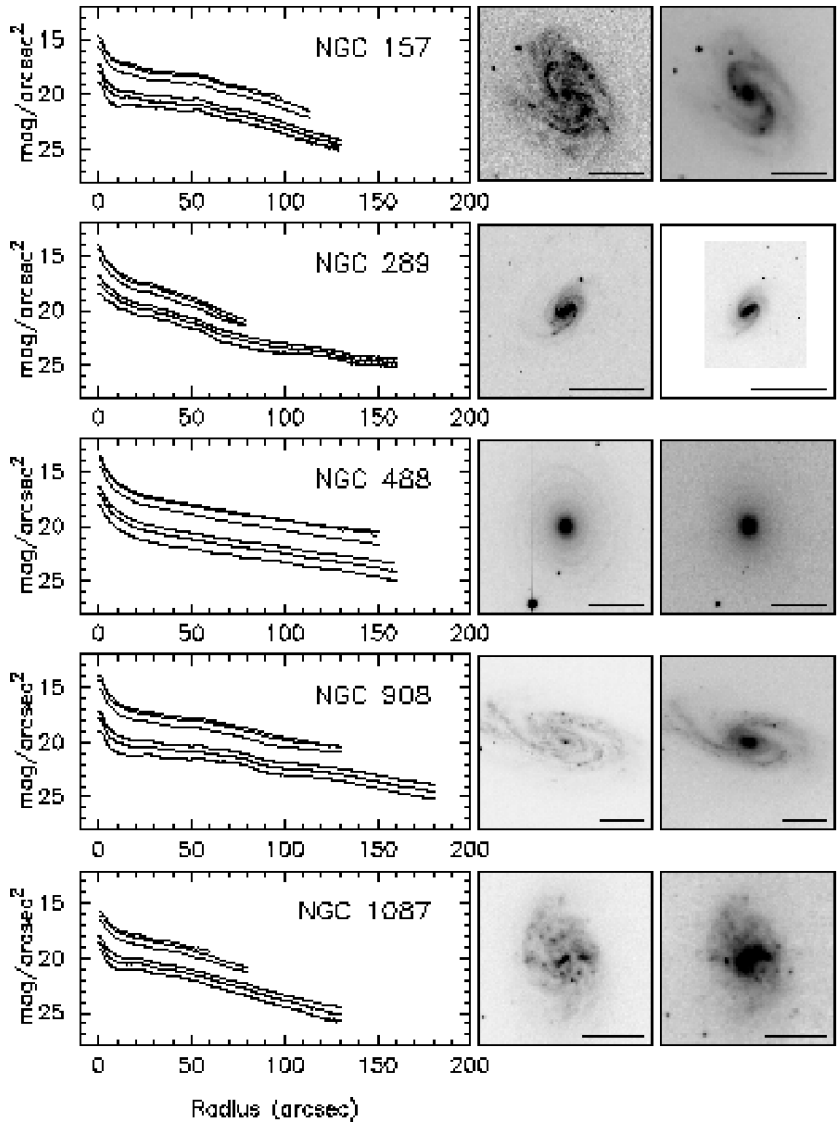


Figure 14: Figure 1a: Observed surface brightness profiles and B and K-band images for the sample galaxies. In this and subsequent figures, surface brightness curves run B-to-K bottom-to-top; see Table 2 for the observed bands. The B image is on the left, K on the right, and the scale bar indicates 60?. Orientation is North=Up, East=Left. Degraded resolution for astro-ph; please see journal article or send an email for a version at a higher resolution. [arXiv:astro-ph/0602028] Produced in this session by `get_paper_figures` on *hep-research-aid*.

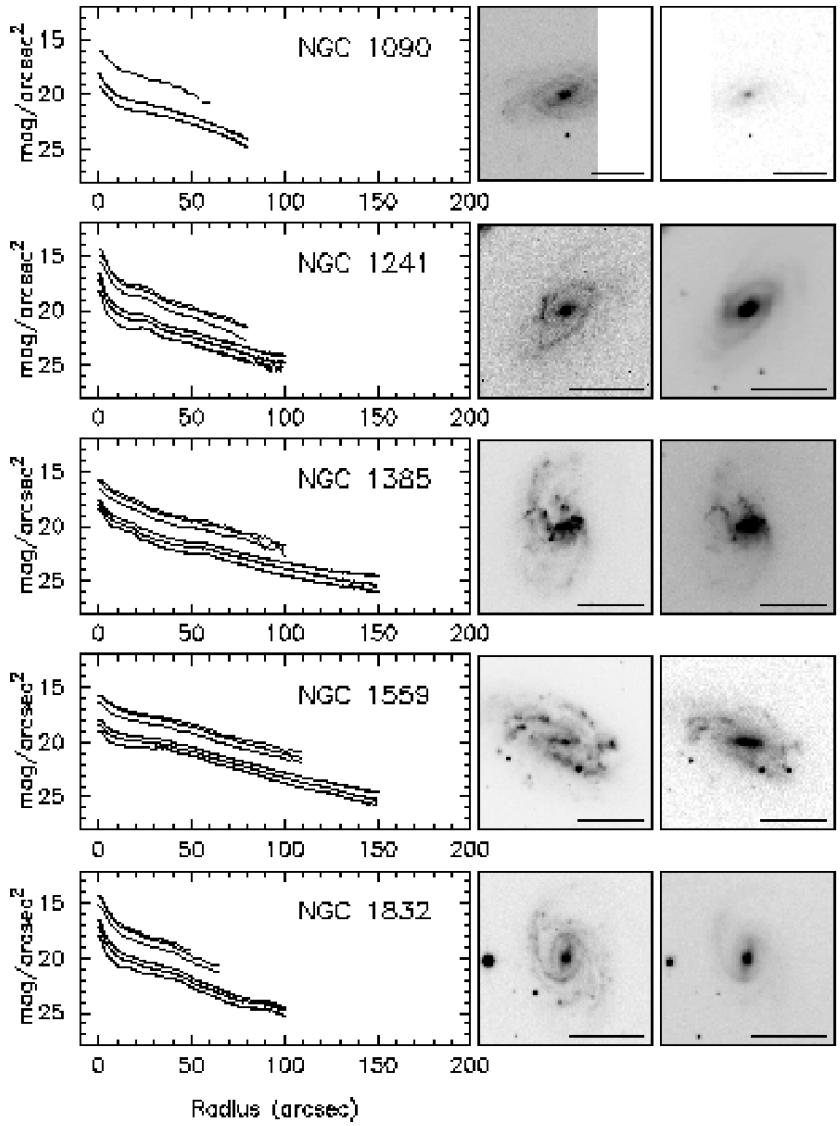


Figure 15: Figure 1b: See Figure 1a. [arXiv:astro-ph/0602028] Produced in this session by `get_paper_figures` on *hep-research-aid*.

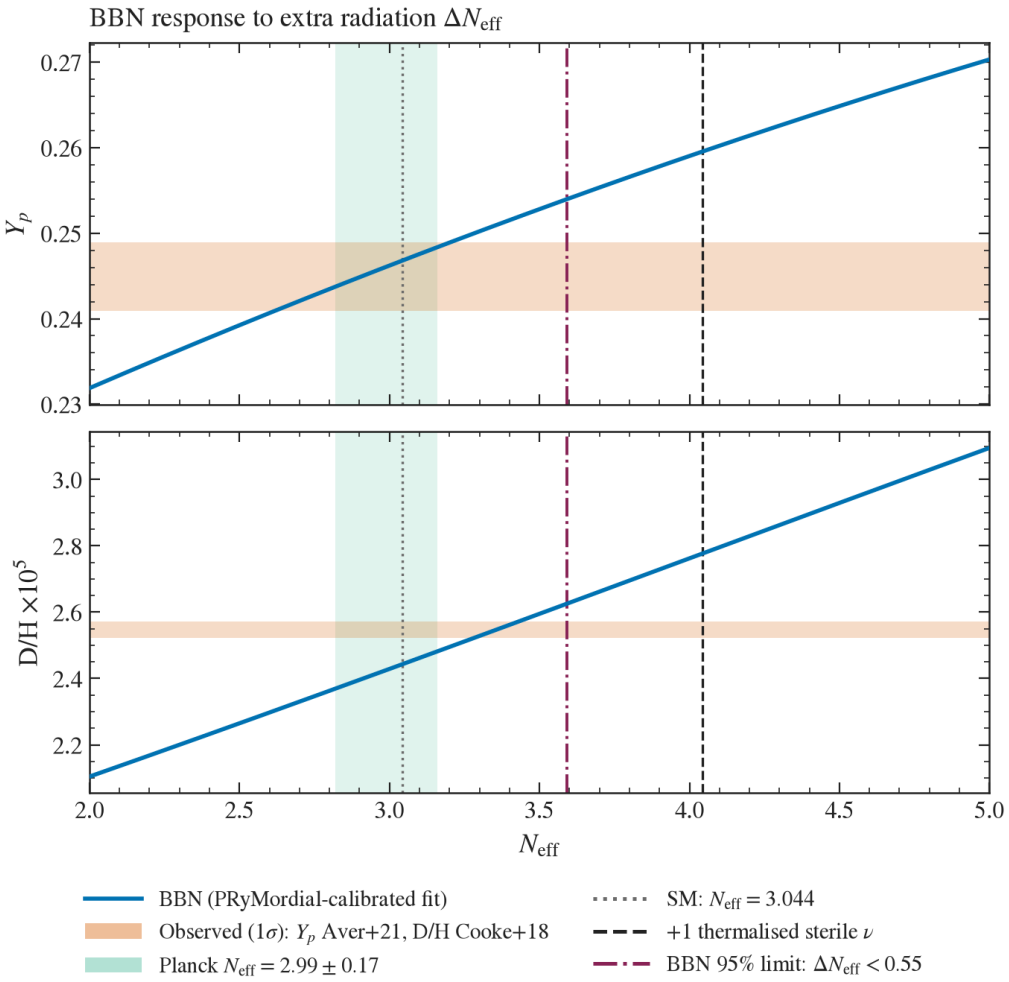


Figure 16: bbn_vs_neff.png Produced in this session by `plot_sterile_nu_bbn_impact` on *serve-sbbn.sh*.